

# PROBABILITY

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## PROB PRACTISE

**2.1** List the elements of each of the following sample spaces:

- (a) the set of integers between 1 and 50 divisible by 8;
- (b) the set  $S = \{x \mid x^2 + 4x - 5 = 0\}$ ;
- (c) the set of outcomes when a coin is tossed until a tail or three heads appear;
- (d) the set  $S = \{x \mid x \text{ is a continent}\}$ ;
- (e) the set  $S = \{x \mid 2x - 4 \geq 0 \text{ and } x < 1\}$ .

- $8 \times 6 = 48$   
 $S = \{8, 16, 24, 32, 40, 48\}$

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**\*\* (b) The set  $S = \{x \mid x^2 + 4x - 5 = 0\}$  \*\***

We need to solve the quadratic equation to find the values of  $x$ . We can factor it:

1. Find two numbers that multiply to  $-5$  and add to  $4$ . These are  $5$  and  $-1$ .
2.  $(x + 5)(x - 1) = 0$
3. Solving for  $x$  gives  $x = -5$  and  $x = 1$ .  
 $S = \{-5, 1\}$

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**\*\* (c) The set of outcomes when a coin is tossed until a tail or three heads appear \*\***

In this experiment, we stop as soon as we see a **T** (Tail) or if we hit a total of **three H** (Heads) in a row.

- **T** (Stopped immediately on first toss)
- **HT** (Stopped on second toss)
- **HHT** (Stopped on third toss)
- **HHH** (Stopped because three heads appeared)  
 $S = \{T, HT, HHT, HHH\}$

**\*\* (d) The set  $S = \{x \mid x \text{ is a continent}\}$ \*\***

This is a straightforward list of the Earth's recognized continents.

$S = \{\text{Africa, Antarctica, Asia, Australia, Europe, North America, South America}\}$

**\*\* (e) The set  $S = \{x \mid 2x - 4 \geq 0 \text{ and } x < 1\}$ \*\***

Let's solve the first inequality:

1.  $2x - 4 \geq 0$
2.  $2x \geq 4$
3.  $x \geq 2$

Now we look at the full condition:  $x \geq 2$  and  $x < 1$ .

- There is no number that is simultaneously greater than or equal to 2 AND less than 1.  
 $S = \emptyset$  (or the empty set  $\{\}$ )

**2.7** Four students are selected at random from a chemistry class and classified as male or female. List the elements of the sample space  $S_1$ , using the letter  $M$  for male and  $F$  for female. Define a second sample space  $S_2$  where the elements represent the number of females selected.

### 1. Sample Space $S_1$ (All Possible Sequences)

This sample space lists every unique combination of gender for the four students in the order they are selected.

- **0 Females:** MMMM
- **1 Female:** FMMM, MFMM, MMFM, MMMF
- **2 Females:** FFMM, FMFM, FMMF, MFFM, MFMF, MMFF
- **3 Females:** FFFM, FFMF, FMFF, MFFF
- **4 Females:** FFFF

**Resulting Set:**

$S_1 = \{\text{MMMM, FMMM, MFMM, MMFM, MMMF, FFMM, FMFM, FMMF, MFFM, FMFM, MMFF, FFFM, FFMF, FMFF, MFFF, FFFF}\}$

## 2. Sample Space $S_2$ (Number of Females)

In this version, we don't care about the order; we only care about the **total count of females** selected. The number of females in a group of four students can range from zero to four.

- If all are male, the count is **0**.
- If one is female, the count is **1**.
- If two are female, the count is **2**.
- If three are female, the count is **3**.
- If all are female, the count is **4**.

**Resulting Set:**

$$S_2 = \{0, 1, 2, 3, 4\}$$

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## Multiplication Or Fundamental Rule of counting

If an operation can be performed in  $n_1$  ways, and if for each of these a second operation can be performed in  $n_2$  ways, and for each of the first two a third operation can be performed in  $n_3$  ways, and so forth, then the sequence of  $k$  operations can be performed in  $n_1 n_2 \cdots n_k$  ways.

- How many sample points are there in the sample space when a pair of dice is thrown once?
- How many three digit numbers can be formed from the digits 2, 4, 6, and 8 if: (i) repetitions are not allowed      (ii) repetitions allowed

### Case (i): Repetitions are not allowed

The first digit has 4 choices, the second has 3, and the third has 2.

- $4 \times 3 \times 2 = \mathbf{24 \text{ numbers}}$

### Case (ii): Repetitions allowed

Each of the three positions has 4 choices.

- $4 \times 4 \times 4 = \mathbf{64 \text{ numbers}}$



## taken “r” at a time

$${}_nP_r = \frac{n!}{(n-r)!}.$$

- In one year, three awards (research, teaching, and service) will be given to a class of 25 graduate students in a statistics department. If each student can receive at most one award, how many possible selections are there?

- **Graduate Student Awards:**

3 different awards given to 25 students, with at most one award per student. This is a permutation of 25 items taken 3 at a time ( ${}_{25}P_3$ ):

$${}_{25}P_3 = \frac{25!}{(25-3)!} = 25 \times 24 \times 23 = \mathbf{13,800 \text{ selections}}$$

- A president and a treasurer are to be chosen from a student club consisting of 50 people. How many different choices of officers are possible if
  - (a) there are no restrictions;
  - (b) A will serve only if he is president;
  - (c) B and C will serve together or not at all;
  - (d) D and E will not serve together?

### Why use the complement?

It's easier to count what's **forbidden** and subtract from the total.

What's forbidden? → A is the treasurer.

If A is fixed as treasurer, how many ways can we pick the president? Anyone except A → 49 ways.

So 49 arrangements are forbidden.

---

### Final Answer

$$\underbrace{2450}_{\text{total}} - \underbrace{49}_{\text{A is treasurer}} = \boxed{2401}$$

If a is fixed as treasurer , how many ways we can pick the president – 49 p 1 means 49 ways

Hence the forbidden case amounts to 49

$$2450 - 49 = 2401$$

### (c) B and C will serve together or not at all

They must **both** serve or **neither** serves. Subtract the cases where exactly one of them serves.

Cases where B is president, C is not treasurer: B is president + someone other than C as treasurer = 48 ways. But let me count systematically.

Cases with exactly one of B/C serving:

- B serves, C doesn't: B is president (C not treasurer): 48 ways; B is treasurer (C not president): 48 ways → 96
- C serves, B doesn't: same → 96

Total invalid = 192

$$2450 - 192 = \boxed{2258}$$

---

### (d) D and E will not serve together

Subtract cases where D and E **both** serve (one as president, one as treasurer):

$$2 \times 1 = 2 \text{ ways (DE or ED)}$$

$$2450 - 2 = \boxed{2448}$$

## Circular Permutations

- The number of permutations of  $n$  objects arranged in a circle is  $(n-1)!$ .

**2.43** In how many ways can 5 different trees be planted in a circle?

4!

## Permutations of $n$ objects when they are not all different.

The number of distinct permutations of  $n$  things of which  $n_1$  are of one kind,  $n_2$  of a second kind,  $\dots$ ,  $n_k$  of a  $k$ th kind is

$$\frac{n!}{n_1!n_2!\cdots n_k!}.$$

### 1. Permutations of the digits of 9995

The digits are: 9, 9, 9, 5 (four digits, with 9 repeated 3 times)

$$\frac{4!}{3!} = \frac{24}{6} = \boxed{4}$$

Example 1: Permutations of the digits of 9995: 9995, 9959, 9599, 5999

## 2. Arrangements of STATISTICS

Letters: S, T, A, T, I, S, T, I, C, S

Count each letter:

- S appears 3 times
- T appears 3 times
- I appears 2 times
- A appears 1 time
- C appears 1 time

Total letters = 10

$$\frac{10!}{3! \cdot 3! \cdot 2! \cdot 1! \cdot 1!} = \frac{3,628,800}{6 \times 6 \times 2 \times 1 \times 1} = \frac{3,628,800}{72} = \boxed{50,400}$$

## 3. Arranging 2 Red, 3 Blue, 4 Green chips

Total chips = 2 + 3 + 4 = 9

Since chips of the same color are identical:

$$\frac{9!}{2! \cdot 3! \cdot 4!} = \frac{362,880}{2 \times 6 \times 24} = \frac{362,880}{288} = \boxed{1,260}$$

- In a college football training session, the defensive coordinator needs to have 10 players standing in a row. Among these 10 players, there are 1 freshman, 2 sophomores, 4 juniors, and 3 seniors. How many different ways can they be arranged in a row if only their class level will be distinguished?

**Permutation of objects where all objects are not different**

### Solution

Since we only distinguish by **class level** (not individual players), we treat all players of the same class as identical.

**Count per class:**

- Freshman: 1
- Sophomores: 2
- Juniors: 4
- Seniors: 3

**Total players = 10**

$$\frac{10!}{1! \cdot 2! \cdot 4! \cdot 3!}$$

**Calculating:**

$$= \frac{3,628,800}{1 \times 2 \times 24 \times 6} = \frac{3,628,800}{288} = \boxed{12,600}$$

There are **12,600** different arrangements possible.

## Combinations

- Selection of “r” objects from “n” different objects and when the order is not important.

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

- In how many ways a committee of 3 students can be selected from 4 students.
- From a group of 10 boys and 6 girls a committee of 3 boys and 2 girls are to be selected. In how many ways can this done?

**1. Committee of 3 from 4 students**

$${}_4C_3 = \frac{4!}{3! \cdot 1!} = \boxed{4}$$

**2. Committee of 3 boys and 2 girls**

$${}_{10}C_3 \times {}_6C_2$$

$$= \frac{10!}{3! \cdot 7!} \times \frac{6!}{2! \cdot 4!}$$

$$= 120 \times 15$$

$$= \boxed{1,800}$$



**2.22** In a medical study, patients are classified in 8 ways according to whether they have blood type  $AB^+$ ,  $AB^-$ ,  $A^+$ ,  $A^-$ ,  $B^+$ ,  $B^-$ ,  $O^+$ , or  $O^-$ , and also according to whether their blood pressure is low, normal, or high. Find the number of ways in which a patient can be classified.

**2.33** If a multiple-choice test consists of 5 questions, each with 4 possible answers of which only 1 is correct,

- (a) in how many different ways can a student check off one answer to each question?
- (b) in how many ways can a student check off one answer to each question and get all the answers wrong?

**2.37** In how many ways can 4 boys and 5 girls sit in a row if the boys and girls must alternate?

## 2.22 — Blood Type & Blood Pressure Classification

Using the **multiplication rule**:

$$8 \times 3 = \boxed{24 \text{ ways}}$$

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## 2.33 — Multiple Choice Test (5 questions, 4 choices each)

(a) Ways to answer all 5 questions

Each question has 4 choices:

$$4 \times 4 \times 4 \times 4 \times 4 = 4^5 = \boxed{1,024}$$

(b) Ways to get ALL answers wrong

Each question has 3 wrong choices (out of 4):

$$3 \times 3 \times 3 \times 3 \times 3 = 3^5 = \boxed{243}$$

---

## 2.37 — 4 Boys and 5 Girls alternating in a row

Since there are 5 girls and 4 boys, the only valid alternating pattern is:

G B G B G B G B G

- Arrange 5 girls in their 5 spots:  $5!$
- Arrange 4 boys in their 4 spots:  $4!$

$$5! \times 4! = 120 \times 24 = \boxed{2,880}$$

**2.45** How many distinct permutations can be made from the letters of the word *INFINITY*?

**2.47** How many ways are there to select 3 candidates from 8 equally qualified recent graduates for openings in an accounting firm?

**2.48** How many ways are there that no two students will have the same birth date in a class of size 60?

Letters: I, N, F, I, N, I, T, Y

Count each:

- I appears 3 times
- N appears 2 times
- F appears 1 time
- T appears 1 time
- Y appears 1 time

Total letters = 8

$$\frac{8!}{3! \cdot 2! \cdot 1! \cdot 1! \cdot 1!} = \frac{40,320}{6 \times 2} = \frac{40,320}{12} = \boxed{3,360}$$

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**2.47 — Select 3 from 8 graduates**

Order doesn't matter → **combination**

$${}_8C_3 = \frac{8!}{3! \cdot 5!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = \boxed{56 \text{ ways}}$$

↓



### 2.48 — No two students share a birth date (class of 60)

There are 365 days in a year. We need to assign a distinct birth date to each of 60 students → this is a permutation:

$${}_{365}P_{60} = \frac{365!}{(365 - 60)!} = \frac{365!}{305!}$$
$$= 365 \times 364 \times 363 \times \cdots \times 306$$

$$= \boxed{\frac{365!}{305!}}$$



## Examples (1 – 3)

1. A coin is tossed twice. What is the probability that at least 1 head occurs?
2. A die is tossed once. What is the probability of getting:  
**(a)** an even number   **(b)** a number less than 3   **(c)** a 4 or higher number   **(d)** a 7   **(e)** A number from 1 to 6
3. A die is loaded in such a way that an even number is twice as likely to occur as an odd number. If  $E$  is the event that a number less than 4 occurs on a single toss of the die, find  $P(E)$ .

#### 1. Coin Tossed Twice — P(at least 1 head)

Sample Space: {HH, HT, TH, TT} → 4 outcomes

"At least 1 head" = {HH, HT, TH} → 3 outcomes

$$P(\text{at least 1 head}) = \frac{3}{4} = \boxed{0.75}$$

#### 2. Die Tossed Once

## 2. Die Tossed Once

Sample Space: {1, 2, 3, 4, 5, 6} → 6 outcomes

(a) Even number → {2, 4, 6}

$$P = \frac{3}{6} = \boxed{\frac{1}{2}}$$

(b) Number less than 3 → {1, 2}

$$P = \frac{2}{6} = \boxed{\frac{1}{3}}$$

(c) Number 4 or higher → {4, 5, 6}

$$P = \frac{3}{6} = \boxed{\frac{1}{2}}$$

(d) A 7 → {} (impossible)

$$P = \frac{0}{6} = \boxed{0}$$

(e) Number from 1 to 6 → {1,2,3,4,5,6} (certain)

$$P = \frac{6}{\downarrow} = \boxed{1}$$

### 3. Loaded Die — $P(E)$ , where $E$ : number less than 4

Even numbers are twice as likely as odd numbers.

Let probability of odd number =  $w$ , then even =  $2w$

$$\underbrace{w + w + w}_{\text{odds: 1,3,5}} + \underbrace{2w + 2w + 2w}_{\text{evens: 2,4,6}} = 1$$

$$3w + 6w = 1 \implies 9w = 1 \implies w = \frac{1}{9}$$

So:

- $P(\text{odd number}) = \frac{1}{9}$
- $P(\text{even number}) = \frac{2}{9}$

Event  $E = \{1, 2, 3\}$  (numbers less than 4)

$$P(E) = P(1) + P(2) + P(3) = \frac{1}{9} + \frac{2}{9} + \frac{1}{9} = \boxed{\frac{4}{9}}$$

## Example 5 – 6

5. A card is drawn at random from the well shuffled pack of 52 playing cards. Find the probability that the card:

**(a)** is a Jack      **(b)** is not a Jack

6. In a poker hand consisting of 5 cards, find the probability of holding 2 aces and 3 jacks.

### 5. Card drawn from 52 playing cards

(a) P(Jack)

There are 4 Jacks in a deck of 52 cards

$$P(\text{Jack}) = \frac{4}{52} = \boxed{\frac{1}{13}}$$

(b) P(not a Jack)

$$P(\text{not a Jack}) = 1 - \frac{1}{13} = \boxed{\frac{12}{13}}$$

### 6. Poker Hand — 2 Aces and 3 Jacks

Total ways to pick any 5 cards from 52:

$${}_{52}C_5 = \frac{52!}{5! \cdot 47!} = 2,598,960$$

Favorable outcomes:

- Choose 2 Aces from 4:  ${}_4C_2 = 6$
- Choose 3 Jacks from 4:  ${}_4C_3 = 4$

$$\text{Favorable} = {}_4C_2 \times {}_4C_3 = 6 \times 4 = 24$$

Probability:

$$P = \frac{24}{2,598,960} = \boxed{\frac{1}{108,290}}$$

- A statistics class for engineers consists of 25 industrial, 10 mechanical, 10 electrical, and 8 civil engineering students. If a person is randomly selected by the instructor to answer a question, find the probability that the student chosen is (a) an industrial engineering major and (b) a civil engineering or an electrical engineering major.

### Setup

Total students =  $25 + 10 + 10 + 8 = 53$  students

(a)  $P(\text{Industrial Engineering})$

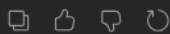
$$P(\text{Industrial}) = \frac{25}{53} = \boxed{0.4717}$$

(b)  $P(\text{Civil OR Electrical Engineering})$

These are **mutually exclusive** events (a student can't be in both), so:

$$P(\text{Civil} \cup \text{Electrical}) = P(\text{Civil}) + P(\text{Electrical})$$

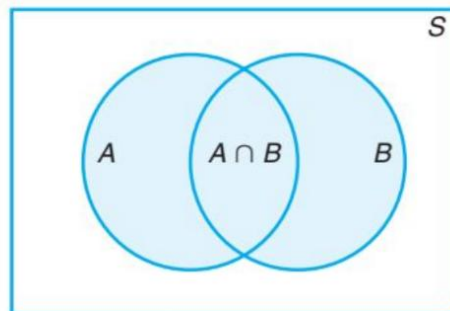
$$= \frac{8}{53} + \frac{10}{53} = \frac{18}{53} = \boxed{0.3396}$$



## Additive Rule: Not - Mutually Exclusive Events

If  $A$  and  $B$  are two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$



For three events  $A$ ,  $B$ , and  $C$ ,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) \\ - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

## Additive Rule: Mutually Exclusive Events

If  $A$  and  $B$  are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B).$$

If  $A_1, A_2, \dots, A_n$  are mutually exclusive, then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n).$$

■

8. John is going to graduate from an industrial engineering department in a university by the end of the semester. After being interviewed at two companies he likes, he assesses that his probability of getting an offer from company  $A$  is 0.8, and his probability of getting an offer from company  $B$  is 0.6. If he believes that the probability that he will get offers from both companies is 0.5, what is the probability that he will get at least one offer from these two companies?
9. What is the probability of getting a total of 7 or 11 when a pair of fair dice is tossed?
10. If the probabilities are, respectively, 0.09, 0.15, 0.21, and 0.23 that a person purchasing a new automobile will choose the color green, white, red, or blue, what is the probability that a given buyer will purchase a new automobile that comes in one of those colors?

### 8. Probability of at least one job offer

Given:

- $P(A) = 0.8, P(B) = 0.6, P(A \cap B) = 0.5$

Using the Addition Rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.8 + 0.6 - 0.5 = \boxed{0.9}$$

## 10. Probability of choosing green, white, red, or blue

These are **mutually exclusive** color choices:

$$\begin{aligned} P(\text{green} \cup \text{white} \cup \text{red} \cup \text{blue}) \\ = 0.09 + 0.15 + 0.21 + 0.23 = \boxed{0.68} \end{aligned}$$

We choose one, so its or condition that's why union

11. If the probabilities that an automobile mechanic will service 3, 4, 5, 6, 7, or 8 or more cars on any given workday are, respectively, 0.12, 0.19, 0.28, 0.24, 0.10, and 0.07, what is the probability that he will service at least 5 cars on his next day at work?
12. Suppose the manufacturer's specifications for the length of a certain type of computer cable are  $2000 \pm 10$  millimeters. In this industry, it is known that small cable is just as likely to be defective (not meeting specifications) as large cable. That is, the probability of randomly producing a cable with length exceeding 2010 millimeters is equal to the probability of producing a cable with length smaller than 1990 millimeters. The probability that the production procedure meets specifications is known to be 0.99.
  - (a) What is the probability that a cable selected randomly is too large?
  - (b) What is the probability that a randomly selected cable is larger than 1990 millimeters?

### 11. P(at least 5 cars)

"At least 5" means 5, 6, 7, or 8+

$$\begin{aligned} P(X \geq 5) &= P(5) + P(6) + P(7) + P(8+) \\ &= 0.28 + 0.24 + 0.10 + 0.07 = \boxed{0.69} \end{aligned}$$

### 12. Computer Cable Specifications ( $2000 \pm 10$ mm)

Given:

- $P(\text{meets specifications}) = 0.99$
- $P(\text{too small}) = P(\text{too large}) \rightarrow \text{symmetric}$

So:  $P(\text{defective}) = 1 - 0.99 = 0.01$

Since defects are equally split:

$$P(\text{too small}) = P(\text{too large}) = \frac{0.01}{2} = 0.005$$

(a)  $P(\text{cable is too large}) = \text{length} > 2010\text{mm}$

$$P(\text{too large}) = 0.005$$

(b)  $P(\text{cable is larger than } 1990\text{mm})$

"Larger than 1990mm" includes:

- Cables that meet specs (1990 to 2010mm)
- Cables that are too large ( $> 2010\text{mm}$ )

$$P(X > 1990) = P(\text{meets spec}) + P(\text{too large})$$

$$= 0.99 + 0.005 = 0.995$$



**2.49** Find the errors in each of the following statements:

- The probabilities that an automobile salesperson will sell 0, 1, 2, or 3 cars on any given day in February are, respectively, 0.19, 0.38, 0.29, and 0.15.
- The probability that it will rain tomorrow is 0.40, and the probability that it will not rain tomorrow is 0.52.
- The probabilities that a printer will make 0, 1, 2, 3, or 4 or more mistakes in setting a document are, respectively, 0.19, 0.34,  $-0.25$ , 0.43, and 0.29.
- On a single draw from a deck of playing cards, the probability of selecting a heart is  $1/4$ , the probability of selecting a black card is  $1/2$ , and the probability of selecting both a heart and a black card is  $1/8$ .



**2.51** A box contains 500 envelopes, of which 75 contain \$100 in cash, 150 contain \$25, and 275 contain \$10. An envelope may be purchased for \$25. What is the sample space for the different amounts of money? Assign probabilities to the sample points and then find the probability that the first envelope purchased contains less than \$100.

### 2.51 Envelope Selection

A box contains 500 envelopes: 75 with \$100, 150 with \$25, and 275 with \$10.

- **Sample Space ( $S$ ):**  $\{\$10, \$25, \$100\}$
- **Probabilities:**
  - $P(\$100) = 75/500 = 0.15$
  - $P(\$25) = 150/500 = 0.30$
  - $P(\$10) = 275/500 = 0.55$
- **Probability (less than \$100):** This includes the \$10 and \$25 envelopes.
  - $P(< \$100) = 0.30 + 0.55 = 0.85$

**2.53** The probability that an American industry will locate in Shanghai, China, is 0.7, the probability that

it will locate in Beijing, China, is 0.4, and the probability that it will locate in either Shanghai or Beijing or both is 0.8. What is the probability that the industry will locate

- (a) in both cities?
- (b) in neither city?

### 2.53 Industrial Location

Given:  $P(S) = 0.7$  (Shanghai),  $P(B) = 0.4$  (Beijing), and  $P(S \cup B) = 0.8$  (Either or Both).

- (a) In both cities?

Use the additive rule:  $P(S \cup B) = P(S) + P(B) - P(S \cap B)$ .

$$0.8 = 0.7 + 0.4 - P(S \cap B) \rightarrow P(S \cap B) = 1.1 - 0.8 = 0.3$$

- (b) In neither city?

This is the complement of "either or both."

$$P(\text{neither}) = 1 - P(S \cup B) = 1 - 0.8 = 0.2$$

**2.56** An automobile manufacturer is concerned about a possible recall of its best-selling four-door sedan. If there were a recall, there is a probability of 0.25 of a defect in the brake system, 0.18 of a defect in the transmission, 0.17 of a defect in the fuel system, and 0.40 of a defect in some other area.

- (a) What is the probability that the defect is the brakes or the fueling system if the probability of defects in both systems simultaneously is 0.15?
- (b) What is the probability that there are no defects in either the brakes or the fueling system?

### 2.56 Automobile Recall

Given probabilities of defects: Brakes ( $B$ ) = 0.25, Transmission = 0.18, Fuel System ( $F$ ) = 0.17, Other = 0.40.

- (a) Probability the defect is brakes OR fueling system if both simultaneous = 0.15?
  - $P(B \cup F) = P(B) + P(F) - P(B \cap F)$
  - $P(B \cup F) = 0.25 + 0.17 - 0.15 = 0.27$
- (b) Probability of no defects in either brakes or fueling system?
  - $P(\text{neither}) = 1 - P(B \cup F) = 1 - 0.27 = 0.73$

Would you like me to explain the "Additive Rule" of probability in more detail to help with

## Examples # 13 – 14

- A small town has one fire engine and one ambulance available for emergencies. The probability that the fire engine is available when needed is 0.98, and the probability that the ambulance is available when called is 0.92. In the event of an injury resulting from a burning building, find the probability that both the ambulance and the fire engine will be available, assuming they operate independently.
- A bag contains 5 red and 7 black balls. A ball is drawn at random from the bag, the color is noted and the ball is replaced. A second ball is then drawn. Find the probability that the first ball is red and the second is black.

### 1. Fire Engine AND Ambulance Available

Given:

- $P(\text{fire engine available}) = 0.98$
- $P(\text{ambulance available}) = 0.92$
- Events are independent

$$P(\text{both available}) = P(F) \times P(A)$$

$$= 0.98 \times 0.92 = \boxed{0.9016}$$

## 2. Ball Drawing (With Replacement)

Bag contains: 5 red + 7 black = 12 balls total

Since the ball is replaced, the two draws are independent.

$$P(\text{1st red}) = \frac{5}{12} \quad P(\text{2nd black}) = \frac{7}{12}$$

$$P(\text{1st red} \cap \text{2nd black}) = \frac{5}{12} \times \frac{7}{12}$$

$$= \frac{35}{144} = \boxed{0.2431}$$



(15). A die is rolled two times. Find the probability of obtaining a 5 on the first thrown and an even number on the second thrown.

(16). The probability that Ahsan will be alive in 30 years is 0.4 and the probability that Bilawal will be alive in 30 years is 0.8. What is the probability that: **(a)** both will be alive in 30 years

**(b)** both of them die **(c)** Ahsan will be alive and B dead.

(17). A town has two fire engines operating independently. The probability that a specific engine is available when needed is

0.96.

**(a)** What is the probability that neither is available when needed?

**(b)** What is the probability that a fire engine is available when needed?

### 15. Die Rolled Twice

P(5 on first) and P(even on second) — independent events

$$P(5) = \frac{1}{6} \quad P(\text{even}) = \frac{3}{6} = \frac{1}{2}$$

$$P(5 \text{ then even}) = \frac{1}{6} \times \frac{1}{2} = \boxed{\frac{1}{12}}$$

---

### 16. Ahsan (A) and Bilawal (B) Alive in 30 years

Given:  $P(A) = 0.4$ ,  $P(B) = 0.8 \rightarrow$  Independent events

So:  $P(A \text{ dies}) = 0.6$ ,  $P(B \text{ dies}) = 0.2$

(a) Both alive

$$P(A \cap B) = 0.4 \times 0.8 = \boxed{0.32}$$

(b) Both die

$$P(A' \cap B') = 0.6 \times 0.2 = \boxed{0.12}$$

(c) Ahsan alive, Bilawal dead

$$P(A \cap B') = 0.4 \times 0.2 = \boxed{0.08}$$

(17). A town has two fire engines operating independently. The probability that a specific engine is available when needed is 0.96.

- (a) What is the probability that neither is available when needed?
- (b) What is the probability that a fire engine is available when needed?

### 17. Two Independent Fire Engines

Given:  $P(\text{available}) = 0.96 \rightarrow P(\text{not available}) = 0.04$

(a)  $P(\text{neither available})$

$$P(\text{neither}) = 0.04 \times 0.04 = \boxed{0.0016}$$

(b)  $P(\text{at least one available})$

$$P(\text{at least one}) = 1 - P(\text{neither})$$

$$= 1 - 0.0016 = \boxed{0.9984}$$



(18). Suppose that we have a fuse box containing 20 fuses, of which 5 are defective. If 2 fuses are selected at random and removed from the box in succession without replacing the first, what is the probability that both fuses are defective?

## The Product Rule: Dependent Events

If in an experiment the events  $A$  and  $B$  can both occur, then

$$P(A \cap B) = P(A)P(B|A), \text{ provided } P(A) > 0.$$

The conditional probability of  $B$ , given  $A$ , denoted by  $P(B|A)$ , is defined by

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \text{ provided } P(A) > 0.$$

Two events  $A$  and  $B$  are **independent** if and only if

$$P(B|A) = P(B) \text{ or } P(A|B) = P(A),$$

assuming the existences of the conditional probabilities. Otherwise,  $A$  and  $B$  are **dependent**.

Without replacing first – dependent

### 18. Two Defective Fuses (Without Replacement)

Given: 20 fuses total, 5 defective

$$P(\text{both defective}) = P(\text{1st defective}) \times P(\text{2nd defective} \mid \text{1st defective})$$

$$= \frac{5}{20} \times \frac{4}{19} = \frac{20}{380} = \boxed{\frac{1}{19} \approx 0.0526}$$

Without replacement – dependent

fuses are defective:

(19). Two cards are drawn in succession from a deck of 52 playing cards without replacement. What is the probability that both cards are spades.

(20). A box contains 8 tickets bearing the numbers 1, 2, 3, 4, 5, 6, 8, 10.

### 19. Two Spades (Without Replacement)

Given: 52 cards, 13 spades

$$P(\text{both spades}) = \frac{13}{52} \times \frac{12}{51}$$

$$= \frac{156}{2652} = \boxed{\frac{1}{17} \approx 0.0588}$$

(20). A box contains 8 tickets bearing the numbers 1, 2, 3, 4, 5, 6, 8, 10. One ticket is drawn and kept aside. Then a second ticket is drawn. What is the probability that both the tickets show even numbers.

### 20. Both Tickets Show Even Numbers (Without Replacement)

Tickets: {1, 2, 3, 4, 5, 6, 8, 10} → 8 total

Even numbers: {2, 4, 6, 8, 10} → 5 even

Odd numbers: {1, 3, 5} → 3 odd

$$P(\text{both even}) = P(\text{1st even}) \times P(\text{2nd even} \mid \text{1st even})$$

$$= \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \boxed{\frac{5}{14} \approx 0.357}$$



$$P(A \cap B) = P(A)P(B|A), \text{ provided } P(A) > 0.$$

The conditional probability of  $B$ , given  $A$ , denoted by  $P(B|A)$ , is defined by

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad \text{provided } P(A) > 0.$$

## Example # 22

**2.76** In an experiment to study the relationship of hypertension and smoking habits, the following data are collected for 180 individuals:

|      | Nonsmokers | Moderate<br>Smokers | Heavy<br>Smokers |
|------|------------|---------------------|------------------|
| $H$  | 21         | 36                  | 30               |
| $NH$ | 48         | 26                  | 19               |

where  $H$  and  $NH$  in the table stand for *Hypertension* and *Nonhypertension*, respectively. If one of these individuals is selected at random, find the probability that the person is

- experiencing hypertension, given that the person is a heavy smoker;
- a nonsmoker, given that the person is experiencing no hypertension.



### Redo 2.76 Using Conditional Probability Formula

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Data Table (Total = 180)

|       | Nonsmokers | Moderate Smokers | Heavy Smokers | Total |
|-------|------------|------------------|---------------|-------|
| H     | 21         | 36               | 30            | 87    |
| NH    | 48         | 26               | 19            | 93    |
| Total | 69         | 62               | 49            | 180   |

(a)  $P(H \mid \text{Heavy Smoker})$

$$P(H \mid \text{Heavy}) = \frac{P(H \cap \text{Heavy})}{P(\text{Heavy})}$$

$$= \frac{30/180}{49/180} = \frac{30}{49} = \boxed{0.6122}$$

(b)  $P(\text{Nonsmoker} \mid NH)$

$$P(\text{Nonsmoker} \mid NH) = \frac{P(\text{Nonsmoker} \cap NH)}{P(NH)}$$

$$= \frac{48/180}{93/180} = \frac{48}{93} = \boxed{0.5161}$$

(21). In a certain college 25% of the students passed Mathematics, 15% of the students passed statistics and 10% of the students passed both mathematics and Statistics. A students is selected at random.

**(a)** if he passed statistics, what is the probability that he passed mathematics.

**(b)** if he passed mathematics, what is the probability that he passed statistics.

(22). Suppose a pair of dice is tossed once. If it is known that one die shows a 3. what is the probability that other die shows a 6.

### 21. College Students — Conditional Probability

Given:

- $P(M) = 0.25, P(S) = 0.15, P(M \cap S) = 0.10$

---

**(a)  $P(\text{Mathematics} \mid \text{passed Statistics})$**

$$P(M \mid S) = \frac{P(M \cap S)}{P(S)} = \frac{0.10}{0.15} = \boxed{\frac{2}{3} \approx 0.667}$$

---

**(b)  $P(\text{Statistics} \mid \text{passed Mathematics})$**

$$P(S \mid M) = \frac{P(M \cap S)}{P(M)} = \frac{0.10}{0.25} = \boxed{0.4}$$

---

(22). Suppose a pair of dice is tossed once. If it is known that one die shows a 3. what is the probability that other die shows a 6.

## 22. Pair of Dice — One Die Shows 3

**Given:** One die shows 3. What is  $P(\text{other die shows 6})$ ?

**Reduced Sample Space** — outcomes where at least one die is 3:

$$\{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (1, 3), (2, 3), (4, 3), (5, 3), (6, 3)\} = 11 \text{ outcomes}$$

**Favorable outcomes** where other die = 6:

$$\{(3, 6), (6, 3)\} = 2 \text{ outcomes}$$

$$P(\text{other is 6} \mid \text{one is 3}) = \frac{2}{11} = \boxed{0.1818}$$



- A card is drawn is random from a deck of ordinary playing cards. What is the probability that it is a diamond, a face card or a king?

### Solution:

Let     A = the card drawn is diamond  
         B = the card drawn is face card, &  
         C = the card drawn is a king  
P (A U B U C) =

### P(Diamond OR Face Card OR King)

Using Addition Rule for 3 events:

$$P(D \cup F \cup K) = P(D) + P(F) + P(K) - P(D \cap F) - P(D \cap K) - P(F \cap K) + P(D \cap F \cap K)$$

#### Count Each:

| Event                                                | Cards | Probability |
|------------------------------------------------------|-------|-------------|
| Diamond (D)                                          | 13    | 13/52       |
| Face cards (F) — J, Q, K of all suits                | 12    | 12/52       |
| Kings (K)                                            | 4     | 4/52        |
| Diamond AND Face ( $D \cap F$ ) — J♦, Q♦, K♦         | 3     | 3/52        |
| Diamond AND King ( $D \cap K$ ) — K♦                 | 1     | 1/52        |
| Face AND King ( $F \cap K$ ) — Kings are face cards! | 4     | 4/52        |
| Diamond AND Face AND King ( $D \cap F \cap K$ ) — K♦ | 1     | 1/52        |

#### Applying the Formula:

$$= \frac{13}{52} + \frac{12}{52} + \frac{4}{52} - \frac{3}{52} - \frac{1}{52} - \frac{4}{52} + \frac{1}{52}$$

Face AND King ( $F \cap K$ ) → Kings are face cards:

4

4/52

Diamond AND Face AND King ( $D \cap F \cap K$ ) →  $K\spadesuit$

1

1/52

Applying the Formula:

$$= \frac{13}{52} + \frac{12}{52} + \frac{4}{52} - \frac{3}{52} - \frac{1}{52} - \frac{4}{52} + \frac{1}{52}$$

$$= \frac{13 + 12 + 4 - 3 - 1 - 4 + 1}{52} = \frac{22}{52} = \boxed{\frac{11}{26} \approx 0.4231}$$



## Example # 25

- Two coins are tossed. What is the conditional probability that two heads results, given that there is at least one head?

### Two Coins Tossed — Conditional Probability

Sample Space: {HH, HT, TH, TT} → 4 outcomes

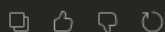
#### Define Events:

- A = at least one head = {HH, HT, TH} →  $P(A) = 3/4$
- B = two heads = {HH} →  $P(B) = 1/4$
- $A \cap B$  = two heads AND at least one head = {HH} →  $P(A \cap B) = 1/4$

#### Apply Formula:

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

$$= \frac{1/4}{3/4} = \boxed{\frac{1}{3} \approx 0.333}$$



## Example # 25

- Two events A & B are such that  $P(A) = 1/4$ ,  $P(A|B) = 1/2$ ,  $P(B|A) = 2/3$
- (i) Are A and B independent events?
- (ii) Are A and B mutually exclusive events?
- iii) Find  $P(A \cap B)$

Solution:

---

(iii) Find  $P(A \cap B)$  — solve this first!

Using the conditional probability formula:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\frac{2}{3} = \frac{P(A \cap B)}{1/4}$$

$$P(A \cap B) = \frac{2}{3} \times \frac{1}{4} = \boxed{\frac{1}{6}}$$

---

Find  $P(B)$

Find P(B)

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$\frac{1}{2} = \frac{1/6}{P(B)}$$

$$P(B) = \frac{1/6}{1/2} = \boxed{\frac{1}{3}}$$

---

(i) Are A and B Independent?

For independence:  $P(A \cap B)$  must equal  $P(A) \times P(B)$

$$P(A) \times P(B) = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12}$$

$$P(A \cap B) = \frac{1}{6} \neq \frac{1}{12}$$

✗ A and B are NOT independent

---

(ii) Are A and B Mutually Exclusive?

For mutually exclusive:  $P(A \cap B)$  must equal 0

$$P(A \cap B) = \frac{1}{6} \neq 0$$

✗ A and B are NOT mutually exclusive



CONDITIONAL PROBABILITY:

Q) In producing screws, let A means "Screw too slim" & B means "screw too short". Let  $P(A) = 0.1$  and let the conditional probability that a slim screw is also too short  $P(B|A) = 0.2$ . What is the probability that screw that we pick randomly from lot produced will be both slim & too short?

$A \cap B$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\therefore P(A \cap B) = P(A) \times P(B|A)$$
$$= 0.1 \times 0.2$$
$$= 0.02$$

= 2% chance. Ans

$$P(A) = 0.1$$

$$P(B|A) = 0.2$$



3/6



Q) Let  $A$  and  $B$  be two events with  $P(A) = \frac{3}{8}$ ,  $P(B) = \frac{7}{8}$  &  $P(A \cap B) = \frac{1}{2}$ . Find  $P(A|B)$  and  $P(B|A)$ .

Sol<sup>n</sup>  $\Rightarrow$   $P(A|B) = \frac{P(A \cap B)}{P(B)}$

$\checkmark$   $= \frac{1/2}{7/8} = \frac{1}{2} \times \frac{8}{7}$

$\boxed{\frac{4}{7}}$  Ans

$P(B|A) = \frac{P(A \cap B)}{P(A)}$

$= \frac{1/2}{3/8} = \frac{1}{2} \times \frac{8}{3}$

$\boxed{\frac{4}{3}}$  Ans

## Engineering

Sachin Kumar



PROBLEMS

2 ADVANCED MATHEMATICS  
BASIC PROBABILITY  
PICK BALLS FROM BULB  
PROBLEMS 13:

3 ADVANCED MATHEMATICS  
AXIOMATIC APPROACH TO PROBABILITY  
PROBLEMS 11:

ADVANCED MATHEMATICS  
CONDITIONAL PROBABILITY  
PROBLEMS 12:

5 ADVANCED MATHEMATICS  
PROBABILITY  
BAYES' THEOREM  
PROBLEMS 7:

6 ADVANCED MATHEMATICS  
PROBABILITY  
INDEPENDENT EVENT  
PROBLEMS 3:

ADVANCED MATHEMATICS  
PROBABILITY



Note:

$P(A \cup B)$  is given.

to find  $P(A \cap B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

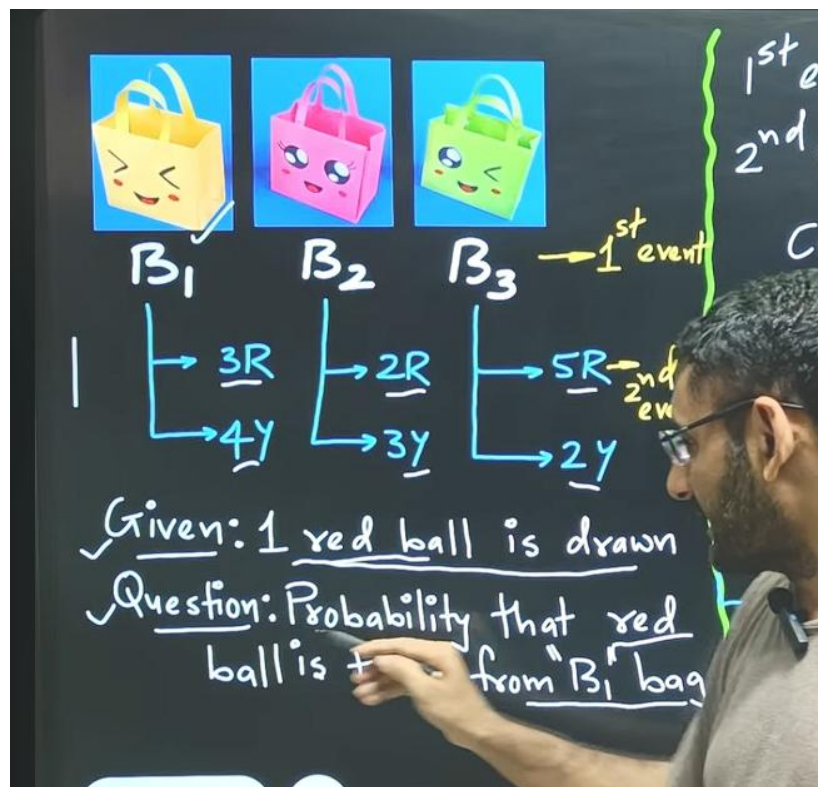
Q) In a certain assembly plant, three m/c,  $B_1, B_2$  and  $B_3$  produce 30%, 45% & 25% of the products. It is known from the past experience that 2%, 3% & 2% of the products made by each machine are defective. Suppose that a finished product is randomly selected. What is the probability that is defective.

$$\begin{array}{l|l} P(B_1) = \frac{30}{100} = 0.3 & P(A|B_1) = \frac{2}{100} = 0.02 \\ P(B_2) = \frac{45}{100} = 0.45 & P(A|B_2) = \frac{3}{100} = 0.03 \\ P(B_3) = \frac{25}{100} = 0.25 & P(A|B_3) = \frac{2}{100} = 0.02 \end{array}$$

$$\begin{aligned} P(A) &= P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3) \\ &= (0.3)(0.02) + (0.45)(0.03) + (0.25)(0.02) \\ &= 0.0245 \\ &= 2.45\% \end{aligned}$$

TOTAL PROBAB

BAYES THEORM:

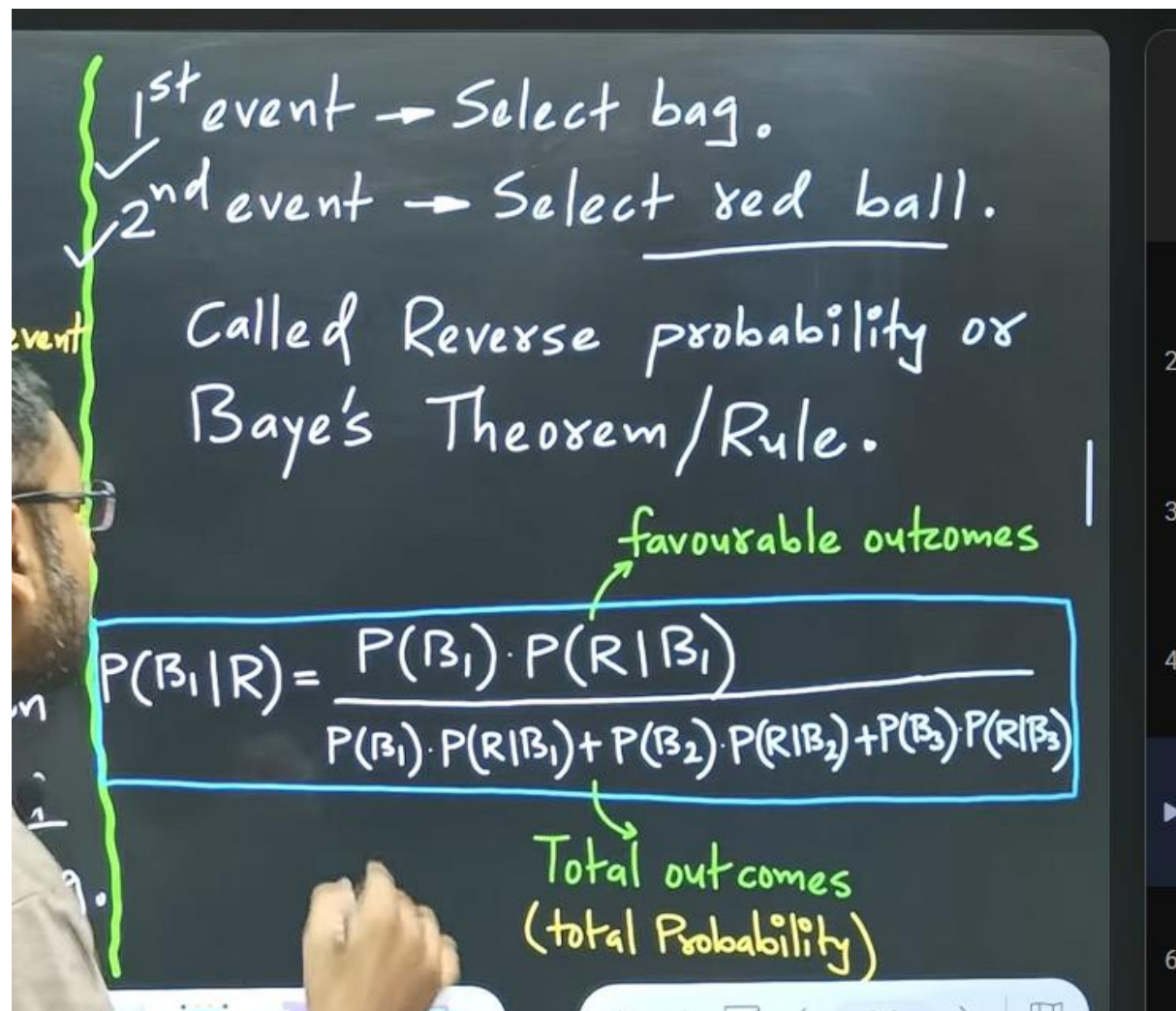


1<sup>st</sup> event  
2<sup>nd</sup> event

$B_1$   $B_2$   $B_3$  → 1<sup>st</sup> event

3R 4Y 2R 3Y 5R 2Y → 2<sup>nd</sup> event

Given: 1 red ball is drawn  
Question: Probability that red ball is from  $B_1$  bag



This means jo humara 2<sup>nd</sup> event hai usme sai hume 1<sup>st</sup> event select karna hai

Jo red ball hume nikalna hai wo b1 bag sai nikle .

Red ball is to be taken from bag 'b1'

2<sup>nd</sup> event sai 1<sup>st</sup> event mai jarhe aesi event ko bayes probability kehte



nd event

on

d

g.

favourable outcomes

$$P(B_1|R) = \frac{P(B_1) \cdot P(R|B_1)}{P(B_1) \cdot P(R|B_1) + P(B_2) \cdot P(R|B_2) + P(B_3) \cdot P(R|B_3)}$$

Total outcomes  
(total Probability)

Given: 1 red ball is drawn

Question: Probability that red ball is taken from B<sub>1</sub>

Q) Bag B<sub>1</sub> contains 5 Red & 3 yellow balls, bag B<sub>2</sub> contains 7 red & 2 yellow balls and bag B<sub>3</sub> contains 3 Red & 8 yellow balls. One Red balls is drawn then find the probability that this (red ball) is drawn from bag B<sub>1</sub>.

Sol<sup>n</sup>

$P(B_1) = \frac{1}{3}$

$P(B_2) = \frac{1}{3}$

$P(B_3) = \frac{1}{3}$

$P(R|B_1) = \frac{5}{8}$

$P(R|B_2) = \frac{7}{9}$

$P(R|B_3) = \frac{3}{11}$

$$P(B_1|R) = \frac{P(B_1) \cdot P(R|B_1)}{P(B_1) \cdot P(R|B_1) + P(B_2) \cdot P(R|B_2) + P(B_3) \cdot P(R|B_3)}$$

$$= \frac{\frac{1}{3} \times \frac{5}{8}}{\frac{1}{3} \times \frac{5}{8} + \frac{1}{3} \times \frac{7}{9} + \frac{1}{3} \times \frac{3}{11}} = \frac{\frac{5}{24}}{\frac{5}{24} + \frac{7}{27} + \frac{1}{11}}$$

$$= \frac{0.2083}{0.5575}$$

Q) Bag  $B_1$  contains 5 Red & 3 yellow balls, bag  $B_2$  contains 7 red & 2 yellow balls and bag  $B_3$  contains 3 Red & 8 yellow balls. One Red balls is drawn then find the probability that this (red ball) is drawn from bag  $B_1$ .

$$P(B_1|R) = \frac{P(B_1) \cdot P(R|B_1)}{P(B_1) \cdot P(R|B_1) + P(B_2) \cdot P(R|B_2) + P(B_3) \cdot P(R|B_3)}$$

$$= \frac{\frac{1}{3} \times \frac{5}{8}}{\left(\frac{1}{3} \times \frac{5}{8}\right) + \left(\frac{1}{3} \times \frac{7}{9}\right) + \left(\frac{1}{3} \times \frac{3}{11}\right)} = \frac{\frac{5}{24}}{\frac{5}{24} + \frac{7}{27} + \frac{1}{11}}$$

$$= \frac{0.2083}{0.2083 + 0.2592 + 0.0909} = \frac{0.2083}{0.5575}$$

$$= 0.3736$$

7.36% Ans

When 1<sup>st</sup> event is not depending on 2<sup>nd</sup> event called independent event.

## RANDOM VARIABLES:

— Random Variable is a Function.  
 — Random Variable is denoted by "X".

Example: 3 coins are tossed then possible outcomes



— Random Variable is a Function.  
 — Random Variable is denoted by "X".

Example: 3 coins are tossed then possible outcomes  
 $\checkmark$   
 $= 2^3$   
 $= 8$

$S = \{$ 

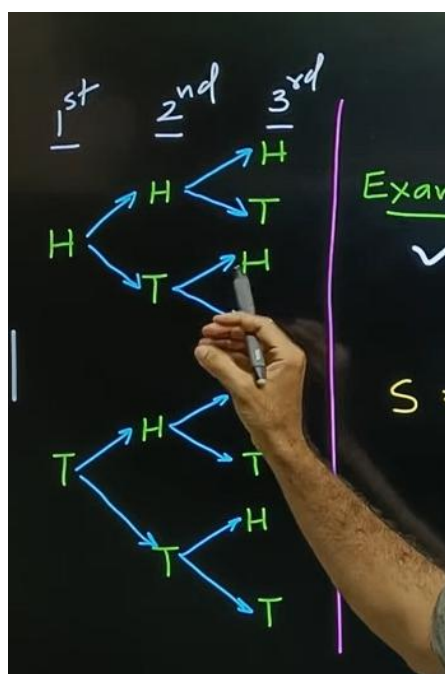
|     |   |   |
|-----|---|---|
| HHH | 3 | ✓ |
| HHT | 2 | ✓ |
| HTH | 2 | ✓ |
| HTT | 1 | ✓ |
| THH | 2 | ✓ |
| THT | 1 | ✓ |
| TTH | 1 | ✓ |
| TTT | 0 | ✓ |

 $\}$

$X = \{0, 1, 2, 3\}$  Random Variables.

When values or numbers are allocated to possible outcomes then these numbers or values are called as Random variables.

We are allocating numbers to possible outcomes such as number of heads



## DISCRETE & CONTINUOUS RANDOM VARIABLES:

### Discrete Variables

- Such variables whose values are countable.
- It will not be in fraction.
- It must be a whole number.

e.g.: 1, 2, 3, 4, 5, 6... Jumping Values.

- Numbers of boys in class - 15
- Numbers of girls in class - 8

1.1, 1.2 cant come

- Numbers of boys in class - 15
- Numbers of girls in class - 8
- Numbers of members in family - 5

|      |     |     |   |     |
|------|-----|-----|---|-----|
| X    | 1   | 2   | 3 | 4   |
| f(x) | 0.6 | 0.7 | 1 | 0.9 |

discrete

We are given jumping values in X

## Continuous Variables.

- Such variables whose values can be measured.
- It may be in fraction or may be a whole number.

e.g.: 1, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8...  
Continuous values.

- Height of students - 4.5<sup>ft</sup>, 4.8<sup>ft</sup>, 5<sup>ft</sup>,...
- Weight of students - 40kg, 41.5kg, 42.2
- Temperature of room - 23°C, 24.5°C, 22.3°C...

|      |     |     |     |     |            |
|------|-----|-----|-----|-----|------------|
| X    | 1.1 | 1.2 | 1.3 | 1.8 | continuous |
| f(x) | 0.6 | 0.9 | 1   | 0.5 |            |



## DISCRETE PROBABILITY DISTRIBUTION

Discrete Probability Distribution.

"X"  
1, 2, 3, ..., n  
Such Values which are countable or whole numbers.

"P(x)"  
0.6, 0.3  
when discrete values are arranged with their respective probabilities.

Properties:

- ①  $0 \leq P(x) \leq 1$
- ②  $\sum P(x) = 1$

| X        | P(X)     |
|----------|----------|
| $x_1$    | $P(x_1)$ |
| $x_2$    | $P(x_2)$ |
| $x_3$    | $P(x_3)$ |
| $\vdots$ | $\vdots$ |
| $x_n$    | $P(x_n)$ |

discrete { } Probability distribution.

# Properties:

- ①  $0 \leq P(x) \leq 1$  ✓
- ②  $\sum P(x) = 1$  ✓

Q) Suppose that a coin is tossed,  
Let '0' denotes the occurrence  
of head & '1' denotes tail.

Then random variable assumed  
values are zero & one. Draw its  
Probability distribution.

Soln

coin  
Head ✓ Tail ✓  
 $X = 0$  ✓  $1$  ✓  
 $P(X) = \frac{1}{2}$   $\frac{1}{2}$

$$P(H) = \frac{\text{favourable}}{\text{Total}} = \frac{1}{2}$$
$$P(T) = \frac{1}{2}$$

Properties

$$0 \leq P(X) \leq 1$$
$$\sum P(X) = 1$$

| X | P(x)          |
|---|---------------|
| 0 | $\frac{1}{2}$ |
| 1 |               |

Ans

Q) Suppose a fair Die is rolled.  
Hence assumed random variables  
between 1 to 6. Find its  
discrete probability distribution.

Sol<sup>n</sup>

|        |               |               |               |               |               |               |
|--------|---------------|---------------|---------------|---------------|---------------|---------------|
|        | Die           |               |               |               |               |               |
| X =    | 1             | 2             | 3             | 4             | 5             | 6             |
| P(X) = | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |

Properties

$$0 \leq P(X) \leq 1$$

$$\sum P(X) = 1$$

| X | P(X)          |
|---|---------------|
| 1 | $\frac{1}{6}$ |
| 2 | $\frac{1}{6}$ |
| 3 | $\frac{1}{6}$ |
| 4 | $\frac{1}{6}$ |
| 5 | $\frac{1}{6}$ |
| 6 | $\frac{1}{6}$ |

Ans

$$P(1) = \frac{\text{favourable}}{\text{Total}}$$

$$= \frac{1}{6}$$

$$P(2) = \frac{1}{6}$$

$$P(3) = \frac{1}{6}$$

$$P(4) = \frac{1}{6}$$

$$P(5) = \frac{1}{6}$$

$$P(6) = \frac{1}{6}$$

Q) A discrete random variables having following probability distribution. Find

|   |   |     |    |     |    |        |
|---|---|-----|----|-----|----|--------|
| 0 | 1 | 2   | 3  | 4   | 5  | X ✓    |
| 0 | k | 0.2 | 2k | 0.3 | 2k | P(X) ✓ |

(a) Value of 'k' (b)  $P(X < 3)$ , (c)  $P(X \geq 3)$   
(d)  $P(2 < X < 5)$ .



Sol<sup>n</sup> (a)  $\sum P(x) = 1$  ✓

$$P(x_1) + P(x_2) + P(x_3) + P(x_4) + P(x_5) + P(x_6)$$

$$0 + k + 0.2 + 2k + 0.3 + 2k = 1$$

$$5k + 0.5 = 1$$

$$5k = 0.5$$

$$\boxed{k = 0.1} \text{ Ans } \checkmark$$

(b)  $P(x < 3) = P(x_1) + P(x_2) + P(x_3)$

$$= 0 + k + 0.2$$

$$= 0.1 + 0.2$$

$$\boxed{= 0.3} \text{ Ans } \checkmark$$

Q) A discrete random variables having following

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |          |
|-------|-------|-------|-------|-------|-------|----------|
| 0     | 1     | 2     | 3     | 4     | 5     | $x$ ✓    |
| 0 ✓   | $k$   | 0.2   | $2k$  | 0.3   | $2k$  | $P(x)$ ✓ |

(a) Value

(d)  $P(2 < x)$

Sol<sup>n</sup> (a)  $\sum P(x) = 1$  ✓

Q) A discrete random variables having following probability distribution. Find

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ | $x$      |
|-------|-------|-------|-------|-------|-------|----------|
| 0     | 1     | 2     | 3     | 4     | 5     | $x$ ✓    |
| 0     | $k$   | 0.2   | $2k$  | 0.3   | $2k$  | $P(x)$ ✓ |

Find (a) Value of 'k' (b)  $P(x < 3)$ , (c)  $P(x \geq 3)$  (d)  $P(2 < x < 5)$ .

Sol<sup>n</sup> (a)  $\sum P(x) = 1$  ✓  
 $P(x_1) + P(x_2) + P(x_3) + P(x_4) + P(x_5) + P(x_6) = 1$   
 $0 + k + 0.2 + 2k + 0.3 + 2k = 1$   
 $5k + 0.5 = 1$   
 $5k = 0.5$  ✓  
 $k = 0.1$  Ans

(b)  $P(x < 3) = P(x_1) + P(x_2)$   
 $= 0 + k$   
 $= 0.1$  Ans

(c)  $P(x \geq 3) = P(x_4) + P(x_5) + P(x_6)$   
 $= 2k + 0.3 + 2k$   
 $= 2(0.1) + 0.3 + 2(0.1)$   
 $= 0.2 + 0.3 + 0.2$   
 $= 0.7$  Ans

(d)  $P(2 < x < 5) = P(x_4) + P(x_5)$   
 $= 2k + 0.3$   
 $= 2(0.1) + 0.3$   
 $= 0.2 + 0.3$   
 $= 0.5$  Ans

## Mathematical Expectation

Discrete Probability Distribution.

| $x$      | $P(x)$   |
|----------|----------|
| $x_1$    | $P(x_1)$ |
| $x_2$    | $P(x_2)$ |
| $x_3$    | $P(x_3)$ |
| $\vdots$ | $\vdots$ |
| $x_n$    | $P(x_n)$ |

discrete values  $\{x_1, x_2, \dots, x_n\}$  and Probabilities  $\{P(x_1), P(x_2), \dots, P(x_n)\}$

$\sum P(x) = 1$   
 $P(x_1) + P(x_2) + \dots + P(x_n) = 1$  ✓

→ If above both conditions are satisfied then mathematical expectation of 'X' or expected value of 'X' is given by "E(X)".

$E(X) = x_1 P(x_1) + x_2 P(x_2) + \dots + x_n P(x_n)$   
 $E(X) = \sum X \cdot P(X)$



$E(x) = x_1 P(x_1) + x_2 P(x_2) + \dots + x_n P(x_n)$  ✓

Q) ✓  $\begin{array}{c|c} X & P(X) \\ \hline 0 & 3/10 \\ 1 & 6/10 \\ 2 & 1/10 \end{array}$  Find  $E(x)$  and  $E(x^2)$  for given discrete probability distribution

Sol<sup>n</sup>  $\Rightarrow E(x) = x_1 P(x_1) + x_2 P(x_2) + x_3 P(x_3)$   
 $= 0 \left(\frac{3}{10}\right) + 1 \left(\frac{6}{10}\right) + 2 \left(\frac{1}{10}\right)$   
 $= 0.6 + 0.2$   
 $= 0.8$  Ans ✓

$E(x^2) = x_1^2 P(x_1) + x_2^2 P(x_2) + x_3^2 P(x_3)$   
 $= 0^2 \left(\frac{3}{10}\right) + 1^2 \left(\frac{6}{10}\right) + 2^2 \left(\frac{1}{10}\right)$   
 $= 0.6 + 0.4 = 1$

Q) A bag contains 4 white and 3 red balls. A person draws 2 balls randomly being promised Rs. 14 for each white ball & Rs. 7 for each red ball be drawn. Find his expectation.

Sol<sup>n</sup>  $\Rightarrow 4 \text{ white} + 3 \text{ red} = 7 \text{ balls}$

$P(2 \text{ white balls}) = \frac{{}^4C_2 \cdot {}^3C_0}{{}^7C_2} = \frac{6 \cdot 1}{21} = \frac{2}{7}$   
 $2 \times 14 = 28$

$P(2 \text{ red ball}) = \frac{{}^4C_0 \cdot {}^3C_2}{{}^7C_2} = \frac{1 \cdot 3}{21} = \frac{1}{7}$   
 $2 \times 7 = 14$

$P(1 \text{ white \& 1 red}) = \frac{{}^4C_1 \cdot {}^3C_1}{{}^7C_2} = \frac{4 \cdot 3}{21} = \frac{4}{7}$   
 $14 + 7 = 21$

expectation  $E(x) = 28 \cdot \frac{2}{7} + 14 \cdot \frac{1}{7} + 21 \cdot \frac{4}{7}$

We want to find expectation , for that we need to find the discrete probability distribution



expectations

$$\begin{aligned}
 E(X) &= X_1 P(X_1) + X_2 P(X_2) + X_3 P(X_3) \\
 &= 28\left(\frac{2}{7}\right) + 14\left(\frac{1}{7}\right) + 21\left(\frac{4}{7}\right) \\
 &= 8 + 2 + 12 \\
 &= 22 \text{ Ans}
 \end{aligned}$$

## RANDOM VARIABLES;

### Topic 1: What is a Random Variable?

A **random variable** is a function that assigns a real number to each outcome in a sample space.

**Example:** You test 3 electronic components. Each is either Non-defective (N) or Defective (D).

Sample space: {NNN, NND, NDN, DNN, NDD, DND, DDN, DDD}

Let  $X$  = number of defectives. Then  $X$  maps each outcome to 0, 1, 2, or 3. That's a random variable.

**Two types:**

- **Discrete** — countable values (0, 1, 2, 3...)
- **Continuous** — any value in an interval (heights, temperatures, time)

A special case: if  $X = 1$  (defective) or  $X = 0$  (not defective), it's called a **Bernoulli random variable**.

### Topic 2: Discrete vs Continuous Sample Space

| Type       | Definition                              | Example                       |
|------------|-----------------------------------------|-------------------------------|
| Discrete   | Finite or countably infinite outcomes   | Number of defectives: 0,1,2,3 |
| Continuous | Infinite outcomes like points on a line | Temperature, time, height     |

### Topic 3: Discrete Probability Distribution (PMF)

The Probability Mass Function (PMF)  $f(x)$  satisfies:

1.  $f(x) \geq 0$
2.  $\sum f(x) = 1$
3.  $P(X = x) = f(x)$

**Example from slides:** 3 helmets returned randomly to Smith, Jones, Brown.  $M$  = number of correct matches.

| Sample Space | m     |       |       |
|--------------|-------|-------|-------|
| SJB          | 3     |       |       |
| SBJ          | 1     |       |       |
| BJS          | 1     |       |       |
| JSB          | 1     |       |       |
| JBS          | 0     |       |       |
| BSJ          | 0     |       |       |
| m            | 0     | 1     | 3     |
| $P(M=m)$     | $1/3$ | $1/2$ | $1/6$ |

Check:  $1/3 + 1/2 + 1/6 = 2/6 + 3/6 + 1/6 = 1 \checkmark$

## Random Variable -

Real valued  $f^n \rightarrow$  Outcome  
Exp.  $\uparrow$

Exp  $\rightarrow \{HH, TT, TH, HT\}$

'X'  $\rightarrow$  Random Variable  
'Number of heads'  $\leftarrow$

$X = \{0, 1, 2\} \leftarrow$

Distribution (PD)

$x_1$   $x_2$   $x_3$   
 $0$   $1$   $2$   
 $\downarrow$   $\downarrow$   $\downarrow$   
 $P(x_1)$   $1/4$   $P_1$   $2/4$   $P_2$   $1/4$   $P_3$

Note  $P_1 + P_2 + P_3 = 1$

$$\sum P_i = 1 \quad \{0 \leq P_i \leq 1\}$$

### Types

D.R.V  $\rightarrow$   $\left. \begin{array}{l} \text{eg} \rightarrow \text{two coins} \\ \rightarrow \text{two die} \\ \rightarrow \text{Bag} \end{array} \right\} \begin{array}{l} \text{small} \\ \text{finite} \end{array}$

C.R.V  $\rightarrow$

## Discrete and Continuous Random variable

a) Discrete Random Variable. A random variable is said to be discrete if it takes only a finite number of values.

For example, in an experiment of tossing a pair of coins, if we define the random variable  $X$  as the number of heads obtained, then the values of random variable  $X$  are 0, 1, 1, 2 corresponding to the outcomes TT, HT, TH and HH respectively.

b) Continuous Random variable. A random variable is said to be continuous if it assume any possible value between certain limits. In such a case it takes any value in an interval.

For example, the height of a person chosen at random from a population of 1000 persons lies between 140 cm and 160 cm. Similarly age, weight etc. are continuous variable.

C.R.V → Height of persons from 100cm to 110cm } large Infinite.

IN DRV, its fixed if we throw 2 coin, its 0, 1, 2, but in crv its not fixed. some people have 100, some have 100.5

1) A coin is biased so that the head is 3 times as likely to occur as a tail. If the coin is tossed twice, find the probability distribution for the number of tails.

Coin is tossed twice. this means  $X$  - number of tails.  $X$  can be 0, 1, 2

$$x + 3x = 1$$

$$x = 1/4$$

$$1/4 \quad 3/4$$

$$P(T) \quad P(H)$$

✓  
small

When tail is coming 0, this means pehli bar bhi head, doosri bar bhi head. HH

SO  $X = 0 - P(0)$  is  $3/4 * 3/4$

Note  $p_1 + p_2 + p_3 = 1$

$$\sum p_i = 1 \quad \{0 \leq p_i \leq 1\}$$

$X \rightarrow$  No. tail

|        |        |        |        |
|--------|--------|--------|--------|
| $X$    | 0      | 1      | 2      |
| $P(X)$ | $9/16$ | $6/16$ | $1/16$ |

$$P(HH) = \frac{3}{4} \times \frac{3}{4} = P(X=0)$$

$$P(HT) + P(TH) = \frac{3}{4} \times \frac{1}{4} \times 2 = \frac{6}{16}$$

$$P(TT) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

$$x + 3x = 1$$

$$x = 1/4$$

|        |        |
|--------|--------|
| $1/4$  | $3/4$  |
| $P(T)$ | $P(H)$ |

$$P(X < 2) = P(0) + P(1)$$

$$+ 1/16 = 1$$

Q4) If  $X$  denotes Random Variable then  $P(X)$ ,

$$P(X=x) = \begin{cases} 0.1 & \text{if } x=0 \\ kx & \text{if } x=1 \text{ or } 2 \\ k(5-x) & \text{if } x=3 \text{ or } 4 \\ 0 & \text{otherwise} \end{cases}$$

a) Find the value of  $k$ .

b)  $P(X < 3)$  ?      c)  $P(0 < X < 2)$  ?

SUBSCRIBE



$$p(x) = \begin{cases} 0.1 & x=0 \\ h x & x=1 \text{ or } 2 \\ h(5-x) & x=3 \text{ or } 4 \\ 0 & \text{otherwise} \end{cases}$$

(i)

We have to find k value for that I need to correctly write distribution

|        |     |     |      |      |     |
|--------|-----|-----|------|------|-----|
| $x$    | 0   | 1   | 2    | 3    | 4   |
| $p(x)$ | 0.1 | $h$ | $2h$ | $2h$ | $h$ |

$6h$

$$6h + 0.1 = 1$$

$$6h = 0.9$$

$$h = \frac{9}{60} = \frac{3}{20}$$

$$\begin{array}{cccccc}
 x & 0 & 1 & 2 & 3 & 4 \\
 P(x) & 0.1 + \frac{1}{2} + \frac{2}{2} + \frac{2}{2} + \frac{1}{2} \\
 & 0.1 & \frac{3}{20} & \frac{3}{10} & \frac{3}{10} & \frac{3}{20}
 \end{array}$$

$$(i) \frac{3}{20}$$

$$\begin{aligned}
 (ii) P(X < 3) &= P(0) + P(1) + P(2) \\
 &= \frac{3}{20} + 0.1 = \frac{9}{20} + \frac{1}{10}
 \end{aligned}$$

$$\begin{aligned}
 (iii) P(0 < X < 2) &= \frac{9+2}{20} = \frac{11}{20} \\
 &= P(1) = \frac{3}{20}
 \end{aligned}$$

### Example:

A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.

Let  $X$  be a random variable whose values  $x$  are the possible numbers of defective computers purchased by the school. Then  $x$  can only take the numbers 0, 1, and

2. Now

$$f(0) = P(X = 0) = \frac{\binom{3}{0} \binom{17}{2}}{\binom{20}{2}} = \frac{68}{95}, \quad f(1) = P(X = 1) = \frac{\binom{3}{1} \binom{17}{1}}{\binom{20}{2}} = \frac{51}{190},$$

$$f(2) = P(X = 2) = \frac{\binom{3}{2} \binom{17}{0}}{\binom{20}{2}} = \frac{3}{190}.$$

Thus, the probability distribution of  $X$  is

|        |                 |                  |                 |
|--------|-----------------|------------------|-----------------|
| $x$    | 0               | 1                | 2               |
| $f(x)$ | $\frac{68}{95}$ | $\frac{51}{190}$ | $\frac{3}{190}$ |

## 🔥 GIVEN

- Total laptops = 20
- Defective = 3
- Good = 17
- Selected = 2

👉 Random variable  $X$  = number of defective laptops chosen

So:

$$X = 0, 1, 2$$

We use combination here bcs we are selecting 2 laptops out of 20

### ♦ TOTAL WAYS (Denominator) [Get Plus x](#)

Total ways to pick 2 laptops out of 20:

$$= \binom{20}{2}$$

### ♦ CASE 1: $f(0) = P(X = 0)$

👉 Means: no defective selected

So we choose both from good laptops (17)

Favorable ways:

$$\binom{17}{2}$$

Probability:

$$P(X = 0) = \frac{\binom{17}{2}}{\binom{20}{2}}$$

### ♦ Solve:

$$\binom{17}{2} = \frac{17 \cdot 16}{2} = 136$$

$$\binom{20}{2} = \frac{20 \cdot 19}{2} = 190$$

$$P(X = 0) = \frac{136}{190} = \frac{68}{95}$$

✓ Matches book

◆ CASE 2:  $f(1) = P(X = 1)$

👉 Means: 1 defective, 1 good

Step-by-step:

- Choose 1 defective from 3:

$$\binom{3}{1}$$

- Choose 1 good from 17:

$$\binom{17}{1}$$

Multiply (independent selections):

$$\binom{3}{1} \binom{17}{1}$$

Probability:

$$P(X = 1) = \frac{\binom{3}{1} \binom{17}{1}}{\binom{20}{2}}$$

◆ Solve:

◆ CASE 3:  $f(2) = P(X = 2)$

👉 Means: both are defective

Favorable ways:

$$\binom{3}{2}$$

Probability:

$$P(X = 2) = \frac{\binom{3}{2}}{\binom{20}{2}}$$

◆ Solve:

$$\binom{3}{2} = 3$$

$$P(X = 2) = \frac{3}{190}$$

✓ Matches book

## 🔥 FINAL PMF

$$f(x) = \begin{cases} \frac{68}{95} & x = 0 \\ \frac{51}{190} & x = 1 \\ \frac{3}{190} & x = 2 \end{cases}$$

## 🔥 WHY nCr (NOT nPr)?

👉 Because:

- Order does NOT matter
- You're selecting laptops, not arranging them

So:

✓ Use nCr (combinations)

✗ Not nPr



### Cumulative Distribution Function

There are many problems where we may wish to compute the probability that the observed value of a random variable  $X$  will be less than or equal to some real number  $x$ . Writing  $F(x) = P(X \leq x)$  for every real number  $x$ , we define  $F(x)$  to be the **cumulative distribution function** of the random variable  $X$ .

The **cumulative distribution function**  $F(x)$  of a discrete random variable  $X$  with probability distribution  $f(x)$  is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t), \quad \text{for } -\infty < x < \infty.$$



**Example:**

Find the cumulative distribution function of the random variable  $X$  in Example 3.9. Using  $F(x)$ , verify that  $f(2) = 3/8$ .

Direct calculations of the probability distribution of Example 3.9 give  $f(0) = 1/16$ ,  $f(1) = 1/4$ ,  $f(2) = 3/8$ ,  $f(3) = 1/4$ , and  $f(4) = 1/16$ . Therefore,

$$F(0) = f(0) = \frac{1}{16},$$

$$F(1) = f(0) + f(1) = \frac{5}{16},$$

$$F(2) = f(0) + f(1) + f(2) = \frac{11}{16},$$

$$F(3) = f(0) + f(1) + f(2) + f(3) = \frac{15}{16},$$

$$F(4) = f(0) + f(1) + f(2) + f(3) + f(4) = 1.$$

Hence,

$$F(x) = \begin{cases} 0, & \text{for } x < 0, \\ \frac{1}{16}, & \text{for } 0 \leq x < 1, \\ \frac{5}{16}, & \text{for } 1 \leq x < 2, \\ \frac{11}{16}, & \text{for } 2 \leq x < 3, \\ \frac{15}{16}, & \text{for } 3 \leq x < 4, \\ 1 & \text{for } x \geq 4. \end{cases}$$

$$F(4) = f(0) + f(1) + f(2) + f(3) + f(4) = 1.$$

Hence,

$$F(x) = \begin{cases} 0, & \text{for } x < 0, \\ \frac{1}{16}, & \text{for } 0 \leq x < 1, \\ \frac{5}{16}, & \text{for } 1 \leq x < 2, \\ \frac{11}{16}, & \text{for } 2 \leq x < 3, \\ \frac{15}{16}, & \text{for } 3 \leq x < 4, \\ 1 & \text{for } x \geq 4. \end{cases}$$

Now

$$f(2) = F(2) - F(1) = \frac{11}{16} - \frac{5}{16} = \frac{3}{8}.$$

**Example (from Example 3.9 in slides):** 4 cars sold,  $f(x) = (1/16)C(4,x)$

$f(0)=1/16, f(1)=4/16, f(2)=6/16, f(3)=4/16, f(4)=1/16$

| x | F(x)  |
|---|-------|
| 0 | 1/16  |
| 1 | 5/16  |
| 2 | 11/16 |
| 3 | 15/16 |
| 4 | 1     |

**Key trick:**  $f(2) = F(2) - F(1) = 11/16 - 5/16 = 6/16 = 3/8 \checkmark$

The CDF graph looked like a staircase – plot then jump at each value of X

**Quick Question 4.1 (Practice 3.13 from slides)** X = imperfections per 10m of fabric. PMF:

| x    | 0    | 1    | 2    | 3    | 4    |
|------|------|------|------|------|------|
| f(x) | 0.41 | 0.37 | 0.16 | 0.05 | 0.01 |

Find F(2).

**Answer:**  $F(2) = f(0)+f(1)+f(2) = 0.41+0.37+0.16 = 0.94$

### Example 4.1:

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

### Solution:

Let  $X$  represent the number of good components in the sample. The probability distribution of  $X$  is

$$f(x) = \frac{\binom{4}{x} \binom{3}{3-x}}{\binom{7}{3}}, \quad x = 0, 1, 2, 3.$$

Simple calculations yield  $f(0) = 1/35$ ,  $f(1) = 12/35$ ,  $f(2) = 18/35$ , and  $f(3) = 4/35$ . Therefore,

$$\mu = E(X) = (0) \left( \frac{1}{35} \right) + (1) \left( \frac{12}{35} \right) + (2) \left( \frac{18}{35} \right) + (3) \left( \frac{4}{35} \right) = \frac{12}{7} = 1.7.$$

Thus, if a sample of size 3 is selected at random over and over again from a lot of 4 good components and 3 defective components, it will contain, on average, 1.7 good components. 

**Example 4.2:**

A salesperson for a medical device company has two appointments on a given day. At the first appointment, he believes that he has a 70% chance to make the deal, from which he can earn \$1000 commission if successful. On the other hand, he thinks he only has a 40% chance to make the deal at the second appointment, from which, if successful, he can make \$1500. What is his expected commission based on his own probability belief? Assume that the appointment results are independent of each other.

**Solution:**

First, we know that the salesperson, for the two appointments, can have 4 possible commission totals: \$0, \$1000, \$1500, and \$2500. We then need to calculate their associated probabilities. By independence, we obtain

$$\begin{aligned} f(\$0) &= (1 - 0.7)(1 - 0.4) = 0.18, & f(\$2500) &= (0.7)(0.4) = 0.28, \\ f(\$1000) &= (0.7)(1 - 0.4) = 0.42, & \text{and } f(\$1500) &= (1 - 0.7)(0.4) = 0.12. \end{aligned}$$

Therefore, the expected commission for the salesperson is

$$\begin{aligned} E(X) &= (\$0)(0.18) + (\$1000)(0.42) + (\$1500)(0.12) + (\$2500)(0.28) \\ &= \$1300. \end{aligned}$$



2

**Example 4.3:**

Let  $X$  be the random variable that denotes the life in hours of a certain electronic device. The probability density function is

$$f(x) = \begin{cases} \frac{20,000}{x^3}, & x > 100, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected life of this type of device.

**Solution:**

$$\mu = E(X) = \int_{100}^{\infty} x \frac{20,000}{x^3} dx = \int_{100}^{\infty} \frac{20,000}{x^2} dx = 200.$$

Therefore, we can expect this type of device to last, *on average*, 200 hours.



Therefore, we can expect this type of device to last, *on average*, 200 hours. 

#### Example 4.4:

Suppose that the number of cars  $X$  that pass through a car wash between 4:00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

| $x$        | 4              | 5              | 6             | 7             | 8             | 9             |
|------------|----------------|----------------|---------------|---------------|---------------|---------------|
| $P(X = x)$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |

Let  $g(X) = 2X - 1$  represent the amount of money, in dollars, paid to the attendant by the manager. Find the attendant's expected earnings for this particular time period.

#### Solution:

$$\begin{aligned}
 E[g(X)] &= E(2X - 1) = \sum_{x=4}^9 (2x - 1)f(x) \\
 &= (7) \left(\frac{1}{12}\right) + (9) \left(\frac{1}{12}\right) + (11) \left(\frac{1}{4}\right) + (13) \left(\frac{1}{4}\right) \\
 &\quad + (15) \left(\frac{1}{6}\right) + (17) \left(\frac{1}{6}\right) = \$12.67.
 \end{aligned}$$

#### Example 4.5:

Let  $X$  be a random variable with density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of  $g(X) = 4X + 3$ .

#### Solution:

$$E(4X + 3) = \int_{-1}^2 \frac{(4x + 3)x^2}{3} dx = \frac{1}{3} \int_{-1}^2 (4x^3 + 3x^2) dx = 8. \quad \text{J}$$

**Example 4.6:**

Let  $X$  and  $Y$  be the random variables with joint probability distribution indicated in Table 3.1 on page 96. Find the expected value of  $g(X, Y) = XY$ . The table is reprinted here for convenience.

| $f(x, y)$     |   | $x$            |                 |                | Row             |
|---------------|---|----------------|-----------------|----------------|-----------------|
|               |   | 0              | 1               | 2              | Totals          |
| $y$           | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|               | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|               | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| Column Totals |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |

**Solution:**

$$\begin{aligned}
 E(XY) &= \sum_{x=0}^2 \sum_{y=0}^2 xyf(x, y) \\
 &= (0)(0)f(0, 0) + (0)(1)f(0, 1) \\
 &\quad + (1)(0)f(1, 0) + (1)(1)f(1, 1) + (2)(0)f(2, 0) \\
 &= f(1, 1) = \frac{3}{14}.
 \end{aligned}$$

**Example 4.7:**

Find  $E(Y/X)$  for the density function

$$f(x, y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, \ 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

**Solution:**

We have

$$E\left(\frac{Y}{X}\right) = \int_0^1 \int_0^2 \frac{y(1+3y^2)}{4} dx dy = \int_0^1 \frac{y + 3y^3}{2} dy = \frac{5}{8}.$$

## Joint PMF — Discrete

### Topic 1: Joint PMF — Discrete

When you have **two** random variables  $X$  and  $Y$  together,  $f(x,y)$  is their **joint PMF** if:

1.  $f(x,y) \geq 0$  for all  $(x,y)$
2.  $\sum \sum f(x,y) = 1$
3.  $P(X=x, Y=y) = f(x,y)$

And for any region  $A$ :  $P[(X,Y) \in A] = \sum \sum \text{over } A \text{ of } f(x,y)$

Think of it like a 2D table where every cell is a probability, and all cells must sum to 1.

**Example 3.14:**

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If  $X$  is the number of blue pens selected and  $Y$  is the number of red pens selected, find

- (a) the joint probability function  $f(x, y)$ ,
- (b)  $P[(X, Y) \in A]$ , where  $A$  is the region  $\{(x, y) | x + y \leq 1\}$ .

Show that the column and row totals of Table 3.1 give the marginal distribution of  $X$  alone and of  $Y$  alone.

**Solution:**



**Example 3.14 (from slides):** Box has 3 blue, 2 red, 3 green pens. Pick 2.  $X$  = blue selected,  $Y$  = red selected.

Total ways to pick 2 from 8:  $C(8,2) = 28$

The joint PMF formula is:

$$f(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}}, \quad x = 0, 1, 2; y = 0, 1, 2; 0 \leq x + y \leq 2$$

### The Setup

Box has: **3 blue**, **2 red**, **3 green** pens. You pick **2 pens**.

- $X$  = number of **blue** pens selected
- $Y$  = number of **red** pens selected
- So the number of **green** pens selected =  $2 - X - Y$

**Example 3.14 (from slides):** Box has 3 blue, 2 red, 3 green pens. Pick 2.  $X$  = blue selected,  $Y$  = red selected.

Total ways to pick 2 from 8:  $C(8,2) = 28$

The joint PMF formula is:

$$f(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}}, \quad x = 0, 1, 2; y = 0, 1, 2; 0 \leq x + y \leq 2$$

The full table:

| $f(x,y)$   | $x=0$  | $x=1$   | $x=2$  | Row Totals |
|------------|--------|---------|--------|------------|
| $y=0$      | $3/28$ | $9/28$  | $3/28$ | $15/28$    |
| $y=1$      | $3/14$ | $3/14$  | 0      | $3/7$      |
| $y=2$      | $1/28$ | 0       | 0      | $1/28$     |
| Col Totals | $5/14$ | $15/28$ | $3/28$ | 1          |

(b) Find  $P(X+Y \leq 1)$  — region  $A = \{(x,y) \mid x+y \leq 1\}$ :

Pairs satisfying  $x+y \leq 1$ :  $(0,0)$ ,  $(0,1)$ ,  $(1,0)$

$$P = f(0,0) + f(0,1) + f(1,0) = \frac{3}{28} + \frac{3}{14} + \frac{9}{28} = \frac{3+6+9}{28} = \frac{18}{28} = \frac{9}{14}$$

Show that the column and row totals of Table 3.1 give the marginal distribution of  $X$  alone and of  $Y$  alone.

**Solution:**

Table 3.1: Joint Probability Distribution for Example 3.14

| $f(x, y)$     |   | $x$            |                 |                | Row Totals      |
|---------------|---|----------------|-----------------|----------------|-----------------|
|               |   | 0              | 1               | 2              |                 |
| $y$           | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|               | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|               | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| Column Totals |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |

2

For the random variable  $X$ , we see that

$$g(0) = f(0, 0) + f(0, 1) + f(0, 2) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14},$$

$$g(1) = f(1, 0) + f(1, 1) + f(1, 2) = \frac{9}{28} + \frac{3}{14} + 0 = \frac{15}{28},$$

and

$$g(2) = f(2, 0) + f(2, 1) + f(2, 2) = \frac{3}{28} + 0 + 0 = \frac{3}{28},$$

which are just the column totals of Table 3.1. In a similar manner we could show that the values of  $h(y)$  are given by the row totals. In tabular form, these marginal distributions may be written as follows:

| $x$    | 0              | 1               | 2              |
|--------|----------------|-----------------|----------------|
| $g(x)$ | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ |

| $y$    | 0               | 1             | 2              |
|--------|-----------------|---------------|----------------|
| $h(y)$ | $\frac{15}{28}$ | $\frac{3}{7}$ | $\frac{1}{28}$ |

**Example 3.15:**

**Quick Question 1.1** (*Practice 3.37a*)  $f(x,y) = cxy$ , for  $x=1,2,3; y=1,2,3$ . Find  $c$ .

Answer:  $\sum \sum cxy = 1 \Rightarrow c \cdot \sum x \cdot \sum y = c \cdot (1+2+3) \cdot (1+2+3) = c \cdot 6 \cdot 6 = 36c = 1 \rightarrow c = 1/36$

**Quick Question 1.2** (*Practice 3.38a*)  $f(x,y) = (x+y)/30$  for  $x=0,1,2,3; y=0,1,2$ . Find  $P(X \leq 2, Y=1)$ .

Answer: Sum over  $x=0,1,2$  with  $y=1$ :  $f(0,1) + f(1,1) + f(2,1) = 1/30 + 2/30 + 3/30 = 6/30 = 1/5$

## Joint PDF — Continuous

### Topic 2: Joint PDF — Continuous

For continuous  $X$  and  $Y$ ,  $f(x,y)$  is a **joint density function** if:

1.  $f(x,y) \geq 0$
2.  $\iint f(x,y) \, dx \, dy = 1$  (over all space)
3.  $P[(X,Y) \in A] = \iint_{\text{over } A} f(x,y) \, dx \, dy$

## 🔥 GIVEN

Get Plus x

$$\int_0^1 \int_0^1 \frac{2}{5} (2x + 3y) \, dx \, dy$$

👉 We integrate with respect to  $x$  first, then  $y$

### ♦ STEP 1: Take constant outside

$$= \frac{2}{5} \int_0^1 \int_0^1 (2x + 3y) \, dx \, dy$$

### ♦ STEP 2: Inner integral (with respect to $x$ )

We treat  $y$  as constant.

So:

$$\int_0^1 (2x + 3y) \, dx$$

Split it:

$$= \int_0^1 2x \, dx + \int_0^1 3y \, dx$$

### ♦ First part:

$$\int_0^1 2x \, dx = [x^2]_0^1 = 1$$

### ♦ Second part (IMPORTANT):

$$\int_0^1 3y \, dx$$

👉 Since  $y$  is constant:

$$= 3y \int_0^1 dx = 3y(1 - 0) = 3y$$

### ♦ Combine both:

$$\int_0^1 (2x + 3y) \, dx = 1 + 3y$$

Get Plus x

#### ◆ STEP 4: Outer integral (with respect to $y$ )

Split again:

$$= \frac{2}{5} \left[ \int_0^1 1 \, dy + \int_0^1 3y \, dy \right]$$

◆ First:

$$\int_0^1 1 \, dy = 1$$

◆ Second:

$$\int_0^1 3y \, dy = \frac{3}{2}$$

◆ Combine:

$$= \frac{2}{5} \left( 1 + \frac{3}{2} \right) = \frac{2}{5} \cdot \frac{5}{2}$$

### FINAL ANSWER

$$= 1$$

✓ Verified

### KEY THINGS YOU MUST NOTICE

#### 1. Inner vs outer integral

- Inner: treat other variable as constant
- Outer: normal integration



**Example 3.15 (from slides):** Drive-in (X) and walk-in (Y) proportions of time in use.

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

**(a) Verify it integrates to 1:**

$$\int_0^1 \int_0^1 \frac{2}{5}(2x + 3y) \, dx \, dy = \int_0^1 \left[ \frac{2x^2}{5} + \frac{6xy}{5} \right]_0^1 dy = \int_0^1 \left( \frac{2}{5} + \frac{6y}{5} \right) dy = \frac{2}{5} + \frac{3}{5} = 1 \checkmark$$

**(b) Find  $P(0 < X < 1/2, 1/4 < Y < 1/2)$ :**

$$= \int_{1/4}^{1/2} \int_0^{1/2} \frac{2}{5}(2x + 3y) \, dx \, dy = \frac{13}{160}$$

(The slide works this out fully — inner integral gives  $(1/10 + 3y/5)$ , outer integral gives the final answer.)



◆ STEP 1: Take constant out

$$= \frac{2}{5} \int_{1/4}^{1/2} \int_0^{1/2} (2x + 3y) \, dx \, dy$$

◆ STEP 2: Inner integral (w.r.t x)

$$\int_0^{1/2} (2x + 3y) \, dx$$

Split:

$$= \int_0^{1/2} 2x \, dx + \int_0^{1/2} 3y \, dx$$

◆ First part:

$$\int_0^{1/2} 2x \, dx = [x^2]_0^{1/2} = \frac{1}{4}$$

◆ Second part:

$$\int_0^{1/2} 3y \, dx = 3y \cdot \frac{1}{2} = \frac{3y}{2}$$

◆ Combine:

$$= \frac{1}{4} + \frac{3y}{2}$$

◆ STEP 3: Multiply constant



$$= \frac{2}{5} \int_{1/4}^{1/2} \left( \frac{1}{4} + \frac{3y}{2} \right) dy$$

◆ STEP 3: Multiply constant

$$= \frac{2}{5} \int_{1/4}^{1/2} \left( \frac{1}{4} + \frac{3y}{2} \right) dy$$

◆ STEP 4: Outer integral

Split:

$$= \frac{2}{5} \left[ \int_{1/4}^{1/2} \frac{1}{4} dy + \int_{1/4}^{1/2} \frac{3y}{2} dy \right]$$

◆ First part:

$$\int_{1/4}^{1/2} \frac{1}{4} dy = \frac{1}{4} \left( \frac{1}{2} - \frac{1}{4} \right) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$$

◆ Second part:

$$\begin{aligned} \int_{1/4}^{1/2} \frac{3y}{2} dy &= \frac{3}{2} \cdot \frac{y^2}{2} \Big|_{1/4}^{1/2} = \frac{3}{4} \left( \frac{1}{4} - \frac{1}{16} \right) \\ &= \frac{3}{4} \cdot \frac{3}{16} = \frac{9}{64} \end{aligned}$$

◆ Combine both:

$$= \frac{2}{5} \left( \frac{1}{16} + \frac{9}{64} \right)$$

Convert:

$$\begin{aligned} &\frac{1}{16} \downarrow \frac{4}{64} \\ &= \frac{2}{5} \cdot \frac{13}{64} \end{aligned}$$

## JPDF & JPMF PRACTISE

(b)  $f(x, y) = c|x - y|$ , for  $x = -2, 0, 2$ ,  $y = -2, 0$ .

**3.38** If the joint probability distribution of  $X$  and  $Y$  is given by

$$f(x, y) = \frac{x + y}{30}, \quad \text{for } x = 0, 1, 2, 3; \quad y = 0, 1, 2,$$

- find
- (a)  $P(X \leq 2, Y = 1)$ ;
  - (b)  $P(X > 2, Y \leq 1)$ ;
  - (c)  $P(X > Y)$ ;
  - (d)  $P(X + Y = 4)$ .

For these questions , first prepare table

### Problem 3.38

$$f(x, y) = \frac{x+y}{30}, \quad x = 0, 1, 2, 3; \quad y = 0, 1, 2$$

First, build the full table — this makes all parts easy:

| $f(x,y)$ | $y=0$ | $y=1$ | $y=2$ | $g(x)$ |
|----------|-------|-------|-------|--------|
| $x=0$    | 0/30  | 1/30  | 2/30  | 3/30   |
| $x=1$    | 1/30  | 2/30  | 3/30  | 6/30   |
| $x=2$    | 2/30  | 3/30  | 4/30  | 9/30   |
| $x=3$    | 3/30  | 4/30  | 5/30  | 12/30  |
| $h(y)$   | 6/30  | 10/30 | 14/30 | 1      |

#### Part (a): $P(X \leq 2, Y = 1)$

Sum all cells where  $x \leq 2$  AND  $y = 1$ :

$$= f(0, 1) + f(1, 1) + f(2, 1)$$

$$= \frac{1}{30} + \frac{2}{30} + \frac{3}{30} = \boxed{\frac{6}{30} = \frac{1}{5}}$$

#### Part (b): $P(X > 2, Y \leq 1)$

$X > 2$  means  $x = 3$  only.  $Y \leq 1$  means  $y = 0$  or  $y = 1$ :

$$= f(3, 0) + f(3, 1)$$

$$= \frac{3}{30} + \frac{4}{30} = \boxed{\frac{7}{30}}$$

**Part (c):  $P(X > Y)$**

List all (x,y) pairs where  $x > y$ :

| Pair  | Condition          | $f(x,y)$ |
|-------|--------------------|----------|
| (1,0) | $1 > 0 \checkmark$ | $1/30$   |
| (2,0) | $2 > 0 \checkmark$ | $2/30$   |
| (2,1) | $2 > 1 \checkmark$ | $3/30$   |
| (3,0) | $3 > 0 \checkmark$ | $3/30$   |
| (3,1) | $3 > 1 \checkmark$ | $4/30$   |
| (3,2) | $3 > 2 \checkmark$ | $5/30$   |

$$P(X > Y) = \frac{1 + 2 + 3 + 3 + 4 + 5}{30} = \boxed{\frac{18}{30} = \frac{3}{5}}$$

**Part (d):  $P(X + Y = 4)$**

List all (x,y) pairs where  $x + y = 4$ :

| Pair  | $x+y$          | $f(x,y)$ |
|-------|----------------|----------|
| (2,2) | $4 \checkmark$ | $4/30$   |
| (3,1) | $4 \checkmark$ | $4/30$   |

(Note: (4,0) is invalid since  $x$  max is 3. (3,2) = 5  $\times$ )

$$P(X + Y = 4) = \frac{4}{30} + \frac{4}{30} = \boxed{\frac{8}{30} = \frac{4}{15}}$$

**3.40** A fast-food restaurant operates both a drive-through facility and a walk-in facility. On a randomly selected day, let  $X$  and  $Y$ , respectively, be the proportions of the time that the drive-through and walk-in facilities are in use, and suppose that the joint density function of these random variables is

$$f(x, y) = \begin{cases} \frac{2}{3}(x + 2y), & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Find the marginal density of  $X$ .
- (b) Find the marginal density of  $Y$ .
- (c) Find the probability that the drive-through facility is busy less than one-half of the time.

Marginal density, we integrate out here because its continuous joint probability density function. Otherwise if it was discrete i.e. jpmf, then we count rows and columns

#### Problem 3.40

$$f(x, y) = \begin{cases} \frac{2}{3}(x + 2y), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

**Part (a): Marginal density  $g(x)$  — integrate out  $y$**

$$g(x) = \int_0^1 \frac{2}{3}(x + 2y) dy$$

**Step 1:** Pull out  $2/3$ :

$$= \frac{2}{3} \int_0^1 (x + 2y) dy$$

**Step 2:** Integrate term by term ( $x$  is constant):

$$= \frac{2}{3} [xy + y^2]_0^1$$

**Step 3:** Plug in  $y=1$  and  $y=0$ :

$$= \frac{2}{3} [(x + 1) - 0]$$

$$\boxed{g(x) = \frac{2(x + 1)}{3}, \quad 0 \leq x \leq 1}$$

---

**Part (b): Marginal density  $h(y)$  — integrate out  $x$**

$$h(y) = \int_0^1 \frac{2}{3}(x + 2y) dx$$

**Step 1:** Pull out  $2/3$ :

$$= \frac{2}{3} \int_0^1 (x + 2y) dx$$

**Step 2:** Integrate term by term ( $y$  is constant):

$$= \frac{2}{3} \left[ \frac{x^2}{2} + 2yx \right]_0^1$$

**Step 3:** Plug in  $x=1$  and  $x=0$ :

$$= \frac{2}{3} \left[ \frac{1}{2} + 2y \right]$$

$$= \frac{2}{3} \cdot \frac{1 + 4y}{2}$$

$$\boxed{h(y) = \frac{1 + 4y}{3}, \quad 0 \leq y \leq 1}$$



**Part (c):  $P(X < 1/2)$  — drive-through busy less than half the time**

This only involves X, so use the marginal  $g(x)$ :

$$P\left(X < \frac{1}{2}\right) = \int_0^{1/2} \frac{2(x+1)}{3} dx$$

**Step 1:** Pull out 2/3:

$$= \frac{2}{3} \int_0^{1/2} (x+1) dx$$

**Step 2:** Integrate:

$$= \frac{2}{3} \left[ \frac{x^2}{2} + x \right]_0^{1/2}$$

**Step 3:** Plug in  $x=1/2$  and  $x=0$ :

$$= \frac{2}{3} \left[ \frac{(1/2)^2}{2} + \frac{1}{2} \right] = \frac{2}{3} \left[ \frac{1}{8} + \frac{4}{8} \right]$$

$$= \frac{2}{3} \times \frac{5}{8} = \frac{10}{24}$$

$$\boxed{P\left(X < \frac{1}{2}\right) = \frac{5}{12} \approx 0.417}$$

This only involves x , so we used only marginal  $g(x)$ .

## Marginal Distribution

### Topic 3: Marginal Distributions

When you have a joint distribution but want to focus on **just one variable**, you sum/integrate out the other.

**Discrete:**

$$g(x) = \sum_y f(x, y) \quad h(y) = \sum_x f(x, y)$$

**Continuous:**

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

**Key insight:** In a joint table, the **column totals** give  $g(x)$  and the **row totals** give  $h(y)$ . That's all marginals are — the totals at the margins of the table.

**3.50** Suppose that  $X$  and  $Y$  have the following joint probability distribution:

|     |   | $x$  |      |
|-----|---|------|------|
|     |   | 2    | 4    |
| $y$ | 1 | 0.10 | 0.15 |
|     | 3 | 0.20 | 0.30 |
|     | 5 | 0.10 | 0.15 |

- (a) Find the marginal distribution of  $X$ .
- (b) Find the marginal distribution of  $Y$ .

10

**3.54** Determine whether the two random variables of Exercise 3.49 are dependent or independent.

**3.55** Determine whether the two random variables of Exercise 3.50 are dependent or independent.

### The Joint Table

| $f(x,y)$ | $x=2$ | $x=4$ |
|----------|-------|-------|
| $y=1$    | 0.10  | 0.15  |
| $y=3$    | 0.20  | 0.30  |
| $y=5$    | 0.10  | 0.15  |

### Part (a) — Marginal Distribution of $X \rightarrow g(x)$

Sum down each column (sum over all  $y$  values):

$$g(2) = f(2, 1) + f(2, 3) + f(2, 5) = 0.10 + 0.20 + 0.10 = \mathbf{0.40}$$

$$g(4) = f(4, 1) + f(4, 3) + f(4, 5) = 0.15 + 0.30 + 0.15 = \mathbf{0.60}$$

| $x$    | 2    | 4    |
|--------|------|------|
| $g(x)$ | 0.40 | 0.60 |

$$\text{Check: } 0.40 + 0.60 = 1 \checkmark$$

### Part (b) — Marginal Distribution of $Y \rightarrow h(y)$

### Part (b) — Marginal Distribution of $Y \rightarrow h(y)$

Sum across each row (sum over all  $x$  values):

$$h(1) = f(2, 1) + f(4, 1) = 0.10 + 0.15 = \mathbf{0.25}$$

$$h(3) = f(2, 3) + f(4, 3) = 0.20 + 0.30 = \mathbf{0.50}$$

$$h(5) = f(2, 5) + f(4, 5) = 0.10 + 0.15 = \mathbf{0.25}$$

| $y$    | 1    | 3    | 5    |
|--------|------|------|------|
| $h(y)$ | 0.25 | 0.50 | 0.25 |

Table 3.1: Joint Probability Distribution for Example 3.14

| $f(x, y)$     |   | $x$            |                 |                | Row             |
|---------------|---|----------------|-----------------|----------------|-----------------|
|               |   | 0              | 1               | 2              | Totals          |
| $y$           | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|               | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|               | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| Column Totals |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |

For the random variable  $X$ , we see that

$$g(0) = f(0, 0) + f(0, 1) + f(0, 2) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14},$$

$$g(1) = f(1, 0) + f(1, 1) + f(1, 2) = \frac{9}{28} + \frac{3}{14} + 0 = \frac{15}{28},$$

and

$$g(2) = f(2, 0) + f(2, 1) + f(2, 2) = \frac{3}{28} + 0 + 0 = \frac{3}{28},$$

which are just the column totals of Table 3.1. In a similar manner we could show

We

for  $0 \leq y \leq 1$ , and  $h(y) = 0$  elsewhere.

**Example 3.16:**

Show that the column and row totals of Table 3.1 give the marginal distribution of  $X$  alone and of  $Y$  alone.

For the random variable  $X$ , we see that

$$g(0) = f(0,0) + f(0,1) + f(0,2) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14},$$

$$g(1) = f(1,0) + f(1,1) + f(1,2) = \frac{9}{28} + \frac{3}{14} + 0 = \frac{15}{28},$$

and

$$g(2) = f(2,0) + f(2,1) + f(2,2) = \frac{3}{28} + 0 + 0 = \frac{3}{28},$$

which are just the column totals of Table 3.1. In a similar manner we could show that the values of  $h(y)$  are given by the row totals. In tabular form, these marginal distributions may be written as follows:

| $x$    | 0              | 1               | 2              |
|--------|----------------|-----------------|----------------|
| $g(x)$ | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ |

| $y$    | 0               | 1             | 2              |
|--------|-----------------|---------------|----------------|
| $h(y)$ | $\frac{15}{28}$ | $\frac{3}{7}$ | $\frac{1}{28}$ |

We can marginal distribution through joint probability which we had already calculated . columns give  $g(x)$  and row gives  $h(x)$

**Example 3.15:**

A privately owned business operates both a drive-in facility and a walk-in facility. On a randomly selected day, let  $X$  and  $Y$ , respectively, be the proportions of the time that the drive-in and the walk-in facilities are in use, and suppose that the joint density function of these random variables is

$$f(x,y) = \begin{cases} \frac{2}{5}(2x+3y), & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

**Example 3.17:**

Find  $g(x)$  and  $h(y)$  for the joint density function of Example 3.15. By definition,

$G(x)$  and  $h(y)$  is being talked , its marginal probability

In case of finding  $g(x)$  , we will integrate in terms of  $y$  and treat  $x$  as constant.

In case of finding  $h(y)$  , we will integrate in terms of  $x$  and treat  $y$  as constant

## Example 3.17 — Full Integration Walkthrough

Finding  $g(x)$  — Integrate out  $Y$

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 \frac{2}{5} (2x + 3y) dy$$

Step 1: Pull the  $2/5$  out:

$$= \frac{2}{5} \int_0^1 (2x + 3y) dy$$

Step 2: Integrate term by term with respect to  $y$  ( $x$  is treated as a constant):

$$= \frac{2}{5} \left[ 2x \cdot y + 3 \cdot \frac{y^2}{2} \right]_0^1$$

$$= \frac{2}{5} \left[ 2xy + \frac{3y^2}{2} \right]_0^1$$

Step 3: Plug in  $y=1$  and  $y=0$ :

$$= \frac{2}{5} \left[ \left( 2x(1) + \frac{3(1)^2}{2} \right) - \left( 2x(0) + \frac{3(0)^2}{2} \right) \right]$$

$$= \frac{2}{5} \left[ 2x + \frac{3}{2} - 0 \right]$$

Step 4: Distribute the  $2/5$ :

$$= \frac{2}{5} \cdot 2x + \frac{2}{5} \cdot \frac{3}{2} = \frac{4x}{5} + \frac{3}{5}$$

$$\boxed{g(x) = \frac{4x + 3}{5}, \quad 0 \leq x \leq 1}$$



### Finding h(y) — Integrate out X

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 \frac{2}{5} (2x + 3y) dx$$

**Step 1:** Pull the 2/5 out:

$$= \frac{2}{5} \int_0^1 (2x + 3y) dx$$

**Step 2:** Integrate term by term with respect to x (y is treated as a constant):

$$= \frac{2}{5} \left[ 2 \cdot \frac{x^2}{2} + 3y \cdot x \right]_0^1$$

$$= \frac{2}{5} [x^2 + 3xy]_0^1$$

**Step 3:** Plug in x=1 and x=0:

$$= \frac{2}{5} [(1)^2 + 3(1)y - ((0)^2 + 3(0)y)]$$

$$= \frac{2}{5} [1 + 3y - 0]$$

**Step 4:** Distribute:

$$= \frac{2}{5} (1 + 3y)$$

$$\boxed{h(y) = \frac{2(1 + 3y)}{5}, \quad 0 \leq y \leq 1}$$

### Quick Sanity Check

Verify  $g(x)$  integrates to 1:

$$\int_0^1 \frac{4x+3}{5} dx = \frac{1}{5} [2x^2 + 3x]_0^1 = \frac{1}{5}(2+3) = \frac{5}{5} = 1 \checkmark$$

Verify  $h(y)$  integrates to 1:

$$\int_0^1 \frac{2(1+3y)}{5} dy = \frac{2}{5} \left[ y + \frac{3y^2}{2} \right]_0^1 = \frac{2}{5} \left( 1 + \frac{3}{2} \right) = \frac{2}{5} \cdot \frac{5}{2} = 1 \checkmark$$

Both marginals integrate to 1 — correct!

JPDF

### GIVEN

$$f(x, y) = \begin{cases} 4xy, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find:

$$P\left(0 \leq X \leq \frac{1}{2}, \frac{1}{4} \leq Y \leq \frac{1}{2}\right)$$

### STEP 1: Write the integral

$$P = \int_{y=1/4}^{1/2} \int_{x=0}^{1/2} 4xy \, dx \, dy$$

🔥 STEP 1: Write the integral

Get Plus ×

$$P = \int_{y=1/4}^{1/2} \int_{x=0}^{1/2} 4xy \, dx \, dy$$

◆ STEP 2: Inner integral (w.r.t  $x$ )

$$\int_0^{1/2} 4xy \, dx$$

👉 Treat  $y$  as constant:

$$\begin{aligned} &= 4y \int_0^{1/2} x \, dx \\ &= 4y \cdot \left[ \frac{x^2}{2} \right]_0^{1/2} \\ &= 4y \cdot \frac{1}{8} = \frac{y}{2} \end{aligned}$$

Get Plus ×

◆ STEP 3: Outer integral (w.r.t  $y$ )

$$\begin{aligned} &\int_{1/4}^{1/2} \frac{y}{2} \, dy \\ &= \frac{1}{2} \int_{1/4}^{1/2} y \, dy \\ &= \frac{1}{2} \cdot \left[ \frac{y^2}{2} \right]_{1/4}^{1/2} \\ &= \frac{1}{4} \left( \frac{1}{4} - \frac{1}{16} \right) \\ &= \frac{1}{4} \cdot \frac{3}{16} = \frac{3}{64} \end{aligned}$$

🔥 FINAL ANSWER

$$\boxed{\frac{3}{64}}$$

Conditional Distribution

## Topic 4: Conditional Distributions

The conditional distribution of  $Y$  given  $X=x$  is:

$$f(y|x) = \frac{f(x,y)}{g(x)}, \quad \text{provided } g(x) > 0$$

Similarly,  $X$  given  $Y=y$ :

$$f(x|y) = \frac{f(x,y)}{h(y)}, \quad \text{provided } h(y) > 0$$

Think of it like Bayes: you already know one variable's value, so you rescale the joint using the marginal.

Relate it like  $h$  is rows, so when  $x$  is first so  $h(y)$  will be found first

**3.42** Let  $X$  and  $Y$  denote the lengths of life, in years, of two components in an electronic system. If the joint density function of these variables is

$$f(x,y) = \begin{cases} e^{-(x+y)}, & x > 0, y > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

find  $P(0 < X < 1 \mid Y = 2)$ .

Whenever we find conditional, we find  $h(y)$  or  $g(x)$ . In previous question we had done it till making of eqn that's why we proceeded from directly putting limits.

Here we are finding  $h(y)$  because first var is  $x$  in  $P$

In  $h(y)$ , we treat  $y$  as constant

🔥 GIVEN

$$\int_0^{\infty} e^{-x} dx$$

♦ STEP 1: Recall basic rule

$$\int e^{ax} dx = \frac{1}{a} e^{ax}$$

👉 Here:

$$a = -1$$

So:

$$\int e^{-x} dx = -e^{-x}$$

E to power 0 is 1

### Step 1: Find the Marginal h(y)

To use the conditional formula, we need h(y) first.

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^{\infty} e^{-(x+y)} dx$$

Split the exponent using the law  $e^{-(x+y)} = e^{-x} \cdot e^{-y}$ :

$$= \int_0^{\infty} e^{-x} \cdot e^{-y} dx$$

Pull out  $e^{-y}$  since it's constant with respect to x:

$$= e^{-y} \int_0^{\infty} e^{-x} dx$$

Integrate  $e^{-x}$ :

$$= e^{-y} [-e^{-x}]_0^{\infty} = e^{-y} [0 - (-1)]$$

$$\boxed{h(y) = e^{-y}, \quad y > 0}$$

---

**Step 2: Find the Conditional  $f(x|y)$** 

$$f(x | y) = \frac{f(x, y)}{h(y)} = \frac{e^{-(x+y)}}{e^{-y}}$$

Cancel  $e^{-y}$  from top and bottom:

$$= \frac{e^{-x} \cdot e^{-y}}{e^{-y}}$$

$$\boxed{f(x | y) = e^{-x}, \quad x > 0}$$

**Key observation:** The  $e^{-y}$  cancelled completely — the conditional density doesn't depend on  $y$  at all. This is the signature of **statistical independence** (we'll confirm below).

**Step 3: Find  $P(0 < X < 1 | Y = 2)$** 

Since  $f(x|y) = e^{-x}$  regardless of what  $y$  is, plugging in  $Y=2$  changes nothing:

$$P(0 < X < 1 | Y = 2) = \int_0^1 f(x | y = 2) dx = \int_0^1 e^{-x} dx$$

**Integrate:**

$$= [-e^{-x}]_0^1$$

**Plug in limits  $x=1$  and  $x=0$ :**

$$= (-e^{-1}) - (-e^0) = -\frac{1}{e} + 1$$

$$\boxed{P(0 < X < 1 | Y = 2) = 1 - \frac{1}{e} \approx 1 - 0.368 = 0.632}$$



- (a) Now,  $f(0, 1)$ , for example, represents the probability that a red and a green pens are selected. The total number of equally likely ways of selecting any 2 pens from the 8 is  $\binom{8}{2} = 28$ . The number of ways of selecting 1 red from 2 red pens and 1 green from 3 green pens is  $\binom{2}{1}\binom{3}{1} = 6$ . Hence,  $f(0, 1) = 6/28 = 3/14$ . Similar calculations yield the probabilities for the other cases, which are presented in Table 3.1. Note that the probabilities sum to 1. In Chapter

5, it will become clear that the joint probability distribution of Table 3.1 can be represented by the formula

$$f(x, y) = \frac{\binom{3}{x}\binom{2}{y}\binom{3}{2-x-y}}{\binom{8}{2}},$$

for  $x = 0, 1, 2$ ;  $y = 0, 1, 2$ ; and  $0 \leq x + y \leq 2$ .

- (b) The probability that  $(X, Y)$  fall in the region  $A$  is

$$\begin{aligned} P[(X, Y) \in A] &= P(X + Y \leq 1) = f(0, 0) + f(0, 1) + f(1, 0) \\ &= \frac{3}{28} + \frac{3}{14} + \frac{9}{28} = \frac{9}{14}. \end{aligned}$$

Table 3.1: Joint Probability Distribution for Example 3.14

| $f(x, y)$     |   | $x$            |                 |                | Row             |
|---------------|---|----------------|-----------------|----------------|-----------------|
|               |   | 0              | 1               | 2              | Totals          |
| $y$           | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|               | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|               | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| Column Totals |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |

**Example 3.18 (from slides):** Using the pen example (3.14). Find  $f(x|1)$  — conditional distribution of  $X$  given  $Y=1$ .

Step 1: Find  $h(1)$  = row total for  $y=1$ :

$$h(1) = f(0, 1) + f(1, 1) + f(2, 1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}$$

Step 2:  $f(x|1) = f(x,1)/h(1) = f(x,1) \times (7/3)$

$$f(0|1) = \frac{3/14}{3/7} = \frac{1}{2}, \quad f(1|1) = \frac{3/14}{3/7} = \frac{1}{2}, \quad f(2|1) = 0$$

Step 3:  $P(X=0 | Y=1) = f(0|1) = 1/2$

Interpretation: If we know 1 red pen was selected, there's a 50% chance no blue pen was selected.

and the conditional distribution of  $X$ , given that  $Y = 1$ , is

| $x$      | 0             | 1             | 2 |
|----------|---------------|---------------|---|
| $f(x 1)$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 0 |

Finally,

$$P(X = 0 | Y = 1) = f(0|1) = \frac{1}{2}.$$

Therefore, if it is known that 1 of the 2 pen refills selected is red, we have a probability equal to  $1/2$  that the other refill is not blue. 

**Example 3.19:**

### **Example 3.19:**

The joint density for the random variables  $(X, Y)$ , where  $X$  is the unit temperature change and  $Y$  is the proportion of spectrum shift that a certain atomic particle produces, is

$$f(x, y) = \begin{cases} 10xy^2, & 0 < x < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- Find the marginal densities  $g(x)$ ,  $h(y)$ , and the conditional density  $f(y|x)$ .
- Find the probability that the spectrum shifts more than half of the total observations, given that the temperature is increased by 0.25 unit.
- By definition,

For  $g(x)$ , we integrate in respect of  $y$  . and apply  $y$  limits.

### Understanding the Domain First

The domain is  $0 < x < y < 1$ . This is NOT a rectangle — it's a triangle.

The condition  $x < y$  means:

- For  $g(x)$ :  $y$  ranges from  $x$  to  $1$  ( $y$  must be greater than  $x$ )
- For  $h(y)$ :  $x$  ranges from  $0$  to  $y$  ( $x$  must be less than  $y$ )

### Example 3.19 — Full Solution

#### Understanding the Domain First

The domain is  $0 < x < y < 1$ . This is NOT a rectangle — it's a triangle.

The condition  $x < y$  means:

- For  $g(x)$ :  $y$  ranges from  $x$  to  $1$  ( $y$  must be greater than  $x$ )
- For  $h(y)$ :  $x$  ranges from  $0$  to  $y$  ( $x$  must be less than  $y$ )

This is the most important thing to get right before integrating

#### Part (a) — Finding $g(x)$

To find  $g(x)$ , integrate  $f(x,y)$  over  $y$ . Since  $x < y < 1$ ,  $y$  runs from  $x$  to  $1$ .

$$g(x) = \int_x^1 10xy^2 \, dy$$

**Part (a) — Finding  $g(x)$** 

To find  $g(x)$ , integrate  $f(x,y)$  over  $y$ . Since  $x < y < 1$ ,  $y$  runs from  $x$  to  $1$ .

$$g(x) = \int_x^1 10xy^2 \, dy$$

**Step 1:** Pull  $10x$  out (it's constant with respect to  $y$ ):

$$= 10x \int_x^1 y^2 \, dy$$

**Step 2:** Integrate  $y^2$ :

$$= 10x \left[ \frac{y^3}{3} \right]_x^1$$

**Step 3:** Plug in limits  $y=1$  and  $y=x$ :

$$= 10x \left[ \frac{1}{3} - \frac{x^3}{3} \right]$$

**Step 4:** Simplify:

$$= 10x \cdot \frac{1 - x^3}{3}$$

$$\boxed{g(x) = \frac{10}{3} x(1 - x^3), \quad 0 < x < 1}$$

### Part (a) — Finding $h(y)$

To find  $h(y)$ , integrate  $f(x,y)$  over  $x$ . Since  $0 < x < y$ ,  $x$  runs from 0 to  $y$ .

$$h(y) = \int_0^y 10xy^2 \, dx$$

Step 1: Pull  $10y^2$  out (constant with respect to  $x$ ):

$$= 10y^2 \int_0^y x \, dx$$

Step 2: Integrate  $x$ :

$$= 10y^2 \left[ \frac{x^2}{2} \right]_0^y$$

Step 3: Plug in limits  $x=y$  and  $x=0$ :

$$= 10y^2 \left[ \frac{y^2}{2} - 0 \right] = 10y^2 \cdot \frac{y^2}{2}$$

Step 4: Simplify:

$$\boxed{h(y) = 5y^4, \quad 0 < y < 1}$$

### Part (a) — Finding $f(y|x)$

$$f(y|x) = \frac{f(x,y)}{g(x)} = \frac{10xy^2}{\frac{10}{3}x(1-x^3)}$$

Cancel  $10x$  from top and bottom:

$$= \frac{y^2}{\frac{1-x^3}{3}} = \frac{3y^2}{1-x^3}$$

$$\boxed{f(y|x) = \frac{3y^2}{1-x^3}, \quad 0 < x < y < 1}$$

- (b) Find the probability that the spectrum shifts more than half of the total observations, given that the temperature is increased by 0.25 unit.

Given means conditional , and upper we are given and we are given Jpdf function.

**Part (b) — Finding  $P(Y > 1/2 | X = 0.25)$**

"Spectrum shifts more than half" means  $Y > 1/2$ , given  $X = 0.25$ .

**Step 1:** Substitute  $x = 0.25$  into the conditional density:

$$f(y | x = 0.25) = \frac{3y^2}{1 - (0.25)^3} = \frac{3y^2}{1 - \frac{1}{64}} = \frac{3y^2}{\frac{63}{64}}$$

**Step 2:** Simplify the constant:

$$= \frac{3y^2 \times 64}{63} = \frac{192y^2}{63} = \frac{64y^2}{21}$$

**Part (b) — Finding  $P(Y > 1/2 | X = 0.25)$**

"Spectrum shifts more than half" means  $Y > 1/2$ , given  $X = 0.25$ .

**Step 1:** Substitute  $x = 0.25$  into the conditional density:

$$f(y | x = 0.25) = \frac{3y^2}{1 - (0.25)^3} = \frac{3y^2}{1 - \frac{1}{64}} = \frac{3y^2}{\frac{63}{64}}$$

**Step 2:** Simplify the constant:

$$= \frac{3y^2 \times 64}{63} = \frac{192y^2}{63} = \frac{64y^2}{21}$$

**Step 3:** Integrate from  $1/2$  to  $1$  (since  $y > 1/2$  and  $y < 1$ ):

$$\begin{aligned} P\left(Y > \frac{1}{2} \mid X = 0.25\right) &= \int_{1/2}^1 \frac{3y^2}{\frac{63}{64}} dy \\ &= \frac{64}{63} \int_{1/2}^1 3y^2 dy = \frac{64}{63} [y^3]_{1/2}^1 \end{aligned}$$

**Step 4:** Plug in limits:

$$= \frac{64}{63} \left[1 - \frac{1}{8}\right] = \frac{64}{63} \times \frac{7}{8}$$

**Step 5:** Multiply:

$$= \frac{64 \times 7}{63 \times 8} = \frac{448}{504} = \frac{8}{9}$$

$$\boxed{P\left(Y > \frac{1}{2} \mid X = 0.25\right) = \frac{8}{9} \approx 0.889}$$

# Statistical Independence

## Topic 5: Statistical Independence

X and Y are **statistically independent** if and only if:

$$f(x, y) = g(x) \cdot h(y) \quad \text{for all } (x, y)$$

The joint = product of marginals. Knowing one variable tells you nothing about the other.

**Quick test:** Pick any one (x,y) pair. If  $f(x,y) \neq g(x) \cdot h(y)$ , they are **dependent**. You only need one counterexample.

### Example 3.14:

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If  $X$  is the number of blue pens selected and  $Y$  is the number of red pens selected, find

- (a) the joint probability function  $f(x, y)$ ,
- (b)  $P[(X, Y) \in A]$ , where  $A$  is the region  $\{(x, y) | x + y \leq 1\}$ .

Show that the column and row totals of Table 3.1 give the marginal distribution of  $X$  alone and of  $Y$  alone.

**Solution:**

**Example 3.14 (from slides):** Box has 3 blue, 2 red, 3 green pens. Pick 2.  $X$  = blue selected,  $Y$  = red selected.

Total ways to pick 2 from 8:  $C(8, 2) = 28$

The joint PMF formula is:

$$f(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}}, \quad x = 0, 1, 2; \quad y = 0, 1, 2; \quad 0 \leq x + y \leq 2$$



### The Setup

Box has: 3 blue, 2 red, 3 green pens. You pick 2 pens.

- X = number of blue pens selected
- Y = number of red pens selected
- So the number of green pens selected = 2 - X - Y

**Example 3.14 (from slides):** Box has 3 blue, 2 red, 3 green pens. Pick 2. X = blue selected, Y = red selected.

Total ways to pick 2 from 8:  $C(8,2) = 28$

The joint PMF formula is:

$$f(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}}, \quad x = 0, 1, 2; y = 0, 1, 2; 0 \leq x + y \leq 2$$

The full table:

| f(x,y)     | x=0  | x=1   | x=2  | Row Totals |
|------------|------|-------|------|------------|
| y=0        | 3/28 | 9/28  | 3/28 | 15/28      |
| y=1        | 3/14 | 3/14  | 0    | 3/7        |
| y=2        | 1/28 | 0     | 0    | 1/28       |
| Col Totals | 5/14 | 15/28 | 3/28 | 1          |

## VARIANCE AND COVARIANCE

### Topic 1: Variance

$$\sigma^2 = E(X^2) - \mu^2$$

This is the **shortcut formula**. The two steps are always:

1. Find  $\mu = E(X)$
2. Find  $E(X^2)$ , then subtract  $\mu^2$

**Discrete:**  $E(X^2) = \sum x^2 f(x)$

**Continuous:**  $E(X^2) = \int x^2 f(x) dx$

Standard deviation =  $\sigma = \sqrt{\sigma^2}$

Variance square =  $E(x^2) - E(x)^2$

**Example 4.9:**

Let the random variable  $X$  represent the number of defective parts for a machine when 3 parts are sampled from a production line and tested. The following is the probability distribution of  $X$ .

|        |      |      |      |      |
|--------|------|------|------|------|
| $x$    | 0    | 1    | 2    | 3    |
| $f(x)$ | 0.51 | 0.38 | 0.10 | 0.01 |

Using Theorem 4.2, calculate  $\sigma^2$ .

**Solution:**

First, we compute

$$\mu = (0)(0.51) + (1)(0.38) + (2)(0.10) + (3)(0.01) = 0.61.$$

Now,

$$E(X^2) = (0)(0.51) + (1)(0.38) + (4)(0.10) + (9)(0.01) = 0.87.$$

Therefore,

$$\sigma^2 = 0.87 - (0.61)^2 = 0.4979.$$

**Example 4.10:**

The weekly demand for a drinking-water product, in thousands of liters, from a local chain of efficiency stores is a continuous random variable  $X$  having the probability density

$$f(x) = \begin{cases} 2(x-1), & 1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the mean and variance of  $X$ .

**Solution:**

Calculating  $E(X)$  and  $E(X^2)$ , we have

$$\mu = E(X) = 2 \int_1^2 x(x-1) dx = \frac{5}{3}$$

and

$$E(X^2) = 2 \int_1^2 x^2(x-1) dx = \frac{17}{6}.$$

Therefore,

$$\sigma^2 = \frac{17}{6} - \left(\frac{5}{3}\right)^2 = \frac{1}{18}.$$

### Quick Question 1.1 (Practice 4.34)

|      |     |     |     |
|------|-----|-----|-----|
| x    | -2  | 3   | 5   |
| f(x) | 0.3 | 0.2 | 0.5 |

Find the standard deviation of X.

Answer:  $\mu = (-2)(0.3) + (3)(0.2) + (5)(0.5) = -0.6 + 0.6 + 2.5 = 2.5$   
 $E(X^2) = (4)(0.3) + (9)(0.2) + (25)(0.5) = 1.2 + 1.8 + 12.5 = 15.5$   
 $\sigma^2 = 15.5 - (2.5)^2 = 15.5 - 6.25 = 9.25$   
 $\sigma = \sqrt{9.25} \approx 3.04$

## Topic 2: Covariance

$$\sigma_{XY} = E(XY) - \mu_X \mu_Y$$

Covariance tells you how X and Y move together:

- $\sigma_{XY} > 0 \rightarrow$  when X increases, Y tends to increase
- $\sigma_{XY} < 0 \rightarrow$  when X increases, Y tends to decrease
- $\sigma_{XY} = 0 \rightarrow$  no linear relationship (independent variables always give 0)

The three steps are always:

1. Find  $\mu_X$  from marginal g(x)
2. Find  $\mu_Y$  from marginal h(y)
3. Find  $E(XY)$  from joint f(x,y), then apply the formula

**Example 4.13:**

Example 3.14 on page 95 describes a situation involving the number of blue refills  $X$  and the number of red refills  $Y$ . Two refills for a ballpoint pen are selected at random from a certain box, and the following is the joint probability distribution:

| $f(x, y)$ |   | $x$            |                 |                | $h(y)$          |
|-----------|---|----------------|-----------------|----------------|-----------------|
|           |   | 0              | 1               | 2              |                 |
| $y$       | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|           | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|           | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| $g(x)$    |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |

Find the covariance of  $X$  and  $Y$ .

**Solution:**

From Example 4.6, we see that  $E(XY) = 3/14$ . Now

$$\mu_X = \sum_{x=0}^2 xg(x) = (0) \left( \frac{5}{14} \right) + (1) \left( \frac{15}{28} \right) + (2) \left( \frac{3}{28} \right) = \frac{3}{4},$$

**The Formula**

**Discrete:**

$$E(XY) = \sum_x \sum_y xy \cdot f(x, y)$$

**Continuous:**

$$E(XY) = \iint xy \cdot f(x, y) dx dy$$

It's just the expected value formula — but  $g(X, Y) = XY$ , so you multiply each  $(x, y)$  pair by its joint probability.

### Discrete Example (from the pen table)

| $f(x,y)$ | $x=0$  | $x=1$  | $x=2$  |
|----------|--------|--------|--------|
| $y=0$    | $3/28$ | $9/28$ | $3/28$ |
| $y=1$    | $3/14$ | $3/14$ | $0$    |
| $y=2$    | $1/28$ | $0$    | $0$    |

$$E(XY) = \sum_x \sum_y xy \cdot f(x,y)$$

Go through every cell, multiply  $x \times y \times f(x,y)$ :

$$E(XY) = \sum_x \sum_y xy \cdot f(x,y)$$

Go through every cell, multiply  $x \times y \times f(x,y)$ :

| $(x,y)$ | $x \times y$ | $f(x,y)$ | product |
|---------|--------------|----------|---------|
| (0,0)   | 0            | $3/28$   | 0       |
| (0,1)   | 0            | $3/14$   | 0       |
| (0,2)   | 0            | $1/28$   | 0       |
| (1,0)   | 0            | $9/28$   | 0       |
| (1,1)   | 1            | $3/14$   | $3/14$  |
| (1,2)   | 2            | 0        | 0       |
| (2,0)   | 0            | $3/28$   | 0       |
| (2,1)   | 2            | 0        | 0       |
| (2,2)   | 4            | 0        | 0       |

$$E(XY) = 0 + 0 + \dots + \frac{3}{14} + \dots = \frac{3}{14}$$

**Key shortcut:** Any cell where  $x=0$  OR  $y=0$  automatically gives 0. Only cells where BOTH  $x \neq 0$  and  $y \neq 0$  survive.



It was discrete example

**Example 4.13 (from slides) — Discrete joint:**

Using the pen-selection joint table (X=blue, Y=red):

**Step 1 —  $\mu_X$  from marginals  $g(x) = \{5/14, 15/28, 3/28\}$ :**

$$\mu_X = 0 \left( \frac{5}{14} \right) + 1 \left( \frac{15}{28} \right) + 2 \left( \frac{3}{28} \right) = 0 + \frac{15}{28} + \frac{6}{28} = \frac{21}{28} = \frac{3}{4}$$

**Step 2 —  $\mu_Y$  from marginals  $h(y) = \{15/28, 3/7, 1/28\}$ :**

$$\mu_Y = 0 \left( \frac{15}{28} \right) + 1 \left( \frac{3}{7} \right) + 2 \left( \frac{1}{28} \right) = 0 + \frac{12}{28} + \frac{2}{28} = \frac{14}{28} = \frac{1}{2}$$

**Step 3 —  $E(XY)$  (from Example 4.6, already computed):**

$$E(XY) = \frac{3}{14}$$

**Step 4 — Covariance:**

$$\sigma_{XY} = \frac{3}{14} - \left( \frac{3}{4} \right) \left( \frac{1}{2} \right) = \frac{3}{14} - \frac{3}{8} = \frac{12}{56} - \frac{21}{56} = -\frac{9}{56}$$

Negative covariance makes sense — more blue pens selected means fewer red pens possible.



$E(x)$  in joint probability discret is found by  $g(x)$  help .

In continuous , we tend to computer  $g(x)$  and  $h(y)$  so that we can apply x and y limit on it and find  $g(x)$  and  $h(y)$

**Example 4.14:**

The fraction  $X$  of male runners and the fraction  $Y$  of female runners who compete in marathon races are described by the joint density function

$$f(x, y) = \begin{cases} 8xy, & 0 \leq y \leq x \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the covariance of  $X$  and  $Y$ .

Find the correlation coefficient of  $X$  and  $Y$

if i was asked to compute  $g(x)$  here manually You integrate with respect to  $y$ , and  $x$  stays as a constant.

When finding covariance , we take help of marginal density function to calculate  $E(x)$  and  $E(x^2)$ .

IN JOINT PROBABILITY , WE ALWAYS USE MARGINAL TO CALCULATE EXPECTATION

**Solution:**

We first compute the marginal density functions. They are

$$g(x) = \begin{cases} 4x^3, & 0 \leq x \leq 1, \\ 0, & \text{elsewhere,} \end{cases}$$

and

$$h(y) = \begin{cases} 4y(1 - y^2), & 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

From these marginal density functions, we compute

$$\mu_X = E(X) = \int_0^1 4x^4 \, dx = \frac{4}{5} \text{ and } \mu_Y = \int_0^1 4y^2(1 - y^2) \, dy = \frac{8}{15}.$$

**Continuous Example (from Example 4.14)**

$$f(x, y) = 8xy, \quad 0 \leq y \leq x \leq 1$$

$$E(XY) = \int_0^1 \int_y^1 xy \cdot 8xy \, dx \, dy = \int_0^1 \int_y^1 8x^2 y^2 \, dx \, dy$$

**Inner integral** (integrate over x, y is constant):

$$= \int_0^1 8y^2 \left[ \frac{x^3}{3} \right]_y^1 dy = \int_0^1 \frac{8y^2}{3} (1 - y^3) \, dy$$

**Outer integral:**

**Inner integral** (integrate over x, y is constant):

$$= \int_0^1 8y^2 \left[ \frac{x^3}{3} \right]_y^1 dy = \int_0^1 \frac{8y^2}{3} (1 - y^3) \, dy$$

**Outer integral:**

$$= \frac{8}{3} \int_0^1 (y^2 - y^5) \, dy = \frac{8}{3} \left[ \frac{y^3}{3} - \frac{y^6}{6} \right]_0^1 = \frac{8}{3} \left[ \frac{1}{3} - \frac{1}{6} \right] = \frac{8}{3} \cdot \frac{1}{6} = \frac{4}{9}$$



From the joint density function given above, we have

$$E(XY) = \int_0^1 \int_y^1 8x^2y^2 \, dx \, dy = \frac{4}{9}.$$

Then

$$\sigma_{XY} = E(XY) - \mu_X \mu_Y = \frac{4}{9} - \left(\frac{4}{5}\right) \left(\frac{8}{15}\right) = \frac{4}{225}.$$

### Topic 3: Correlation Coefficient

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

Where  $\sigma_X = \sqrt{\sigma_X^2}$  and  $\sigma_Y = \sqrt{\sigma_Y^2}$ .

**Key facts:**

- Always between -1 and +1
- $\rho = +1 \rightarrow$  perfect positive linear relationship
- $\rho = -1 \rightarrow$  perfect negative linear relationship
- $\rho = 0 \rightarrow$  no linear relationship
- Independent variables  $\rightarrow \rho = 0$  (but  $\rho=0$  doesn't guarantee independence)

**Steps:** Find  $\sigma_X^2$  and  $\sigma_Y^2$  using the variance formula, then divide  $\sigma_{XY}$  by  $\sigma_X \sigma_Y$ .

$$\rho_{XY} = \frac{E(XY) - E(X)E(Y)}{\sqrt{E(X^2) - E(X)^2} \sqrt{E(Y^2) - E(Y)^2}}$$

### The Full Breakdown

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \cdot \sigma_Y}$$

Where:

- $\sigma_{XY}$  = covariance of X and Y
- $\sigma_X$  = standard deviation of X =  $\sqrt{\sigma_X^2}$
- $\sigma_Y$  = standard deviation of Y =  $\sqrt{\sigma_Y^2}$

So written fully:

$$\rho_{XY} = \frac{\sigma_{XY}}{\sqrt{\sigma_X^2} \cdot \sqrt{\sigma_Y^2}}$$

$$E(XY) - E(X) \cdot E(Y) / \sqrt{E(X^2) - E(X)^2} \sqrt{E(Y^2) - E(Y)^2}$$

Because

$$E(X^2) = \int_0^1 4x^5 \, dx = \frac{2}{3} \text{ and } E(Y^2) = \int_0^1 4y^3(1-y^2) \, dy = 1 - \frac{2}{3} = \frac{1}{3},$$

we conclude that

$$\sigma_X^2 = \frac{2}{3} - \left(\frac{4}{5}\right)^2 = \frac{2}{75} \text{ and } \sigma_Y^2 = \frac{1}{3} - \left(\frac{8}{15}\right)^2 = \frac{11}{225}.$$

Hence,

$$\rho_{XY} = \frac{4/225}{\sqrt{(2/75)(11/225)}} = \frac{4}{\sqrt{66}}.$$

**4.52** Random variables  $X$  and  $Y$  follow a joint distribution

$$f(x, y) = \begin{cases} 2, & 0 < x \leq y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the correlation coefficient between  $X$  and  $Y$ .

 **GIVEN**

$$f(x, y) = \begin{cases} 2, & 0 < x \leq y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

 Region: triangle where  $0 < x \leq y < 1$

 **STEP 1: MARGINALS**

◆ **Marginal of  $X$**

For a fixed  $x$ ,  $y$  goes from  $x \rightarrow 1$ :

$$g(x) = \int_{y=x}^1 2 \, dy = 2(1 - x), \quad 0 < x < 1$$

◆ **Marginal of  $Y$**

For a fixed  $y$ ,  $x$  goes from  $0 \rightarrow y$ :

$$h(y) = \int_{x=0}^y 2 \, dx = 2y, \quad 0 < y < 1$$



## 🔥 STEP 2: MEANS

### ♦ $E[X]$

$$\begin{aligned} E[X] &= \int_0^1 x \cdot 2(1-x) dx = 2 \int_0^1 (x - x^2) dx \\ &= 2 \left( \frac{1}{2} - \frac{1}{3} \right) = 2 \cdot \frac{1}{6} = \frac{1}{3} \end{aligned}$$

### ♦ $E[Y]$

$$E[Y] = \int_0^1 y \cdot 2y dy = 2 \int_0^1 y^2 dy = 2 \cdot \frac{1}{3} = \frac{2}{3}$$



Expectation of x goes from 0 to 1 bcs of g(x) as they always go 0 to 1

## 🔥 STEP 3: SECOND MOMENTS

### ♦ $E[X^2]$

$$\begin{aligned} E[X^2] &= \int_0^1 x^2 \cdot 2(1-x) dx = 2 \int_0^1 (x^2 - x^3) dx \\ &= 2 \left( \frac{1}{3} - \frac{1}{4} \right) = 2 \cdot \frac{1}{12} = \frac{1}{6} \end{aligned}$$

### ♦ $E[Y^2]$

$$E[Y^2] = \int_0^1 y^2 \cdot 2y dy = 2 \int_0^1 y^3 dy = 2 \cdot \frac{1}{4} = \frac{1}{2}$$

## 🔥 HOW TO ALWAYS DECIDE ORDER (CAM METHOD)

### Step 1:

Write region clearly:

$$0 < x \leq y < 1$$

### Step 2:

Pick order (dx dy or dy dx)

### Step 3:

Fix outer variable, then find inner limits

## 🧠 QUICK INTUITION TRICK

👉 If you see:

$$x \leq y$$

Then:

- Either:
  - $x$  goes  $0 \rightarrow y$
- Or:
  - $y$  goes  $x \rightarrow 1$



## 🔥 STEP 3: DECIDE ORDER

We chose:

$$dx \, dy$$

So:

- Inner  $\rightarrow x$
- Outer  $\rightarrow y$

## ◆ STEP 4: FIND LIMITS

Outer variable:  $y$

From region:

$$0 \leq y \leq 1$$

Inner variable:  $x$

For a fixed  $y$ :

$$0 \leq x \leq y$$

## 🔥 FINAL SETUP

$$E[XY] = \int_{y=0}^1 \int_{x=0}^y xy \cdot 2 \, dx \, dy$$

STEP 5

## 🔥 STEP 5: $E[XY]$

Use joint PDF:

$$E[XY] = \int_0^1 \int_0^y xy \cdot 2 \, dx \, dy$$

- ◆ Inner integral (w.r.t  $x$ ):

$$\int_0^y 2xy \, dx = 2y \int_0^y x \, dx = 2y \cdot \frac{y^2}{2} = y^3$$

- ◆ Outer integral:

$$E[XY] = \int_0^1 y^3 \, dy = \frac{1}{4}$$

STEP 6: CONCLUSION

### 🔥 STEP 6: COVARIANCE

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$= \frac{1}{4} - \frac{1}{3} \cdot \frac{2}{3} = \frac{1}{4} - \frac{2}{9}$$

Common denominator 36:

$$= \frac{9}{36} - \frac{8}{36} = \frac{1}{36}$$

### 🔥 STEP 7: CORRELATION

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

$$= \frac{1/36}{\sqrt{(1/18)(1/18)}}$$

$$= \frac{1/36}{1/18} = \frac{1}{2}$$



### 🔥 FINAL ANSWER

$$\boxed{\rho_{XY} = 0.5}$$

BERNOULLI THEORM:



## Topic 1: The Bernoulli Process

A Bernoulli process is any experiment where:

1. There are **repeated trials**
2. Each trial has exactly **two outcomes** — success or failure
3. Probability of success **p stays constant** trial to trial
4. Trials are **independent**

**Examples that ARE Bernoulli:**

- Flipping a coin 10 times (H or T,  $p=0.5$  always)
- Testing items off a production line (defective or not)

**Examples that are NOT Bernoulli:**

- Drawing cards without replacement —  $p$  changes each draw (13/52, then 12/51...)
- Rolling a die and observing number of spots — more than 2 outcomes

### PROBLEMS

**Problem 1:** Which of the following are binomial experiments? Explain why.

1. Selecting a few households from New York City and observing whether or not they own stocks when it is known that 28% of all households in New York City own stocks.
2. Rolling a die 10 times and observing the number of spots.

**Quick Question 1.1** (*Problem 1 from slides*) Is this a Bernoulli process? "Select households from NYC, observe if they own stocks. 28% own stocks."

**Answer:** YES. Two outcomes (own/don't own),  $p=0.28$  constant, households selected independently. ✓

**Quick Question 1.2** Is this a Bernoulli process? "12 students, 9 use Instagram. Pick 5 and ask if they use it."

3. In a group of 12 students at a college, 9 use Instagram. Five students are selected from this group of 12 and are asked whether or not they use Instagram. Is this experiment a binomial experiment?

**Quick Question 1.2** Is this a Bernoulli process? "12 students, 9 use Instagram. Pick 5 and ask if they use it."

**Answer:** NO. Sampling without replacement from a small group —  $p$  changes each draw. First pick: 9/12, second: 8/11 or 9/11 depending on first result. Not constant. ✗

own stocks when it is known that 28% of all households in New Yo

2. Rolling a die 10 times and observing the number of spots

## 🔥 Short Answer:

👉 NO — rolling a die is NOT a Bernoulli trial

### ♦ WHY?

A Bernoulli trial must have:

1. Only 2 outcomes
  - Success / Failure
2. Same probability each time
3. Independent trials

### ♦ Rolling a die

Each roll has:

$\{1, 2, 3, 4, 5, 6\}$

👉 That's 6 outcomes, not 2

✗ So not Bernoulli

## Binomial Distribution

It's a type of distribution used in discrete.

Used when:

Fixed number of trials  $< 30$

Successes (p) . Failure (q = 1 - p)

$X \sim B(n, p)$  where n = trials and p = favourable outcomes

## Topic 2: The Binomial Distribution Formula

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

How to read  $b(x; n, p)$ : "the probability of getting exactly x successes in n trials with success probability p"

## 🔥 What we were discussing e [Get Plus](#) x

When I said:

“repeated trials”

👉 That actually refers to **Bernoulli trials**, NOT the Bernoulli distribution itself.

## 🔥 Key Distinction (VERY IMPORTANT)

### ♦ Bernoulli Trial (process)

- A **single experiment**
- 2 outcomes: success / failure
- Can be **repeated**

👉 Example:

- One coin flip = one Bernoulli trial
- 10 coin flips = 10 Bernoulli trials

### ♦ Bernoulli Distribution (random variable)

- Describes the outcome of **ONE trial**

$$X = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

Bernoulli distribution describes the outcome of one trial.

## 🔥 So your confusion:

“Wasn’t Bernoulli based on repeated trials?”

👉 Correct idea, but slightly off wording:

- ✓ Bernoulli trials can be repeated
- ✗ But **Bernoulli distribution itself = one trial**

## 🔥 Clean Relationship

- One trial → **Bernoulli distribution**
- Repeat trials → **sequence of Bernoulli trials**
- Count successes → **Binomial distribution**

## 🔥 Core Difference

### ◆ Bernoulli Distribution

- One single trial
- Only 2 outcomes: success (1) or failure (0)

$$X \in \{0, 1\}$$

👉 Example:

Flip a coin once

- Head = 1
- Tail = 0

### ◆ Binomial Distribution

- Multiple Bernoulli trials
- Counts number of successes

$X$  = number of successes in  $n$  trials

## 🔥 Important Insight (Exam Favorite)

👉 Bernoulli is a special case of Binomial

When:

$$n = 1$$

Then:

$$\text{Binomial}(1, p) = \text{Bernoulli}(p)$$

## 🧠 Intuition

- Bernoulli = "Did it happen?"
- Binomial = "How many times did it happen?"

## 🔥 Example (Tie it together)

Roll a die 10 times, success = "get 6"

- Each roll → Bernoulli trial
- Each roll outcome → Bernoulli distribution
- Total number of 6's → Binomial

### Example:

The probability that a certain kind of component will survive a shock test is  $3/4$ . Find the probability that exactly 2 of the next 4 components tested survive.

Assuming that the tests are independent and  $p = 3/4$  for each of the 4 tests, we obtain

$$b\left(2; 4, \frac{3}{4}\right) = \binom{4}{2} \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^2 = \left(\frac{4!}{2! 2!}\right) \left(\frac{3^2}{4^4}\right) = \frac{27}{128}.$$

The number of successes here are 2, not 1 so it's not Bernoulli distribution.

We are given  $p$  – success and  $n$  (number)

#### Key Cumulative Tricks

| What you want        | How to get it                 |
|----------------------|-------------------------------|
| $P(X = x)$           | Read directly from table      |
| $P(X \leq x)$        | Sum from $x=0$ to $x$         |
| $P(X \geq x)$        | $1 - P(X \leq x-1)$           |
| $P(a \leq X \leq b)$ | $P(X \leq b) - P(X \leq a-1)$ |
| $P(X > x)$           | $1 - P(X \leq x)$             |



#### Part (a) — $P(X \geq 10)$ "at least 10 survive"

Strategy:  $P(X \geq 10) = 1 - P(X \leq 9)$

We need to sum  $b(x; 15, 0.4)$  from  $x=0$  to  $x=9$  and subtract from 1.

$$P(X \geq 10) = 1 - \sum_{x=0}^9 b(x; 15, 0.4)$$

| x   | b(x; 15, 0.4) |
|-----|---------------|
| 0   | 0.0005        |
| 1   | 0.0047        |
| 2   | 0.0219        |
| 3   | 0.0634        |
| 4   | 0.1268        |
| 5   | 0.1859        |
| 6   | 0.2066        |
| 7   | 0.1771        |
| 8   | 0.1181        |
| 9   | 0.0612        |
| Sum | 0.9662        |

$$P(X \geq 10) = 1 - 0.9662 = \boxed{0.0338}$$

Only a 3.38% chance that 10 or more survive — recovery is unlikely given  $p=0.4$ .

**Part (b) —  $P(3 \leq X \leq 8)$  "from 3 to 8 survive"**

**Strategy:**  $P(3 \leq X \leq 8) = P(X \leq 8) - P(X \leq 2)$

In one way removing everything below 3

## 🔥 So WHY use the second method?

### ◆ Reason 1: Less work (very important in exams)

Direct method:

- You compute:

$$P(3) + P(4) + P(5) + P(6) + P(7) + P(8)$$

👉 6 terms

CDF method:

- Only compute:

$$P(X \leq 8) \quad \text{and} \quad P(X \leq 2)$$

👉 Usually much faster

### ◆ Reason 2: Tables / calculator [Get Plus x](#) Cumulative values

In distributions like:

- Binomial
- Normal

👉 You often have:

$$P(X \leq x)$$

NOT individual probabilities

### ◆ Reason 3: Cleaner mathematically

Think of it like:

$$P(3 \leq X \leq 8) = P(X \leq 8) - P(X \leq 2)$$

👉 You're removing:

- everything below 3



$$P(3 \leq X \leq 8) = \sum_{x=0}^8 b(x; 15, 0.4) - \sum_{x=0}^2 b(x; 15, 0.4)$$

$P(X \leq 8)$  — sum  $x=0$  to 8:

| x | $b(x; 15, 0.4)$ | Running total |
|---|-----------------|---------------|
| 0 | 0.0005          | 0.0005        |
| 1 | 0.0047          | 0.0052        |
| 2 | 0.0219          | 0.0271        |
| 3 | 0.0634          | 0.0905        |
| 4 | 0.1268          | 0.2173        |
| 5 | 0.1859          | 0.4032        |
| 6 | 0.2066          | 0.6098        |
| 7 | 0.1771          | 0.7869        |
| 8 | 0.1181          | 0.9050        |

We sum till 0 to 8

$P(X \leq 2)$  — sum  $x=0$  to 2:

$$= 0.0005 + 0.0047 + 0.0219 = 0.0271$$

$$P(3 \leq X \leq 8) = 0.9050 - 0.0271 = \boxed{0.8779}$$

**87.79%** chance of between 3 and 8 survivors — this is the most likely range.

### Part (c) — $P(X = 5)$ "exactly 5 survive"

**Strategy:** You can't read  $P(X=5)$  directly from a cumulative table, so:

$$P(X = 5) = P(X \leq 5) - P(X \leq 4)$$

From the running totals above:

- $P(X \leq 5) = 0.4032$
- $P(X \leq 4) = 0.2173$

$$P(X = 5) = 0.4032 - 0.2173 = \boxed{0.1859}$$

Or directly using the formula to verify:

$$b(5; 15, 0.4) = \binom{15}{5} (0.4)^5 (0.6)^{10} = 3003 \times 0.01024 \times 0.06047 = \mathbf{0.1859} \checkmark$$

## Topic 3: Using the Binomial Table

The table gives **individual probabilities**  $b(x; n, p)$  directly — no calculation needed.

**How to read the table:**

- Find your  $n$  on the left
- Find your  $x$  row
- Go across to your  $p$  column
- Read the value

**BUT** for "at least", "at most", "between" — you need to **add or subtract** values.

### Key Cumulative Tricks

| What you want        | How to get it                 |
|----------------------|-------------------------------|
| $P(X = x)$           | Read directly from table      |
| $P(X \leq x)$        | Sum from $x=0$ to $x$         |
| $P(X \geq x)$        | $1 - P(X \leq x-1)$           |
| $P(a \leq X \leq b)$ | $P(X \leq b) - P(X \leq a-1)$ |
| $P(X > x)$           | $1 - P(X \leq x)$             |



---

## 🔥 KEY IDEA

- 👉 You can **ALWAYS** use the formula
- 👉 But you use **tables** when calculation is hard

### ◆ Case 1: $P(X \leq 5)$

- ✓ Using formula (ALWAYS possible)

$$P(X \leq 5) = \sum_{x=0}^5 \binom{n}{x} p^x q^{n-x}$$

- 👉 This is correct
- ✗ But:
  - You must calculate **6 terms**
  - Gets **long and messy**

### ◆ Case 2: $P(X = 5)$

- ✓ Using formula

$$P(X = 5) = \binom{n}{5} p^5 q^{n-5}$$

- ✓ Fast and clean

- ✓ Using table

$$P(X = 5) = P(X \leq 5) - P(X \leq 4)$$

- ✓ Used when:
    - Only CDF table is given
-

### FINAL RULE (MEMORIZE THIS)

| Type                | Best Method                                                                               |  |
|---------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| $P(X = x)$          | Formula  |                                                                                     |
| $P(X \leq x)$       | Table    |                                                                                     |
| Large sums          | Table    |                                                                                     |
| Small/simple values | Formula  |                                                                                     |

**Problem 3:** According to a survey conducted at the local DMV, 50% of drivers who drive to work stated that they regularly exceed the posted speed limit on their way to work. Suppose that this result is true for the population of drivers who drive to work. A random sample of 13 drivers who drive to work is selected. Use the binomial probabilities table to find the probability that the number of drivers in this sample of 13 who regularly exceed the posted speed limit on their way to work is,  
**a.** at most 5 **b.** 6 to 9 **c.** at least 7

## ✓ Binomial Distribution → use when:

Check these 4 conditions:

1. Fixed number of trials → ✓ (13 drivers)
2. Two outcomes → ✓ (speed / no speed)
3. Same probability → ✓ ( $p = 0.5$ )
4. Independent trials → ✓ (assumed)

👉 ALL satisfied → BINOMIAL

## ✗ When NOT binomial?

### ◆ Poisson → use when:

- Counting rare events
- Given average rate ( $\lambda$ )
- No fixed  $n$

👉 Example: number of accidents per day

### ◆ Normal → use when:

- $n$  is large
- Often approximation of binomial
- Need mean & variance



## 🔥 IN THIS QUESTION

- $n = 13$
- $p = 0.5$

👉 Clearly:

$$X \sim \text{Binomial}(13, 0.5)$$

**Quick Question 3.1** (*Problem 3 from slides*) 50% of drivers exceed speed limit.  $n=13$ ,  $p=0.50$ . Use the table provided.

(a)  $P(X \leq 5)$  — "at most 5":

From table,  $n=13$ ,  $p=0.50$ , sum  $x=0$  to 5:  
 $0.0001+0.0016+0.0095+0.0349+0.0873+0.1571 = 0.2905$

(b)  $P(6 \leq X \leq 9)$  — "6 to 9":

$P(X \leq 9) - P(X \leq 5) =$   
 $(0.0001+0.0016+0.0095+0.0349+0.0873+0.1571+0.2095+0.2095+0.1571+0.0873) -$   
 $0.2905 = 0.9539 - 0.2905 = 0.6634$

(c)  $P(X \geq 7)$  — "at least 7":

$1 - P(X \leq 6) = 1 - (0.2905 + 0.2095) = 1 - 0.5000 = 0.5000$

Makes sense — symmetric distribution at  $p=0.5$ !

### **Example:**

A large chain retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 3%.

- (a) The inspector randomly picks 20 items from a shipment. What is the probability that there will be at least one defective item among these 20?
- (b) Suppose that the retailer receives 10 shipments in a month and the inspector randomly tests 20 devices per shipment. What is the probability that there will be exactly 3 shipments each containing at least one defective device among the 20 that are selected and tested from the shipment?
- (a) Denote by  $X$  the number of defective devices among the 20. Then  $X$  follows a  $b(x; 20, 0.03)$  distribution. Hence,

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) = 1 - b(0; 20, 0.03) \\ &= 1 - (0.03)^0(1 - 0.03)^{20-0} = 0.4562. \end{aligned}$$

- (b) In this case, each shipment can either contain at least one defective item or not. Hence, testing of each shipment can be viewed as a Bernoulli trial with  $p = 0.4562$  from part (a). Assuming independence from shipment to shipment and denoting by  $Y$  the number of shipments containing at least one defective item,  $Y$  follows another binomial distribution  $b(y; 10, 0.4562)$ . Therefore,

$$P(Y = 3) = \binom{10}{3} 0.4562^3 (1 - 0.4562)^7 = 0.1602.$$

## Topic 4: Mean and Variance of Binomial

These are beautifully simple — no integration or summation needed:

$$\mu = np \quad \sigma^2 = npq \quad \sigma = \sqrt{npq}$$

That's it. Just multiply.

### Binomial Distribution Formulas

- Mean ( $\mu$ ):

$$\mu = np$$

*(This is the average or expected value of the distribution.)*

- Variance ( $\sigma^2$ ):

$$\sigma^2 = npq$$

*(This measures the spread or variability of the data.)*

- Standard Deviation ( $\sigma$ ):

$$\sigma = \sqrt{npq}$$

*(This is the square root of the variance, representing the spread in the same units as the mean.)*

### Example:

It is conjectured that an impurity exists in 30% of all drinking wells in a certain rural community. In order to gain some insight into the true extent of the problem, it is determined that some testing is necessary. It is too expensive to test all of the wells in the area, so 10 are randomly selected for testing.

- Using the binomial distribution, what is the probability that exactly 3 wells have the impurity, assuming that the conjecture is correct?
- What is the probability that more than 3 wells are impure?

(a) We require

$$b(3; 10, 0.3) = \sum_{x=0}^3 b(x; 10, 0.3) - \sum_{x=0}^2 b(x; 10, 0.3) = 0.6496 - 0.3828 = 0.2668.$$

(b) In this case,  $P(X > 3) = 1 - 0.6496 = 0.3504$ .

Example:



| <i>n</i> | <i>x</i> | <i>p</i> |       |       |       |       |       |       |       |       |       |       |
|----------|----------|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|          |          | .05      | .10   | .20   | .30   | .40   | .50   | .60   | .70   | .80   | .90   | .95   |
| 10       | 0        | .5987    | .3487 | .1074 | .0282 | .0060 | .0010 | .0001 | .0000 | .0000 | .0000 | .0000 |
|          | 1        | .3151    | .3874 | .2684 | .1211 | .0403 | .0098 | .0016 | .0001 | .0000 | .0000 | .0000 |
|          | 2        | .0746    | .1937 | .3020 | .2335 | .1209 | .0439 | .0106 | .0014 | .0001 | .0000 | .0000 |
|          | 3        | .0105    | .0574 | .2013 | .2668 | .2150 | .1172 | .0425 | .0090 | .0008 | .0000 | .0000 |
|          | 4        | .0010    | .0112 | .0881 | .2001 | .2508 | .2051 | .1115 | .0368 | .0055 | .0001 | .0000 |
|          | 5        | .0001    | .0015 | .0264 | .1029 | .2007 | .2461 | .2007 | .1029 | .0264 | .0015 | .0001 |
|          | 6        | .0000    | .0001 | .0055 | .0368 | .1115 | .2051 | .2508 | .2001 | .0881 | .0112 | .0010 |
|          | 7        | .0000    | .0000 | .0008 | .0090 | .0425 | .1172 | .2150 | .2668 | .2013 | .0574 | .0105 |
|          | 8        | .0000    | .0000 | .0001 | .0014 | .0106 | .0439 | .1209 | .2335 | .3020 | .1937 | .0746 |
|          | 9        | .0000    | .0000 | .0000 | .0001 | .0016 | .0098 | .0403 | .1211 | .2684 | .3874 | .3151 |
| 13       | 10       | .0000    | .0000 | .0000 | .0000 | .0001 | .0010 | .0060 | .0282 | .1074 | .3487 | .5987 |
|          | 0        | .5133    | .2542 | .0550 | .0097 | .0013 | .0001 | .0000 | .0000 | .0000 | .0000 | .0000 |
|          | 1        | .3512    | .3672 | .1787 | .0540 | .0113 | .0016 | .0001 | .0000 | .0000 | .0000 | .0000 |
|          | 2        | .1109    | .2448 | .2680 | .1388 | .0453 | .0095 | .0012 | .0001 | .0000 | .0000 | .0000 |
|          | 3        | .0214    | .0997 | .2457 | .2181 | .1107 | .0349 | .0065 | .0006 | .0000 | .0000 | .0000 |
|          | 4        | .0028    | .0277 | .1535 | .2337 | .1845 | .0873 | .0243 | .0034 | .0001 | .0000 | .0000 |
|          | 5        | .0003    | .0055 | .0691 | .1803 | .2214 | .1571 | .0656 | .0142 | .0011 | .0000 | .0000 |
|          | 6        | .0000    | .0008 | .0230 | .1030 | .1968 | .2095 | .1312 | .0442 | .0058 | .0001 | .0000 |
|          | 7        | .0000    | .0001 | .0058 | .0442 | .1312 | .2095 | .1968 | .1030 | .0230 | .0008 | .0000 |
|          | 8        | .0000    | .0000 | .0011 | .0142 | .0656 | .1571 | .2214 | .1803 | .0691 | .0055 | .0003 |
|          | 9        | .0000    | .0000 | .0001 | .0034 | .0243 | .0873 | .1845 | .2337 | .1535 | .0277 | .0028 |
|          | 10       | .0000    | .0000 | .0000 | .0006 | .0065 | .0349 | .1107 | .2181 | .2457 | .0997 | .0214 |
|          | 11       | .0000    | .0000 | .0000 | .0001 | .0012 | .0095 | .0453 | .1388 | .2680 | .2448 | .1109 |
| 12       | .0000    | .0000    | .0000 | .0000 | .0001 | .0016 | .0113 | .0540 | .1787 | .3672 | .3512 |       |
| 13       | .0000    | .0000    | .0000 | .0000 | .0000 | .0001 | .0013 | .0097 | .0550 | .2542 | .5133 |       |

For second part , sum till 3

1-  $P(x \leq 3)$

## Poisson Distribution

#### Definition

### What is the Poisson Distribution?

It models the **number of events occurring in a fixed interval** of time, distance, or area — when events happen independently and at a constant average rate.

Examples of Poisson random variables:

 Calls received per hour

 Days school closed per winter

 Bacteria in a culture

 Typing errors per page

interval or fall in such a small region is negligible.

The number  $X$  of outcomes occurring during a Poisson experiment is called a **Poisson random variable**, and its probability distribution is called the **Poisson distribution**. The mean number of outcomes is computed from  $\mu = \lambda t$ , where

Mean number of outcomes  $\mu = \lambda t$

$P(x; \lambda t)$

Following formula is used for computing Poisson probabilities.

**Poisson  
Distribution**

The probability distribution of the Poisson random variable  $X$ , representing the number of outcomes occurring in a given time interval or specified region denoted by  $t$ , is

$$p(x; \lambda t) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}, \quad x = 0, 1, 2, \dots,$$

where  $\lambda$  is the average number of outcomes per unit time, distance, area, or volume and  $e = 2.71828 \dots$ .

### Example:

During a laboratory experiment, the average number of radioactive particles passing through a counter in 1 millisecond is 4. What is the probability that 6 particles enter the counter in a given millisecond?

: Using the Poisson distribution with  $x = 6$  and  $\lambda t = 4$  and referring to Table A.2, we have

$$p(6; 4) = \frac{e^{-4}4^6}{6!} = \sum_{x=0}^6 p(x; 4) - \sum_{x=0}^5 p(x; 4) = 0.8893 - 0.7851 = 0.1042.$$

### Example:

Ten is the average number of oil tankers arriving each day at a certain port. The facilities at the port can handle at most 15 tankers per day. What is the probability that on a given day tankers have to be turned away?

Let  $X$  be the number of tankers arriving each day. Then, using Table A.2, we have

$$P(X > 15) = 1 - P(X \leq 15) = 1 - \sum_{x=0}^{15} p(x; 10) = 1 - 0.9513 = 0.0487.$$

•  $p(3) = 0.007567$

•  $p(4) = 0.018917$

•  $p(5) = 0.037833$

•  $p(6) = 0.063055$

•  $p(7) = 0.090079 \leftarrow$  shown above

•  $p(8) = 0.112599$

•  $p(9) = 0.125110$

•  $p(10) = 0.125110$

•  $p(11) = 0.113736$

•  $p(12) = 0.094780$

•  $p(13) = 0.072908$

•  $p(14) = 0.052077$

•  $p(15) = 0.034718$

Step — Sum all terms:

$$P(X \leq 15) = 0.0000454 + 0.000454 + 0.002270 + \dots + 0.034718 = 0.9513$$

Final answer:

$$P(X > 15) = 1 - 0.9513 = 0.0487$$

There is a 4.87% probability that tankers will be turned away on a given day.

Want to be notified when Claude responds?

Notify

×

### Mean and Variance:

Both the mean and the variance of the Poisson distribution  $p(x; \lambda t)$  are  $\lambda t$ .

(b) no errors.

**5.58** A certain area of the eastern United States is, on average, hit by 6 hurricanes a year. Find the probability that in a given year that area will be hit by

- (a) fewer than 4 hurricanes;
- (b) anywhere from 6 to 8 hurricanes.

We are given:

- Average hurricanes per year:  $\lambda = 6$
- Let  $X$  = number of hurricanes in a year.
- $X \sim \text{Poisson}(\lambda = 6)$

The Poisson probability formula is:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

**(a) Probability of fewer than 4 hurricanes**

We want:

$$P(X < 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$

Step 1: Compute each term

- $P(X = 0) = \frac{e^{-6}6^0}{0!} = e^{-6}$
- $P(X = 1) = \frac{e^{-6}6^1}{1!} = 6e^{-6}$
- $P(X = 2) = \frac{e^{-6}6^2}{2!} = \frac{36e^{-6}}{2} = 18e^{-6}$
- $P(X = 3) = \frac{e^{-6}6^3}{3!} = \frac{216e^{-6}}{6} = 36e^{-6}$

Step 2: Sum them:

$$P(X < 4) = e^{-6}(1 + 6 + 18 + 36) = e^{-6} \cdot 61$$

Step 3: Approximate using  $e^{-6} \approx 0.00247875$

$$P(X < 4) \approx 61 \cdot 0.00247875 \approx 0.151$$

✓ So, probability fewer than 4 hurricanes  $\approx 0.151$

**(b) Probability of 6 to 8 hurricanes**

We want:

$$P(6 \leq X \leq 8) = P(X = 6) + P(X = 7) + P(X = 8)$$

Step 1: Compute each term

- $P(X = 6) = \frac{e^{-6}6^6}{6!} = \frac{46656e^{-6}}{720} = 64.8e^{-6}$
- $P(X = 7) = \frac{e^{-6}6^7}{7!} = \frac{279936e^{-6}}{5040} = 55.555e^{-6}$
- $P(X = 8) = \frac{e^{-6}6^8}{8!} = \frac{1679616e^{-6}}{40320} = 41.625e^{-6}$

Step 2: Sum them:

$$P(6 \leq X \leq 8) = e^{-6}(64.8 + 55.555 + 41.625) = e^{-6} \cdot 161.98$$

Step 3: Multiply by  $e^{-6} \approx 0.00247875$

$$P(6 \leq X \leq 8) \approx 161.98 \cdot 0.00247875 \approx 0.401$$

✓ So, probability between 6 and 8 hurricanes  $\approx 0.401$

✓ **Final Answer**

- (a)  $P(X < 4) \approx 0.151$
- (b)  $P(6 \leq X \leq 8) \approx 0.401$

**5.57** On average, a textbook author makes two word-processing errors per page on the first draft of her textbook. What is the probability that on the next page she will make

- (a) 4 or more errors?
- (b) no errors?

### (a) Probability of 4 or more errors

We want:

$$P(X \geq 4) = 1 - P(X < 4) = 1 - P(X = 0, 1, 2, 3)$$

Step 1: Compute probabilities for  $X = 0, 1, 2, 3$

- $P(X = 0) = \frac{e^{-2}2^0}{0!} = e^{-2}$
- $P(X = 1) = \frac{e^{-2}2^1}{1!} = 2e^{-2}$
- $P(X = 2) = \frac{e^{-2}2^2}{2!} = 2e^{-2}$
- $P(X = 3) = \frac{e^{-2}2^3}{3!} = \frac{8e^{-2}}{6} = 1.333e^{-2}$

Step 2: Sum them:

$$P(X < 4) = e^{-2}(1 + 2 + 2 + 1.333) = e^{-2} \cdot 6.333$$

Step 3: Multiply by  $e^{-2} \approx 0.1353$

$$P(X < 4) \approx 6.333 \cdot 0.1353 \approx 0.857$$

Step 4: Find  $P(X \geq 4)$

$$P(X \geq 4) = 1 - 0.857 = 0.143$$

✓ So, probability of 4 or more errors  $\approx 0.143$

### (b) Probability of no errors

$$P(X = 0) = \frac{e^{-2}2^0}{0!} = e^{-2} \approx 0.135$$

✓ So, probability of no errors  $\approx 0.135$

## NORMAL DISTRIBUTION

Represented by z

$x = \mu$ , the mean

### What is the Normal Distribution?

It is the most important continuous probability distribution in statistics. Its graph is the famous bell-shaped curve, symmetric around the center. It describes many real-world things like heights, weights, measurement errors, rainfall, etc.

It is also called the Gaussian distribution.

Mean =  $\mu$ , Variance =  $\sigma^2$ , Standard deviation =  $\sigma$ .

**Same  $\sigma$ (S.D), different  $\mu$  (mean)  $\rightarrow$  same shape curves, just shifted left or right.**

values of  $x$  and the curve drawn. In Figure 6.3, we have sketched two normal curves having the same standard deviation but different means. The two curves are identical in form but are centered at different positions along the horizontal axis.

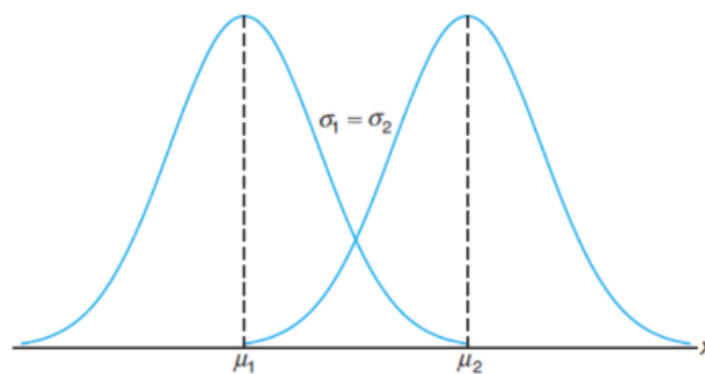


Figure 6.3: Normal curves with  $\mu_1 < \mu_2$  and  $\sigma_1 = \sigma_2$ .

Both curves are identical just both are centered at different position due to mean

**Same  $\mu$ (Mean), different  $\sigma$  (S.D)  $\rightarrow$  curves centered at same point, but larger  $\sigma$  = wider and flatter, smaller  $\sigma$  = taller and narrower.**

Figure 6.3: Normal curves with  $\mu_1 < \mu_2$  and  $\sigma_1 = \sigma_2$ .

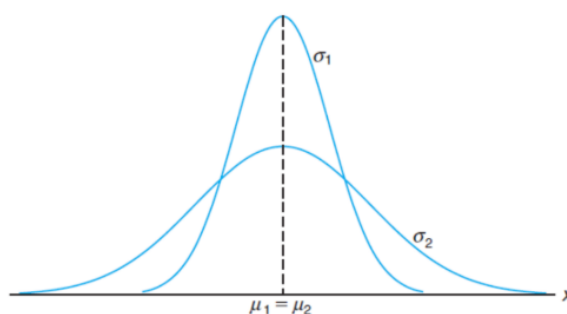


Figure 6.4: Normal curves with  $\mu_1 = \mu_2$  and  $\sigma_1 < \sigma_2$ .



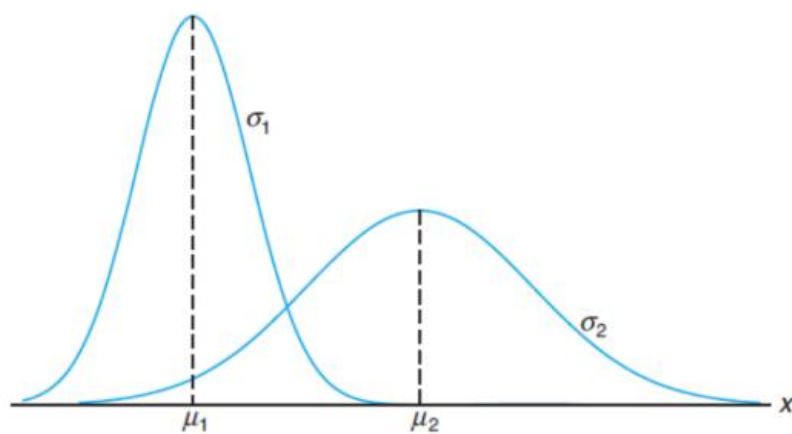
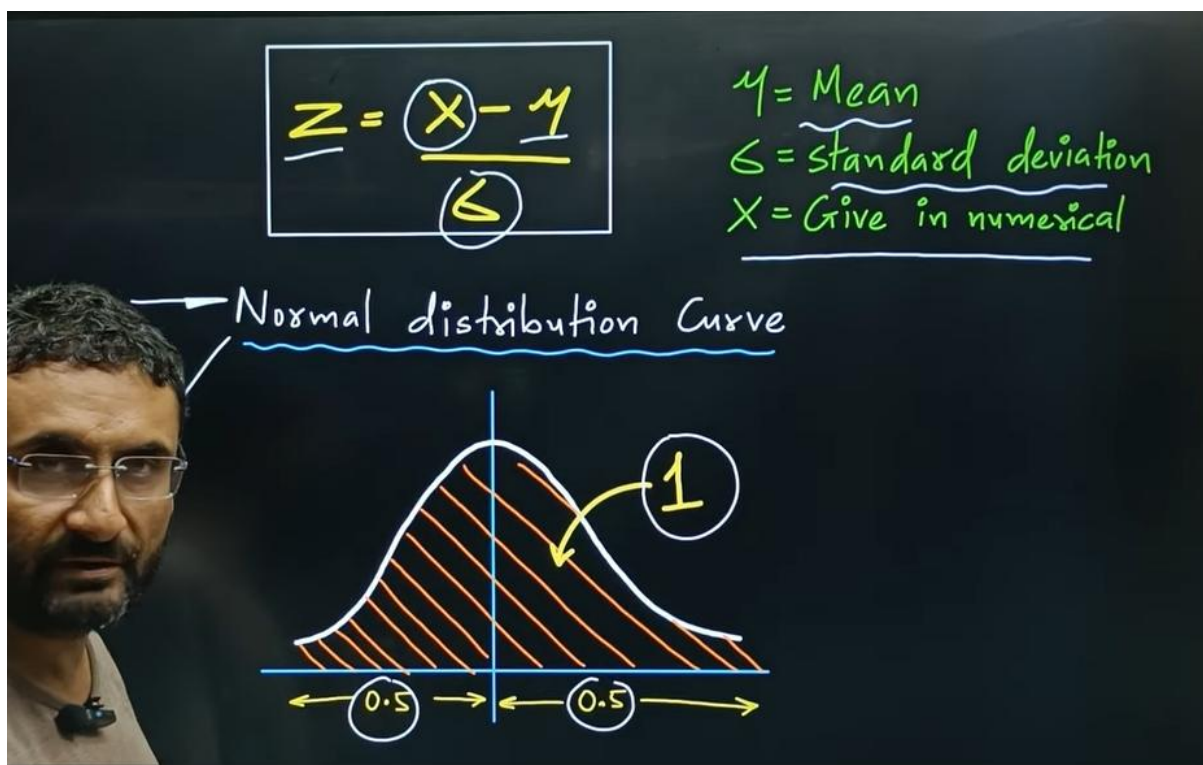


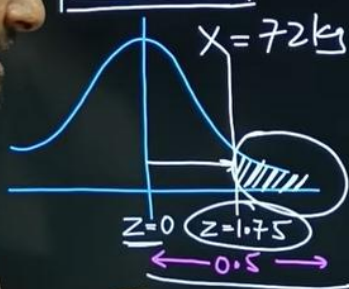
Figure 6.5: Normal curves with  $\mu_1 < \mu_2$  and  $\sigma_1 < \sigma_2$ .



Q) Assume that mean weight of students in a pharmacy institution to be 65 kg with std. deviation of 16 kg. How many students in the institute of 400 students would you expect to be over 72 kg? ( $P(Z=1.75)=0.4599$ )

$$Z = \frac{X - \mu}{\sigma} = \frac{72 - 65}{16}$$

$$Z = 1.75$$



$$\begin{aligned} \mu &= 65 \\ \sigma &= 16 \\ X &= 72 \end{aligned}$$

$$\begin{aligned} P(X > 72) &= P(Z > 1.75) \\ &= 0.5 - P(Z = 1.75) \\ &= 0.5 - 0.4599 \\ &= 0.0401 \end{aligned}$$

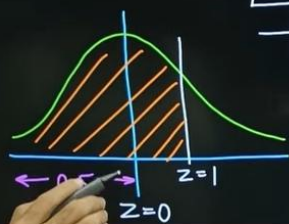
$$\begin{aligned} \text{No. of students over 72 kg} &= 400 \times 0.0401 \\ &= 16.04 \\ &\approx 16 \text{ Ans} \end{aligned}$$

Q) Daily income of a worker following normal distribution with mean of 1000/- and std. deviation 100/-. Find the probability of income ① Less than 11,000/- and ② more than 11,000/-. ( $P(Z=1)=0.3413$ )

Sol<sup>n</sup> ① Less than 11,000/-

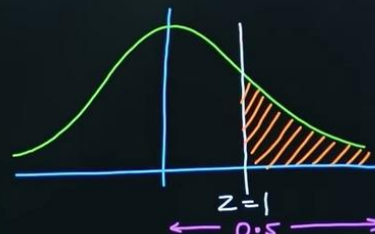
$$\begin{aligned} \mu &= 1000 \\ \sigma &= 100 \\ X &= 11,000 \end{aligned} \quad \begin{aligned} Z &= \frac{X - \mu}{\sigma} \\ \therefore Z &= \frac{11,000 - 1000}{100} \end{aligned}$$

$$Z = 1$$



$$\begin{aligned} P(X < 11,000) &= P(Z < 1) \\ &= 0.5 + P(Z = 1) \\ &= 0.5 + 0.3413 \\ &= 0.8413 \text{ Ans} \end{aligned}$$

② More than 11,000/-



$$\begin{aligned} P(X > 11,000) &= P(Z > 1) \\ &= 0.5 - P(Z = 1) \\ &= 0.5 - 0.3413 \\ &= 0.1587 \text{ Ans} \end{aligned}$$

Q) 2000 students appeared for the exam. Distribution of marks is assumed to be normal with mean 30, std. deviation 6.25. How many students will get marks between 20 and 40. ( $P(z=1.6) = 0.4452$ ).

Sol<sup>y</sup>

$X_1 = 20$   
 $Z_1 = \frac{X - \mu}{\sigma}$   
 $Z_1 = \frac{20 - 30}{6.25}$   
 $Z_1 = -1.6$

$X_2 = 40$   
 $Z_2 = \frac{X - \mu}{\sigma}$   
 $Z_2 = \frac{40 - 30}{6.25}$   
 $Z_2 = 1.6$

$\mu = 30 \checkmark$   
 $\sigma = 6.25 \checkmark$  } given.

$$P(20 < X < 40) = P(-1.6 < Z < 1.6)$$

$$= P(Z = -1.6) + P(Z = 1.6)$$

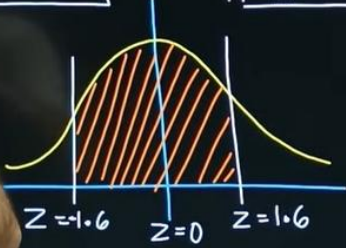
$$= 0.4452 + 0.4452$$

$$= 0.8905$$

No. of students between 20 & 40

$$= 2000 \times 0.8905$$

$$= 1780.8$$

$$= 1781 \text{ Ans}$$


### Area Under the normal curve:

The curve of any continuous probability distribution or density function is constructed so that the area under the curve bounded by the two ordinates  $x = x_1$  and  $x = x_2$  equals the probability that the random variable  $X$  assumes a value between  $x = x_1$  and  $x = x_2$ . Thus, for the normal curve in Figure 6.6,

$$P(x_1 < X < x_2) = \int_{x_1}^{x_2} n(x; \mu, \sigma) dx = \frac{1}{\sqrt{2\pi}\sigma} \int_{x_1}^{x_2} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dx$$

is represented by the area of the shaded region.

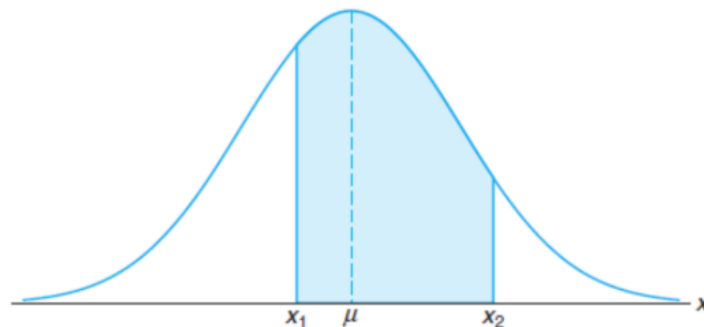


Figure 6.6:  $P(x_1 < X < x_2) = \text{area of the shaded region.}$

where  $Z$  is seen to be a normal random variable with mean 0 and variance 1.

**Definition 6.1:**

The distribution of a normal random variable with mean 0 and variance 1 is called a **standard normal distribution**.

The original and transformed distributions are illustrated in Figure 6.8. Since

**Example 6.2:**

Given a standard normal distribution, find the area under the curve that lies

- (a) to the right of  $z = 1.84$  and
- (b) between  $z = -1.97$  and  $z = 0.86$ .

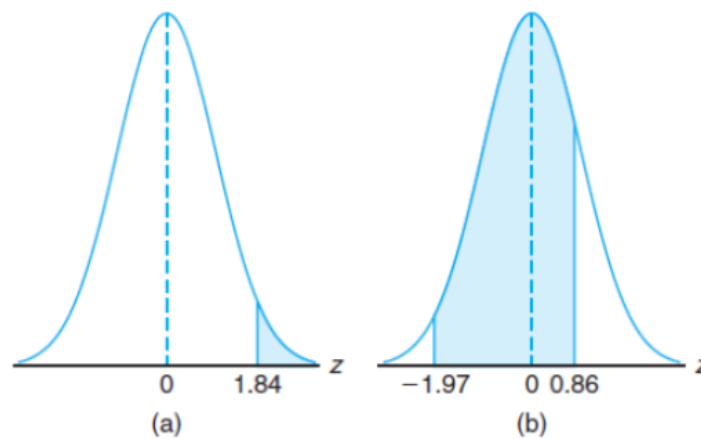


Figure 6.9: Areas for Example 6.2.

**The question is  $P(Z > 1.84)$**

**Method 1 — using 1:**

$$P(Z > 1.84) = 1 - P(Z < 1.84) = 1 - 0.9671 = \mathbf{0.0329}$$

**Method 2 — using 0.5 (your way):**

$$P(Z > 1.84) = 0.5 - P(0 < Z < 1.84) = 0.5 - 0.4671 = \mathbf{0.0329}$$

Same answer ✓



### Why both work:

The right half of the curve from 0 to  $\infty = 0.5$

So the tiny right tail = 0.5 minus the middle chunk from 0 to 1.84.

$$P(0 < Z < 1.84) = P(Z < 1.84) - 0.5 = 0.9671 - 0.5 = 0.4671$$

$$\text{Then } 0.5 - 0.4671 = 0.0329$$

In case of inequality on both sides , we subtract right side from left

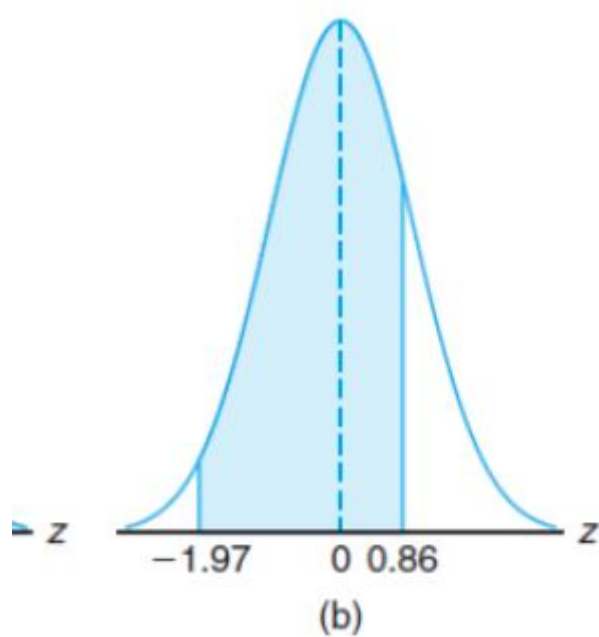
Table A.3 (continued) Areas under the Normal Curve

| z   | .00    | .01    | .02    | .03    | .04    | .05    | .06    | .07    | .08    | .09    |
|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |
| 1.7 | 0.9554 | 0.9564 | 0.9573 | 0.9582 | 0.9591 | 0.9599 | 0.9608 | 0.9616 | 0.9625 | 0.9633 |
| 1.8 | 0.9641 | 0.9649 | 0.9656 | 0.9664 | 0.9671 | 0.9678 | 0.9686 | 0.9693 | 0.9699 | 0.9706 |

### When to prefer which:

- The 1 - table value method is faster because you directly use the table value without an extra subtraction step.
- The 0.5 method is more intuitive visually — you're literally looking at the right half and removing the known middle portion.

Both are 100% correct, use whichever feels natural to you.



in Table A.3 to the left of  $z = 1.84$ , namely,  $1 - 0.9671 = 0.0329$ .

- (b) The area in Figure 6.9(b) between  $z = -1.97$  and  $z = 0.86$  is equal to the area to the left of  $z = 0.86$  minus the area to the left of  $z = -1.97$ . From Table A.3 we find the desired area to be  $0.8051 - 0.0244 = 0.7807$ . ■

**Example 6.3:**

Given a standard normal distribution, find the value of  $k$  such that

- (a)  $P(Z > k) = 0.3015$  and
- (b)  $P(k < Z < -0.18) = 0.4197$ .

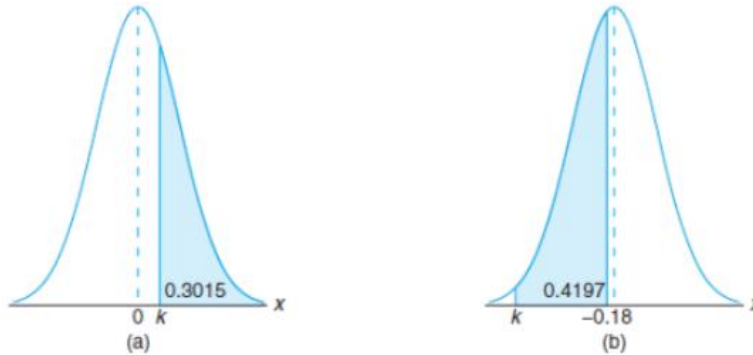


Figure 6.10: Areas for Example 6.3.

**Solution:**

Distributions and the desired areas are shown in Figure 6.10.

- (a) In Figure 6.10(a), we see that the  $k$  value leaving an area of 0.3015 to the right must then leave an area of 0.6985 to the left. From Table A.3 it follows that  $k = 0.52$ .
- (b) From Table A.3 we note that the total area to the left of  $-0.18$  is equal to 0.4286. In Figure 6.10(b), we see that the area between  $k$  and  $-0.18$  is 0.4197, so the area to the left of  $k$  must be  $0.4286 - 0.4197 = 0.0089$ . Hence, from Table A.3, we have  $k = -2.37$ . ■



**Example 6.4:**

Given a random variable  $X$  having a normal distribution with  $\mu = 50$  and  $\sigma = 10$ , find the probability that  $X$  assumes a value between 45 and 62.

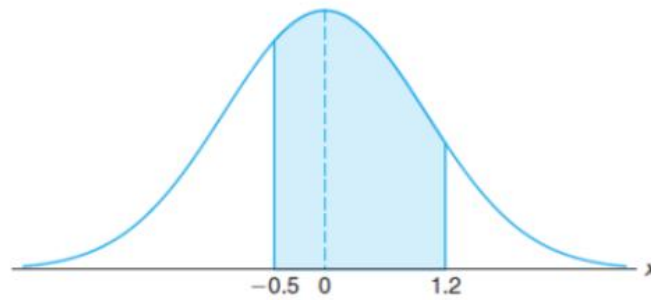


Figure 6.11: Area for Example 6.4.

**Solution:**

The  $z$  values corresponding to  $x_1 = 45$  and  $x_2 = 62$  are

$$z_1 = \frac{45 - 50}{10} = -0.5 \text{ and } z_2 = \frac{62 - 50}{10} = 1.2.$$

Therefore,

$$P(45 < X < 62) = P(-0.5 < Z < 1.2).$$

$P(-0.5 < Z < 1.2)$  is shown by the area of the shaded region in Figure 6.11. This area may be found by subtracting the area to the left of the ordinate  $z = -0.5$  from the entire area to the left of  $z = 1.2$ . Using Table A.3, we have

$$\begin{aligned} P(45 < X < 62) &= P(-0.5 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.5) \\ &= 0.8849 - 0.3085 = 0.5764. \end{aligned}$$

**Example 6.5:**

Given that  $X$  has a normal distribution with  $\mu = 300$  and  $\sigma = 50$ , find the probability that  $X$  assumes a value greater than 362.

**Solution:**

The normal probability distribution with the desired area shaded is shown in Figure 6.12. To find  $P(X > 362)$ , we need to evaluate the area under the normal curve to the right of  $x = 362$ . This can be done by transforming  $x = 362$  to the corresponding  $z$  value, obtaining the area to the left of  $z$  from Table A.3, and then subtracting this area from 1. We find that

$$z = \frac{362 - 300}{50} = 1.24.$$

Hence,

$$P(X > 362) = P(Z > 1.24) = 1 - P(Z < 1.24) = 1 - 0.8925 = 0.1075.$$



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## HYPOTHESIS TESTING

1. Company XYZ manufactures calculators with an average mass of 450g. An engineer believes that average weight to be different and decides to calculate the average mass of 50 calculators. State the null and alternative hypotheses.

Null:  $H_0$   $\mu$

Alternative:  $H_a$

If there was percentage, we deal with proportion

1. Company XYZ manufactures calculators with an average mass of 450g. An engineer believes that average weight to be different and decides to calculate the average mass of 50 calculators. State the null and alternative hypotheses.

Null:  $H_0$   $\mu = 450g$

Alternative:  $H_a$   $\mu \neq 450g$

2. The teachers in a school believes that at least 80% of students will complete high school. A student disagrees with this value and decides to conduct a test. State the null and alternative hypotheses.

$H_0: p \geq 0.80$

$H_a: p < 0.80$   
 $< -5$

3. A teacher wishes to test if the average GPA of students in the high school is different from 2.7. State the null and alternative hypotheses.

$$H_0: \mu = 2.7$$

$$H_a: \mu \neq 2.7$$

4. The percentage of residents who own a vehicle in town XYZ is no more than 75%. A researcher disagrees with the value and decides to survey 100 residents asking them if they own a vehicle. State the null and alternative hypotheses.

$$H_0: p \leq 0.75$$

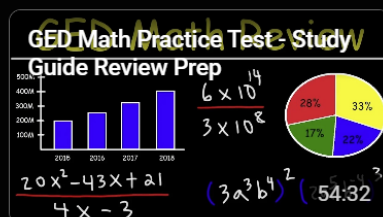
$$H_a: p > 0.75$$

JG

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Type I Error: Rejecting the null hypothesis when it is true.

Type 2 Error: Not rejecting the null hypothesis when it is false.

$$P(\text{type I error} / H_0 \text{ is true}) = \alpha$$

$$P(\text{type II error} / H_0 \text{ is false}) = \beta$$

$$P(\text{rejecting a false } H_0) = 1 - \beta$$

|                      | $H_0$        |               |
|----------------------|--------------|---------------|
|                      | True         | False         |
| Reject $H_0$         | Type I error | ✓             |
| Fail to Reject $H_0$ | ✓            | Type II error |



1. Let's say that the null hypothesis  $H_0$  is: John's used car is safe to drive. (a) Which statement represents a type I error? (b) Which statement represents a type II error? (c) Which type of error has greater consequence?

- (a) John thinks that his car may be safe when, in fact, it is not safe. X
- (b) John thinks that his car may be safe when, in fact, it is safe. ✓
- (c) John thinks that his car may not be safe when, in fact, it is not safe. ✓
- (d) John thinks that his car may not be safe when, in fact, it is safe. X

1. Let's say that the null hypothesis  $H_0$  is: John's used car is safe to drive. (a) Which statement represents a type I error? (b) Which statement represents a type II error? (c) Which type of error has greater consequence?

- False  $H_0$
- (a) John thinks that his car may be safe when, in fact, it is not safe. X Type II error
  - (b) John thinks that his car may be safe when, in fact, it is safe. ✓
  - (c) John thinks that his car may not be safe when, in fact, it is not safe. ✓
  - (d) John thinks that his car may not be safe when, in fact, it is safe. X Type I error

Type I error  $\rightarrow H_0$  is true

Type II error  $\rightarrow H_0$  is false

2. In a criminal court case, the null hypothesis  $H_0$  is that the defendant is presumed innocent. (a) Which statement represents a type I error? (b) Which statement represents a type II error? (c) Which type of error has greater consequence?

- Bad
- (a) The jury believes that the defendant is guilty when, in fact, he is innocent. X
  - (b) The jury believes that the defendant is guilty when, in fact, he is not innocent. ✓
  - (c) The jury believes that the defendant is not guilty when, in fact, he is not innocent. X
  - (d) The jury believes that the defendant is not guilty when, in fact, he is innocent. ✓

$H_0$  is true  $\rightarrow$  Type I error

$H_0$  is false  
 $\downarrow$   
Type II error

Bad

$H_0 : \mu = 100g$   
 $H_a : \mu \neq 100g$

1:43 / 5:40 Two Tailed Tests >

**One Tailed and Two Tailed Tests, Critical Values, & Significance Level - Inferential Statistics**

Lets say test is conducted on 95 % confidence interval .

C is 1- alpha

Alpha is significance level

So  $1 - 0.05$  is 0.95 .

0.05 is significance level

Bcs Alpha is planned on two divisions

$H_0 : \mu = 100g$      $\alpha = 0.05$   
 $H_a : \mu \neq 100g$

95%     $C = 0.95$

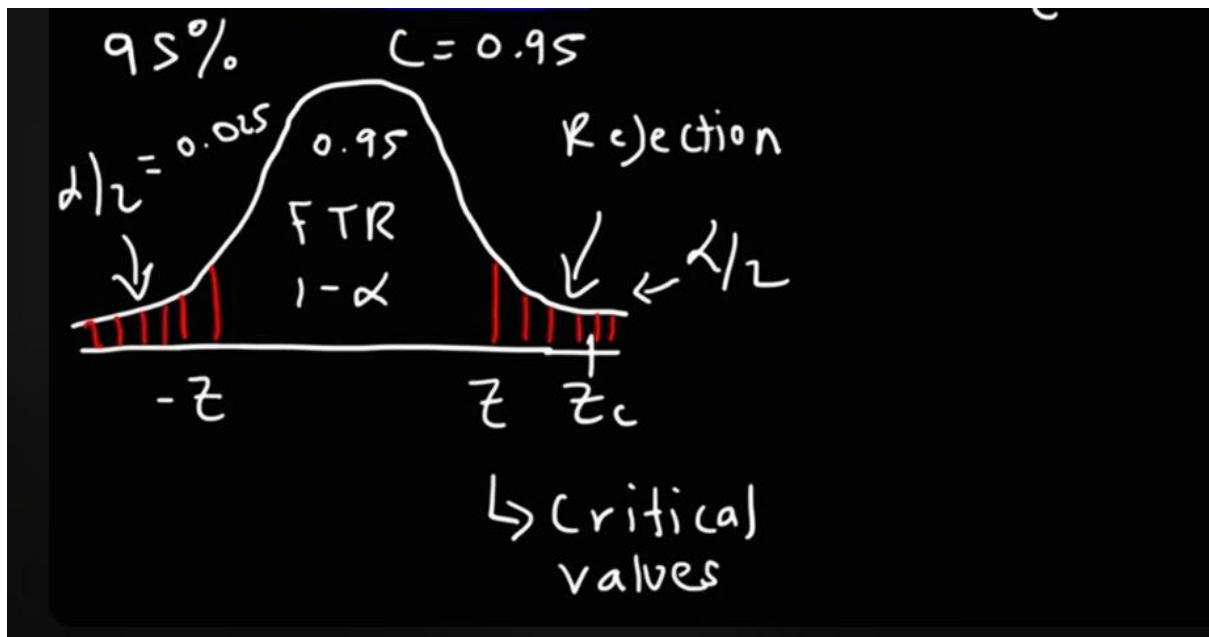
$\alpha/2 = 0.025$      $\alpha/2$

critical values

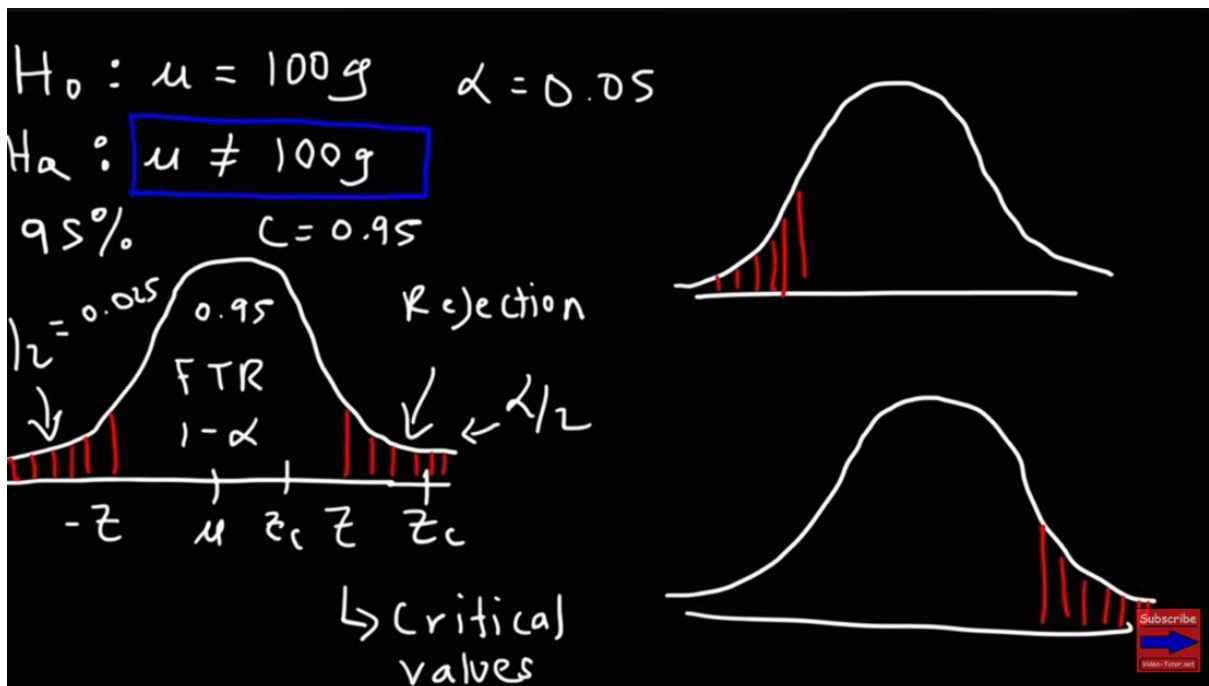


Now in order to determine whether to reject or fail to reject null hypothesis , we need to calculate z value and compare it to critical value

If the calculated z value is greater than critical value , you should reject null hypothesis



If z value is not in shaded region , we should not reject null hypothesis



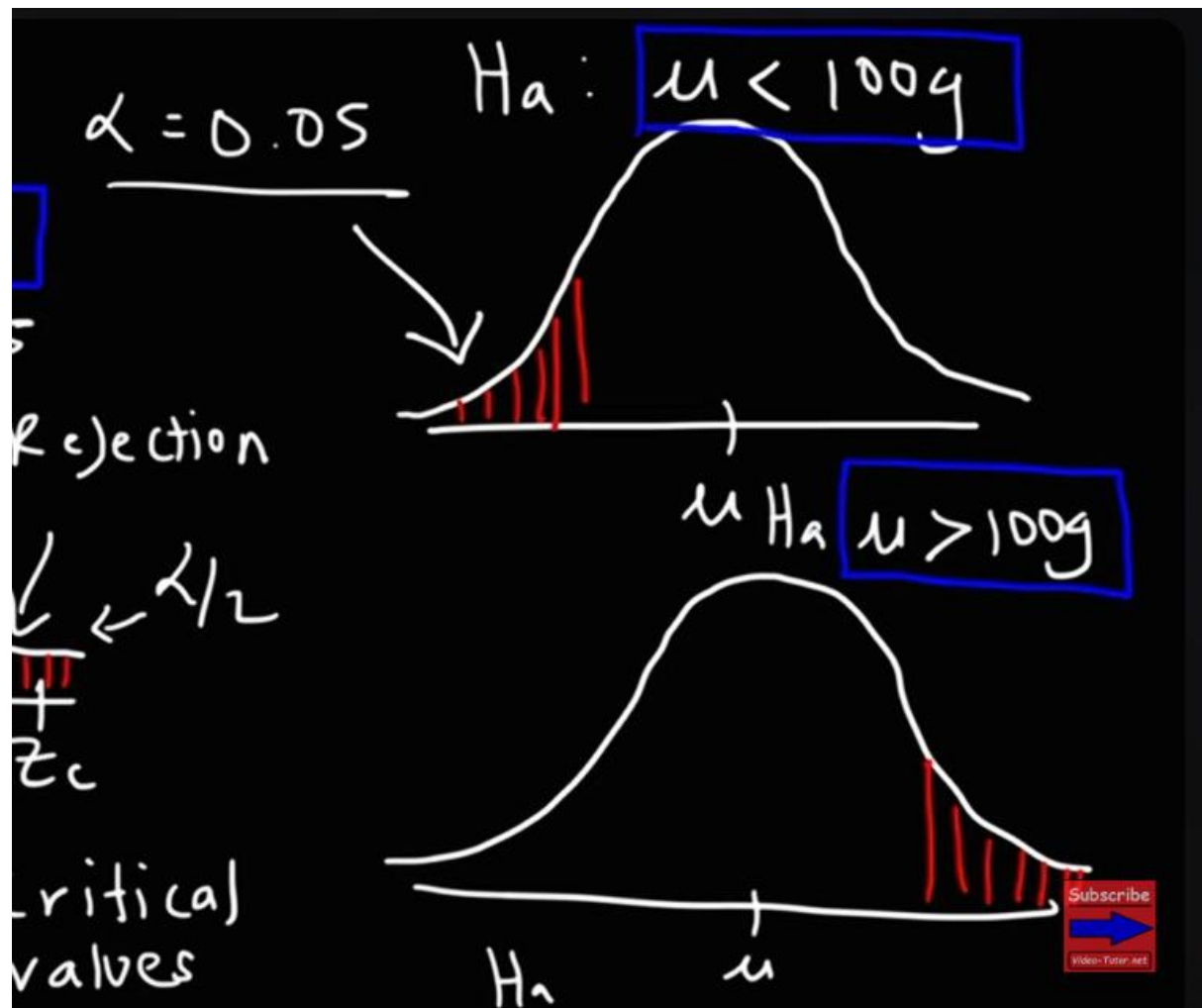
First one is left tailed test , second one is right tailed test

For 5 percent significance level , alpha would be on left hand side ,

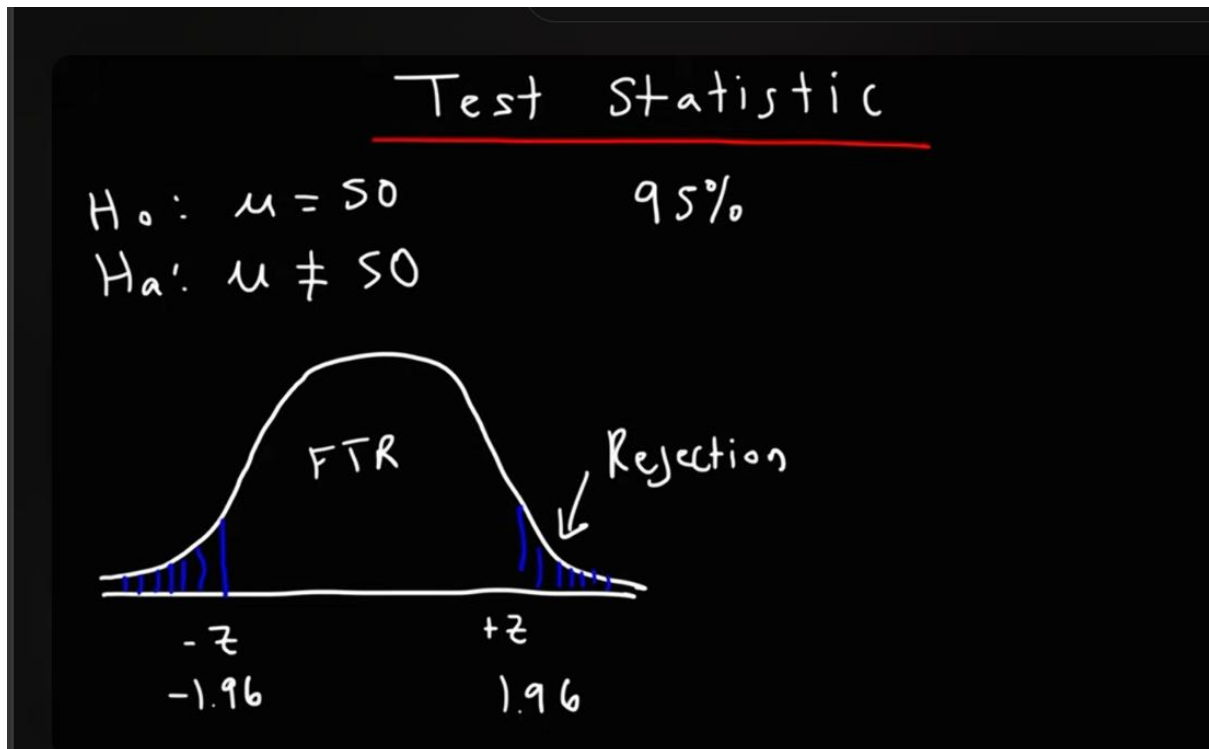
Lets say that alternative hypphthesis is such that mean is less than 100 g

So if its less than some number we need to check left tailed test

And if mean is greater than some number ie mean is greater than 100 , then we use right tailed test







With 95 percent confidence interval critical values would be 1.96

Now whether to know whether you should reject or accept the null hypothesis , you need to calculate another z value , a calculated z value and compare it to critical

$\mu \rightarrow$  pop. mean

$n < 30$

$\sigma$  is unknown

If n is less than 30 and pop standard deviation is unknown , then we need to use **t distribution**

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Sample mean – population mean divided by sample S.D over square root of n

$$n < 30$$
$$\sigma \text{ known}$$

In this case , we can use normal distribution and we **calculate z value**

Z is calculated as sample mean – population mean divided by pop s.d over square root of n

$$z_c = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$



$$n > 30$$

$\sigma$  is unknown

Since sample size is large ,we can apply normal distribution and calculate z value

$$n > 30$$

$\sigma$  is unknown

$$z_c = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$$

This time its sample S.D

$$n > 30$$

$\sigma$  is unknown

$$z_c = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$n > 30$$

$\sigma$  is known

$$z_c = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

THESE ARE THE FORMULAS WHEN DEALING WITH POPULATION MEAN

WHEN DEALING WITH POPULATION PROPORTION

Sample proportion – population proportion over the square root of pq over n

$p \rightarrow$  pop. proportion

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

$$q = 1 - p$$

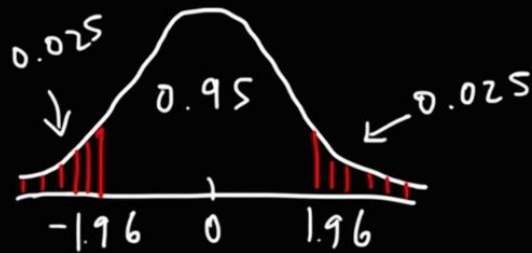
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1. A factory has a machine that dispenses 80 mL of fluid in a bottle. An employee believes the average amount of fluid is not 80 mL. Using 40 samples, he measures the average amount dispensed by the machine to be 78 mL with a standard deviation of 2.5. (a) State the null and alternative hypotheses. (b) At a 95% confidence level, is there enough evidence to support the idea that the machine is not working properly?

$$H_0: \mu = 80$$

$$H_a: \mu \neq 80$$



Shaded area is rejection region

If  $n$  would be 30, then we would be using  $t$  distribution, but here  $n$  is 40, so we use  $z$  test

level, is there enough evidence to support the idea that the machine is not working properly?

$$\bar{x} = 78 \quad s = 2.5 \quad n = 40$$

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

0.025  
= Rejection

1. A factory has a machine that dispenses 80 mL of fluid in a bottle. An employee believes the average amount of fluid is not 80 mL. Using 40 samples, he measures the average amount dispensed by the machine to be 78 mL with a standard deviation of 2.5. (a) State the null and alternative hypotheses. (b) At a 95% confidence level, is there enough evidence to support the idea that the machine is not working properly?

$$H_0: \mu = 80$$

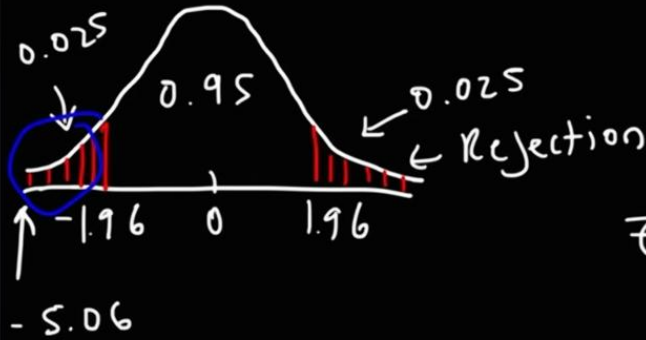
$$H_a: \mu \neq 80$$

$$\bar{x} = 78 \quad s = 2.5 \quad n = 40$$

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{78 - 80}{2.5/\sqrt{40}} = \frac{-2}{0.39528}$$

$$z_c \approx -5.06$$



We reject the null hypothesis

2. A company manufactures car batteries with an average life span of 2 or more years. An engineer believes this value to be less. Using 10 samples, he measures the average life span to be 1.8 years with a standard deviation of 0.15. (a) State the null and alternative hypotheses. (b) At a 99% confidence level, is there enough evidence to discard the null hypothesis?

$$H_0: \mu \geq 2$$

$$H_a: \mu < 2$$



Since mean is less than 2 , it's a left tail test

2. A company manufactures car batteries with an average life span of 2 or more years. An engineer believes this value to be less. Using 10 samples, he measures the average life span to be 1.8 years with a standard deviation of 0.15. (a) State the null and alternative hypotheses. (b) At a 99% confidence level, is there enough evidence to discard the null hypothesis?

$$H_0: \mu \geq 2$$

$$H_a: \mu < 2$$


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Alpha / significance level indicates the shaded region which is 0.01

2. A company manufactures car batteries with an average life span of 2 or more years. An engineer believes this value to be less. Using 10 samples, he measures the average life span to be 1.8 years with a standard deviation of 0.15. (a) State the null and alternative hypotheses. (b) At a 99% confidence level, is there enough evidence to discard the null hypothesis?

$$H_0: \mu \geq 2$$

$$H_a: \mu < 2$$

$n = 10$     $\bar{x} = 1.8$     $s = 0.15$     $\mu_0 = 2$



Use t test , as n is less than 30 and we don't know pop s.d

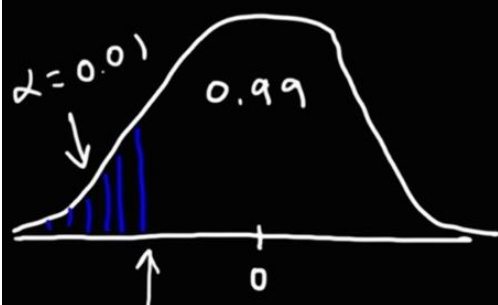


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$$H_0: \mu \geq 2$$

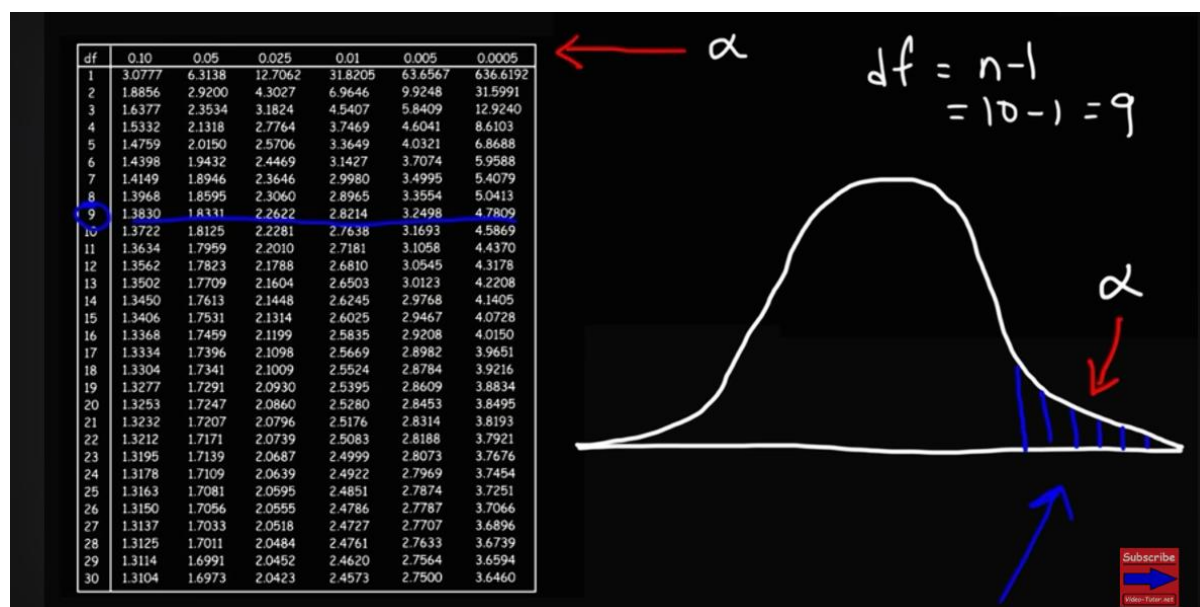
$$H_a: \mu < 2$$

$$n = 10 \quad \bar{x} = 1.8 \quad s = 0.15 \quad \mu_0 = 2$$

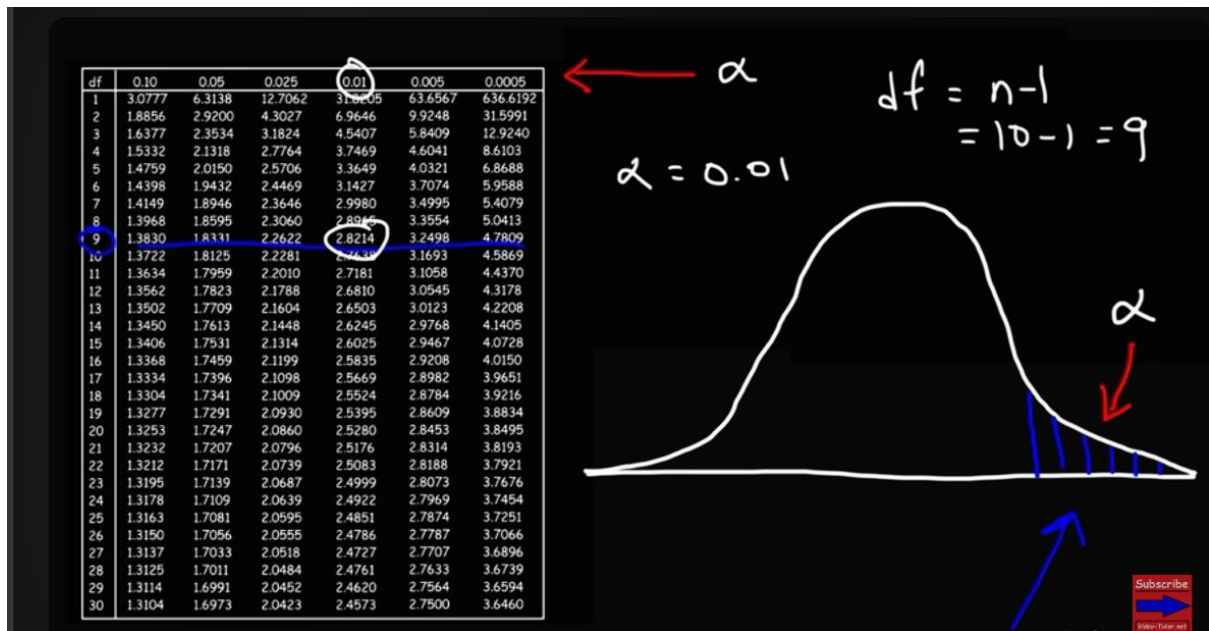


First we need to find the t value which corresponds to critical value before calculating t

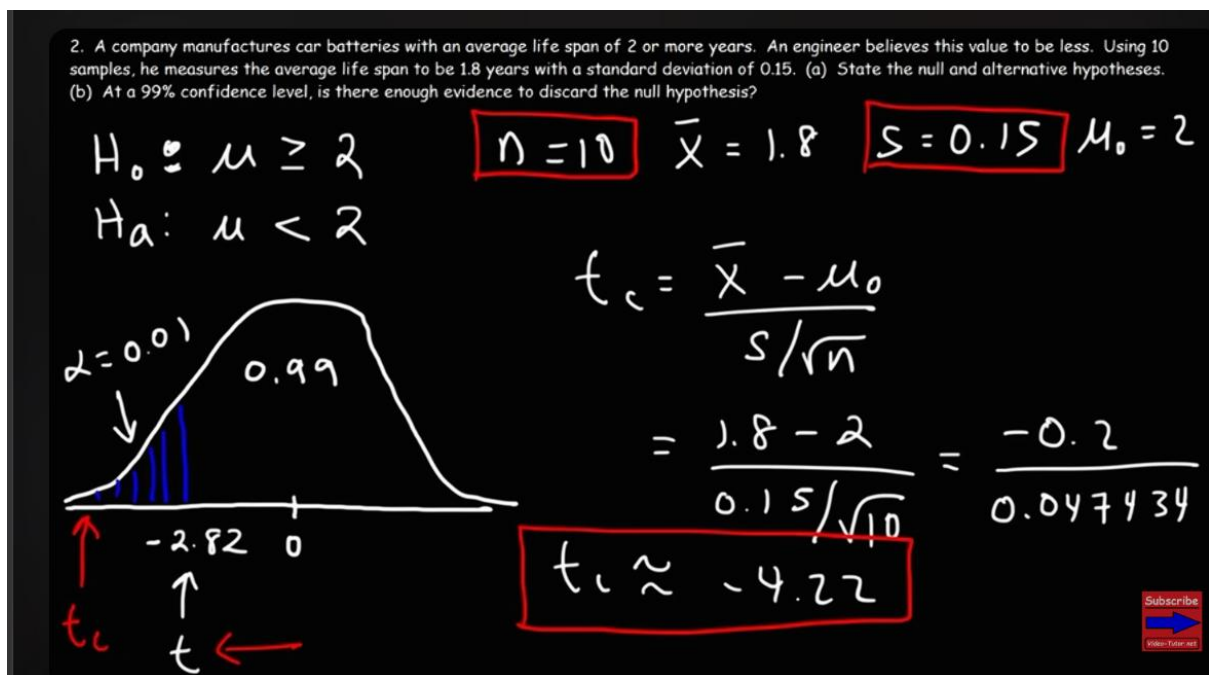
Our n was 10, so df is = n- 1 which is 9



Next thing is our alpha value which is 0.01



Our t value is 2.82 on right and -2.82 on left



Reject null hypothesis

We can say that

With 99 percent of confidence interval, we cannot accept null hypothesis that the average life span of car batteries is 2 or more years.

1. The average weight of all residents in town XYZ is 168 lbs. A nutritionist believes the true mean to be different. She measured the weight of 36 individuals and found the mean to be 169.5 lbs with a standard deviation of 3.9. (a) State the null and alternative hypotheses. (b) At a 95% confidence level, is there enough evidence to discard the null hypothesis? (Use the p-value method)

$$H_0: \mu = 168 \quad n = 36 \quad \bar{x} = 169.5 \quad s = 3.9$$

$$H_a: \mu \neq 168 \quad C = 0.95 \quad \alpha = 1 - C = 0.05$$

p-value

P value corresponds to the area of z or t statistic

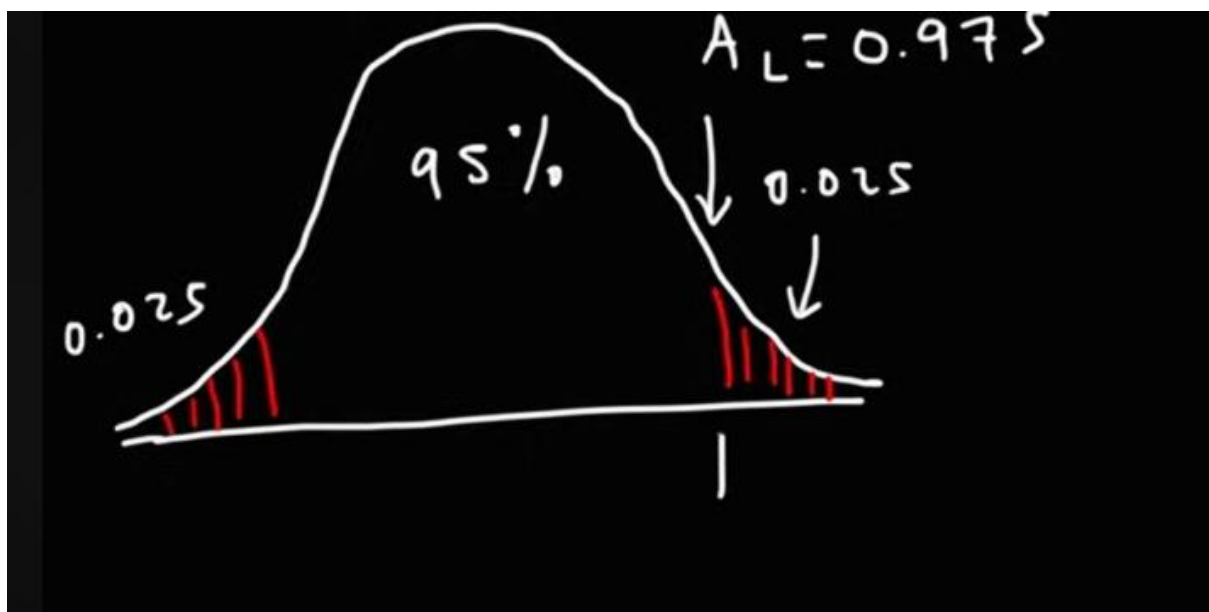
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$$H_0: \mu = 168 \quad n = 36 \quad \bar{x} = 169.5 \quad s = 3.9$$

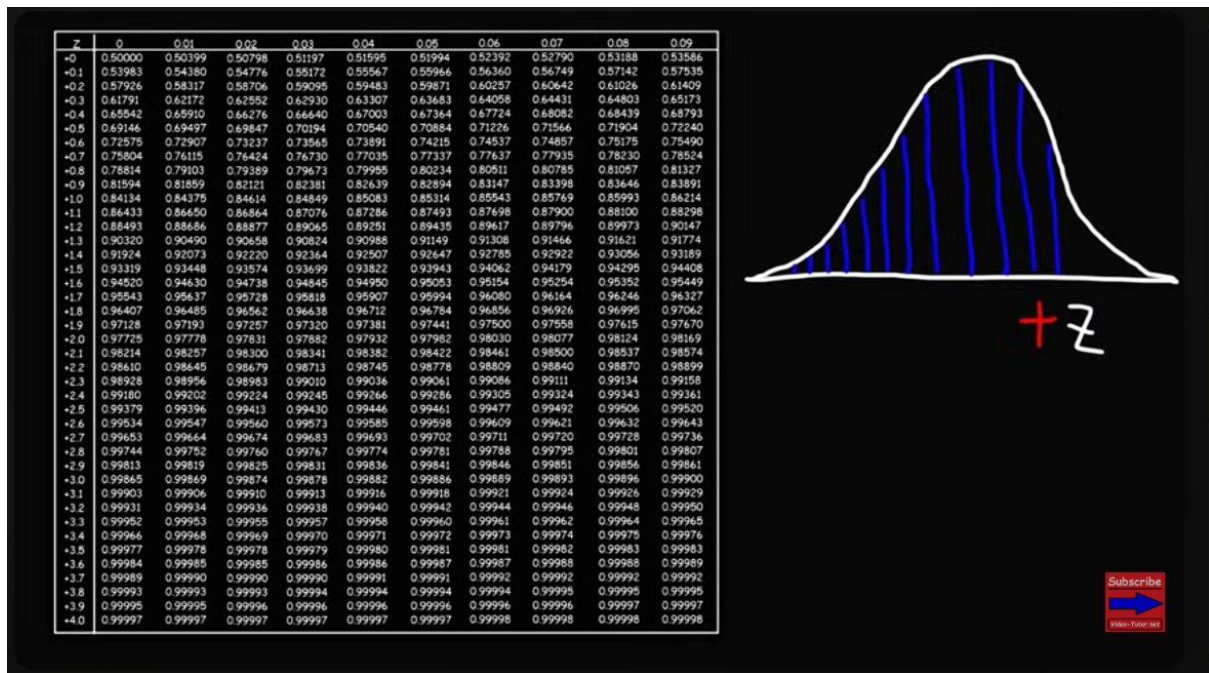
$$H_a: \mu \neq 168 \quad C = 0.95 \quad \alpha = 1 - C = 0.05$$

p-value <  $\alpha \rightarrow$  reject  $H_0$

p-value  $\geq \alpha \rightarrow$  Accept  $H_0$

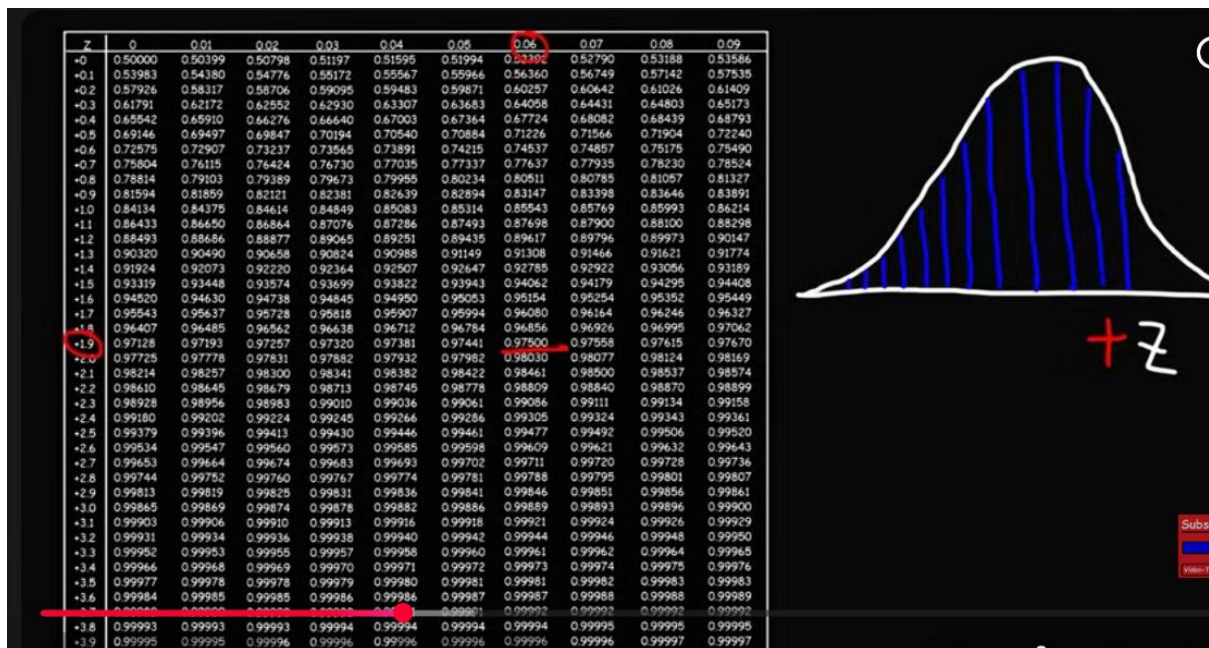


Area of left is 0.975, we will find z value which corresponds to that area



Positive z table

Locate area in table



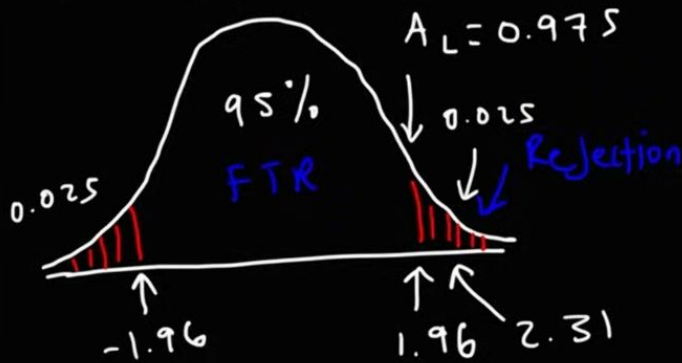




1. The average weight of all residents in town XYZ is 168 lbs. A nutritionist believes the true mean to be different. She measured the weight of 36 individuals and found the mean to be 169.5 lbs with a standard deviation of 3.9. (a) State the null and alternative hypotheses. (b) At a 95% confidence level, is there enough evidence to discard the null hypothesis? (Use the p-value method)

$$H_0: \mu = 168 \quad n = 36 \quad \bar{x} = 169.5 \quad s = 3.9$$

$$H_a: \mu \neq 168 \quad C = 0.95 \quad \alpha = 1 - C = 0.05$$



$$z_c = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{169.5 - 168}{3.9/\sqrt{36}}$$

$$= \frac{1.5}{0.65} = 2.31$$

$$z_c > z_{cv} \rightarrow \text{reject } H_0$$



that's the traditional method we used .

now for p method

1. The average weight of all residents in town XYZ is 168 lbs. A nutritionist believes the true mean to be different. She measured the weight of 36 individuals and found the mean to be 169.5 lbs with a standard deviation of 3.9. (a) State the null and alternative hypotheses. (b) At a 95% confidence level, is there enough evidence to discard the null hypothesis? (Use the p-value method)

$$H_0: \mu = 168 \quad n = 36 \quad \bar{x} = 169.5 \quad s = 3.9$$

$$H_a: \mu \neq 168 \quad C = 0.95 \quad \alpha = 1 - C = 0.05$$



$$z_c = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{169.5 - 168}{3.9/\sqrt{36}}$$

$$= \frac{1.5}{0.65} = 2.31$$

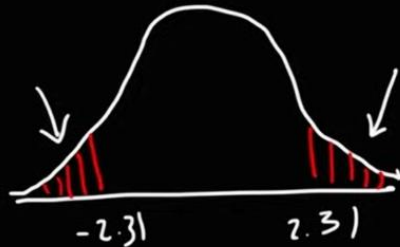


For this , we put calculated z value in normal distribution

1. The average weight of all residents in town XYZ is 168 lbs. A nutritionist believes the true mean to be different. She measured the weight of 36 individuals and found the mean to be 169.5 lbs with a standard deviation of 3.9. (a) State the null and alternative hypotheses. (b) At a 95% confidence level, is there enough evidence to discard the null hypothesis? (Use the p-value method)

$$H_0: \mu = 168 \quad n = 36 \quad \bar{X} = 169.5 \quad s = 3.9$$

$$H_a: \mu \neq 168 \quad C = 0.95 \quad \alpha = 1 - C = 0.05$$



$$z_c = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{169.5 - 168}{3.9/\sqrt{36}}$$

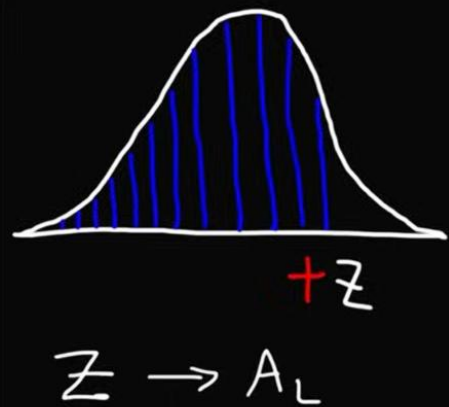
$$= \frac{1.5}{0.65} = 2.31$$



We need to determine area of left and right , those area corresponds to p value

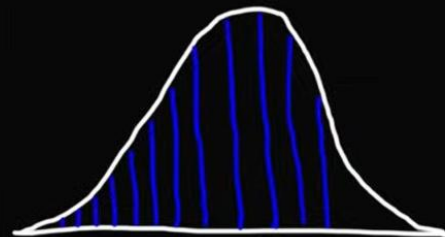
We have z value , we need to find corresponding area value

| z    | 0       | 0.01    | 0.02    | 0.03    | 0.04    | 0.05    | 0.06    | 0.07    | 0.08    | 0.09    |
|------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| -0   | 0.50000 | 0.50399 | 0.50798 | 0.51197 | 0.51596 | 0.51994 | 0.52392 | 0.52790 | 0.53188 | 0.53586 |
| -0.1 | 0.53983 | 0.54380 | 0.54776 | 0.55172 | 0.55567 | 0.55966 | 0.56360 | 0.56749 | 0.57142 | 0.57535 |
| -0.2 | 0.57926 | 0.58317 | 0.58706 | 0.59095 | 0.59483 | 0.59871 | 0.60257 | 0.60642 | 0.61026 | 0.61409 |
| -0.3 | 0.61791 | 0.62172 | 0.62552 | 0.62930 | 0.63307 | 0.63683 | 0.64058 | 0.64431 | 0.64803 | 0.65173 |
| -0.4 | 0.65542 | 0.65910 | 0.66276 | 0.66640 | 0.67003 | 0.67364 | 0.67724 | 0.68082 | 0.68439 | 0.68793 |
| -0.5 | 0.69146 | 0.69497 | 0.69847 | 0.70194 | 0.70540 | 0.70884 | 0.71226 | 0.71566 | 0.71904 | 0.72240 |
| -0.6 | 0.72575 | 0.72907 | 0.73237 | 0.73565 | 0.73891 | 0.74215 | 0.74537 | 0.74857 | 0.75175 | 0.75490 |
| -0.7 | 0.75804 | 0.76115 | 0.76424 | 0.76730 | 0.77035 | 0.77337 | 0.77637 | 0.77935 | 0.78230 | 0.78524 |
| -0.8 | 0.78814 | 0.79103 | 0.79389 | 0.79673 | 0.79955 | 0.80234 | 0.80511 | 0.80785 | 0.81057 | 0.81327 |
| -0.9 | 0.81594 | 0.81859 | 0.82121 | 0.82381 | 0.82639 | 0.82894 | 0.83147 | 0.83398 | 0.83646 | 0.83891 |
| -1.0 | 0.84134 | 0.84375 | 0.84614 | 0.84849 | 0.85083 | 0.85314 | 0.85543 | 0.85769 | 0.85993 | 0.86214 |
| -1.1 | 0.86433 | 0.86650 | 0.86864 | 0.87076 | 0.87286 | 0.87493 | 0.87698 | 0.87900 | 0.88100 | 0.88298 |
| -1.2 | 0.88493 | 0.88686 | 0.88877 | 0.89065 | 0.89251 | 0.89435 | 0.89617 | 0.89796 | 0.89973 | 0.90147 |
| -1.3 | 0.90320 | 0.90490 | 0.90658 | 0.90824 | 0.90988 | 0.91149 | 0.91308 | 0.91466 | 0.91621 | 0.91774 |
| -1.4 | 0.91924 | 0.92073 | 0.92220 | 0.92364 | 0.92507 | 0.92647 | 0.92785 | 0.92922 | 0.93056 | 0.93189 |
| -1.5 | 0.93319 | 0.93448 | 0.93574 | 0.93699 | 0.93822 | 0.93943 | 0.94062 | 0.94179 | 0.94295 | 0.94408 |
| -1.6 | 0.94520 | 0.94630 | 0.94738 | 0.94845 | 0.94950 | 0.95053 | 0.95154 | 0.95254 | 0.95352 | 0.95449 |
| -1.7 | 0.95543 | 0.95637 | 0.95728 | 0.95818 | 0.95907 | 0.95994 | 0.96080 | 0.96164 | 0.96246 | 0.96327 |
| -1.8 | 0.96407 | 0.96485 | 0.96562 | 0.96638 | 0.96712 | 0.96784 | 0.96856 | 0.96926 | 0.96995 | 0.97062 |
| -1.9 | 0.97128 | 0.97193 | 0.97257 | 0.97320 | 0.97381 | 0.97441 | 0.97500 | 0.97558 | 0.97615 | 0.97670 |
| -2.0 | 0.97725 | 0.97778 | 0.97831 | 0.97882 | 0.97932 | 0.97982 | 0.98030 | 0.98077 | 0.98124 | 0.98169 |
| -2.1 | 0.98214 | 0.98257 | 0.98300 | 0.98341 | 0.98382 | 0.98422 | 0.98461 | 0.98500 | 0.98537 | 0.98574 |
| -2.2 | 0.98610 | 0.98645 | 0.98679 | 0.98713 | 0.98745 | 0.98778 | 0.98809 | 0.98840 | 0.98870 | 0.98899 |
| -2.3 | 0.98928 | 0.98956 | 0.98983 | 0.99010 | 0.99036 | 0.99061 | 0.99086 | 0.99111 | 0.99134 | 0.99158 |
| -2.4 | 0.99180 | 0.99202 | 0.99224 | 0.99245 | 0.99266 | 0.99286 | 0.99305 | 0.99324 | 0.99343 | 0.99361 |
| -2.5 | 0.99379 | 0.99396 | 0.99413 | 0.99430 | 0.99446 | 0.99461 | 0.99477 | 0.99492 | 0.99506 | 0.99520 |
| -2.6 | 0.99534 | 0.99547 | 0.99560 | 0.99573 | 0.99585 | 0.99598 | 0.99609 | 0.99621 | 0.99632 | 0.99643 |
| -2.7 | 0.99653 | 0.99664 | 0.99674 | 0.99683 | 0.99693 | 0.99702 | 0.99711 | 0.99720 | 0.99728 | 0.99736 |
| -2.8 | 0.99744 | 0.99752 | 0.99760 | 0.99767 | 0.99774 | 0.99781 | 0.99788 | 0.99795 | 0.99801 | 0.99807 |
| -2.9 | 0.99813 | 0.99819 | 0.99825 | 0.99831 | 0.99836 | 0.99841 | 0.99846 | 0.99851 | 0.99856 | 0.99861 |
| -3.0 | 0.99865 | 0.99869 | 0.99874 | 0.99878 | 0.99882 | 0.99886 | 0.99889 | 0.99893 | 0.99896 | 0.99900 |
| -3.1 | 0.99903 | 0.99906 | 0.99910 | 0.99913 | 0.99916 | 0.99918 | 0.99921 | 0.99924 | 0.99926 | 0.99929 |
| -3.2 | 0.99931 | 0.99934 | 0.99936 | 0.99938 | 0.99940 | 0.99942 | 0.99944 | 0.99946 | 0.99948 | 0.99950 |
| -3.3 | 0.99952 | 0.99953 | 0.99955 | 0.99957 | 0.99958 | 0.99960 | 0.99961 | 0.99962 | 0.99964 | 0.99965 |
| -3.4 | 0.99966 | 0.99968 | 0.99969 | 0.99970 | 0.99971 | 0.99972 | 0.99973 | 0.99974 | 0.99975 | 0.99976 |
| -3.5 | 0.99977 | 0.99978 | 0.99978 | 0.99979 | 0.99980 | 0.99981 | 0.99981 | 0.99982 | 0.99983 | 0.99983 |
| -3.6 | 0.99984 | 0.99985 | 0.99985 | 0.99986 | 0.99986 | 0.99987 | 0.99987 | 0.99988 | 0.99988 | 0.99989 |
| -3.7 | 0.99989 | 0.99990 | 0.99990 | 0.99991 | 0.99991 | 0.99992 | 0.99992 | 0.99992 | 0.99992 | 0.99992 |
| -3.8 | 0.99993 | 0.99993 | 0.99993 | 0.99994 | 0.99994 | 0.99994 | 0.99994 | 0.99995 | 0.99995 | 0.99995 |
| -3.9 | 0.99995 | 0.99995 | 0.99996 | 0.99996 | 0.99996 | 0.99996 | 0.99996 | 0.99996 | 0.99997 | 0.99997 |
| -4.0 | 0.99997 | 0.99997 | 0.99997 | 0.99997 | 0.99997 | 0.99997 | 0.99998 | 0.99998 | 0.99998 | 0.99998 |





| z    | 0       | 0.01    | 0.02    | 0.03    | 0.04    | 0.05    | 0.06    | 0.07    | 0.08    | 0.09    |
|------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| +0   | 0.50000 | 0.50399 | 0.50798 | 0.51197 | 0.51595 | 0.51994 | 0.52392 | 0.52790 | 0.53188 | 0.53586 |
| +0.1 | 0.53983 | 0.54380 | 0.54776 | 0.55172 | 0.55567 | 0.55964 | 0.56360 | 0.56749 | 0.57142 | 0.57535 |
| +0.2 | 0.57926 | 0.58317 | 0.58706 | 0.59095 | 0.59483 | 0.59871 | 0.60257 | 0.60642 | 0.61026 | 0.61409 |
| +0.3 | 0.61791 | 0.62172 | 0.62552 | 0.62930 | 0.63307 | 0.63683 | 0.64058 | 0.64431 | 0.64803 | 0.65173 |
| +0.4 | 0.65542 | 0.65910 | 0.66276 | 0.66640 | 0.67003 | 0.67364 | 0.67724 | 0.68082 | 0.68439 | 0.68793 |
| +0.5 | 0.69146 | 0.69497 | 0.69847 | 0.70194 | 0.70540 | 0.70884 | 0.71226 | 0.71566 | 0.71904 | 0.72240 |
| +0.6 | 0.72575 | 0.72907 | 0.73237 | 0.73565 | 0.73891 | 0.74215 | 0.74537 | 0.74857 | 0.75175 | 0.75490 |
| +0.7 | 0.75804 | 0.76115 | 0.76424 | 0.76730 | 0.77035 | 0.77337 | 0.77637 | 0.77935 | 0.78230 | 0.78524 |
| +0.8 | 0.78814 | 0.79103 | 0.79389 | 0.79673 | 0.79955 | 0.80234 | 0.80511 | 0.80785 | 0.81057 | 0.81327 |
| +0.9 | 0.81594 | 0.81859 | 0.82121 | 0.82381 | 0.82639 | 0.82894 | 0.83147 | 0.83398 | 0.83646 | 0.83891 |
| +1.0 | 0.84134 | 0.84375 | 0.84614 | 0.84849 | 0.85083 | 0.85314 | 0.85543 | 0.85769 | 0.85993 | 0.86214 |
| +1.1 | 0.86433 | 0.86650 | 0.86864 | 0.87076 | 0.87284 | 0.87491 | 0.87698 | 0.87900 | 0.88100 | 0.88298 |
| +1.2 | 0.88493 | 0.88686 | 0.88877 | 0.89065 | 0.89251 | 0.89435 | 0.89617 | 0.89799 | 0.89973 | 0.90147 |
| +1.3 | 0.90320 | 0.90490 | 0.90658 | 0.90824 | 0.90988 | 0.91149 | 0.91308 | 0.91466 | 0.91621 | 0.91774 |
| +1.4 | 0.91924 | 0.92073 | 0.92220 | 0.92364 | 0.92507 | 0.92647 | 0.92785 | 0.92922 | 0.93056 | 0.93189 |
| +1.5 | 0.93319 | 0.93448 | 0.93574 | 0.93699 | 0.93822 | 0.93943 | 0.94062 | 0.94179 | 0.94295 | 0.94408 |
| +1.6 | 0.94520 | 0.94630 | 0.94738 | 0.94845 | 0.94950 | 0.95053 | 0.95154 | 0.95254 | 0.95352 | 0.95449 |
| +1.7 | 0.95543 | 0.95637 | 0.95728 | 0.95818 | 0.95907 | 0.95994 | 0.96080 | 0.96164 | 0.96246 | 0.96327 |
| +1.8 | 0.96407 | 0.96485 | 0.96562 | 0.96638 | 0.96712 | 0.96784 | 0.96856 | 0.96926 | 0.96995 | 0.97062 |
| +1.9 | 0.97128 | 0.97193 | 0.97257 | 0.97320 | 0.97381 | 0.97441 | 0.97500 | 0.97558 | 0.97615 | 0.97670 |
| +2.0 | 0.97725 | 0.97778 | 0.97831 | 0.97882 | 0.97932 | 0.97982 | 0.98030 | 0.98077 | 0.98124 | 0.98169 |
| +2.1 | 0.98214 | 0.98257 | 0.98300 | 0.98341 | 0.98382 | 0.98422 | 0.98461 | 0.98500 | 0.98537 | 0.98574 |
| +2.2 | 0.98610 | 0.98645 | 0.98679 | 0.98713 | 0.98745 | 0.98778 | 0.98809 | 0.98840 | 0.98870 | 0.98899 |
| +2.3 | 0.98928 | 0.98956 | 0.98983 | 0.99010 | 0.99036 | 0.99061 | 0.99086 | 0.99111 | 0.99134 | 0.99158 |
| +2.4 | 0.99180 | 0.99202 | 0.99224 | 0.99245 | 0.99266 | 0.99286 | 0.99305 | 0.99324 | 0.99343 | 0.99361 |
| +2.5 | 0.99379 | 0.99396 | 0.99413 | 0.99430 | 0.99446 | 0.99461 | 0.99477 | 0.99492 | 0.99506 | 0.99520 |
| +2.6 | 0.99534 | 0.99547 | 0.99560 | 0.99573 | 0.99585 | 0.99598 | 0.99609 | 0.99621 | 0.99632 | 0.99643 |
| +2.7 | 0.99653 | 0.99664 | 0.99674 | 0.99683 | 0.99693 | 0.99702 | 0.99711 | 0.99720 | 0.99728 | 0.99736 |
| +2.8 | 0.99744 | 0.99752 | 0.99760 | 0.99767 | 0.99774 | 0.99781 | 0.99788 | 0.99795 | 0.99801 | 0.99807 |
| +2.9 | 0.99813 | 0.99819 | 0.99825 | 0.99831 | 0.99836 | 0.99841 | 0.99846 | 0.99851 | 0.99856 | 0.99861 |
| +3.0 | 0.99865 | 0.99869 | 0.99874 | 0.99878 | 0.99882 | 0.99886 | 0.99889 | 0.99893 | 0.99896 | 0.99900 |
| +3.1 | 0.99903 | 0.99906 | 0.99910 | 0.99913 | 0.99916 | 0.99918 | 0.99921 | 0.99924 | 0.99926 | 0.99929 |
| +3.2 | 0.99931 | 0.99934 | 0.99936 | 0.99938 | 0.99940 | 0.99942 | 0.99944 | 0.99946 | 0.99948 | 0.99950 |
| +3.3 | 0.99952 | 0.99953 | 0.99955 | 0.99957 | 0.99958 | 0.99960 | 0.99961 | 0.99962 | 0.99964 | 0.99965 |
| +3.4 | 0.99966 | 0.99968 | 0.99969 | 0.99970 | 0.99971 | 0.99972 | 0.99973 | 0.99974 | 0.99975 | 0.99976 |
| +3.5 | 0.99977 | 0.99978 | 0.99978 | 0.99979 | 0.99980 | 0.99981 | 0.99981 | 0.99982 | 0.99983 | 0.99983 |
| +3.6 | 0.99984 | 0.99985 | 0.99985 | 0.99986 | 0.99986 | 0.99987 | 0.99987 | 0.99988 | 0.99988 | 0.99989 |
| +3.7 | 0.99989 | 0.99990 | 0.99990 | 0.99991 | 0.99991 | 0.99991 | 0.99992 | 0.99992 | 0.99992 | 0.99992 |
| +3.8 | 0.99993 | 0.99993 | 0.99993 | 0.99994 | 0.99994 | 0.99994 | 0.99994 | 0.99995 | 0.99995 | 0.99995 |
| +3.9 | 0.99995 | 0.99995 | 0.99996 | 0.99996 | 0.99996 | 0.99996 | 0.99996 | 0.99997 | 0.99997 | 0.99997 |
| +4.0 | 0.99997 | 0.99997 | 0.99997 | 0.99997 | 0.99997 | 0.99997 | 0.99998 | 0.99998 | 0.99998 | 0.99998 |



$+z$

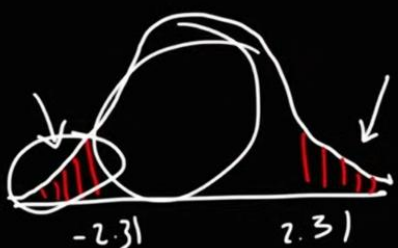
$z \rightarrow A_L$

$2.31 \rightarrow 0.98956$



1. The average weight of all residents in town XYZ is 168 lbs. A nutritionist believes the true mean to be different. She measured the weight of 36 individuals and found the mean to be 169.5 lbs with a standard deviation of 3.9. (a) State the null and alternative hypotheses. (b) At a 95% confidence level, is there enough evidence to discard the null hypothesis? (Use the p-value method)

$H_0: \mu = 168$        $n = 36$        $\bar{x} = 169.5$        $s = 3.9$   
 $H_a: \mu \neq 168$        $C = 0.95$        $\alpha = 1 - C = 0.05$



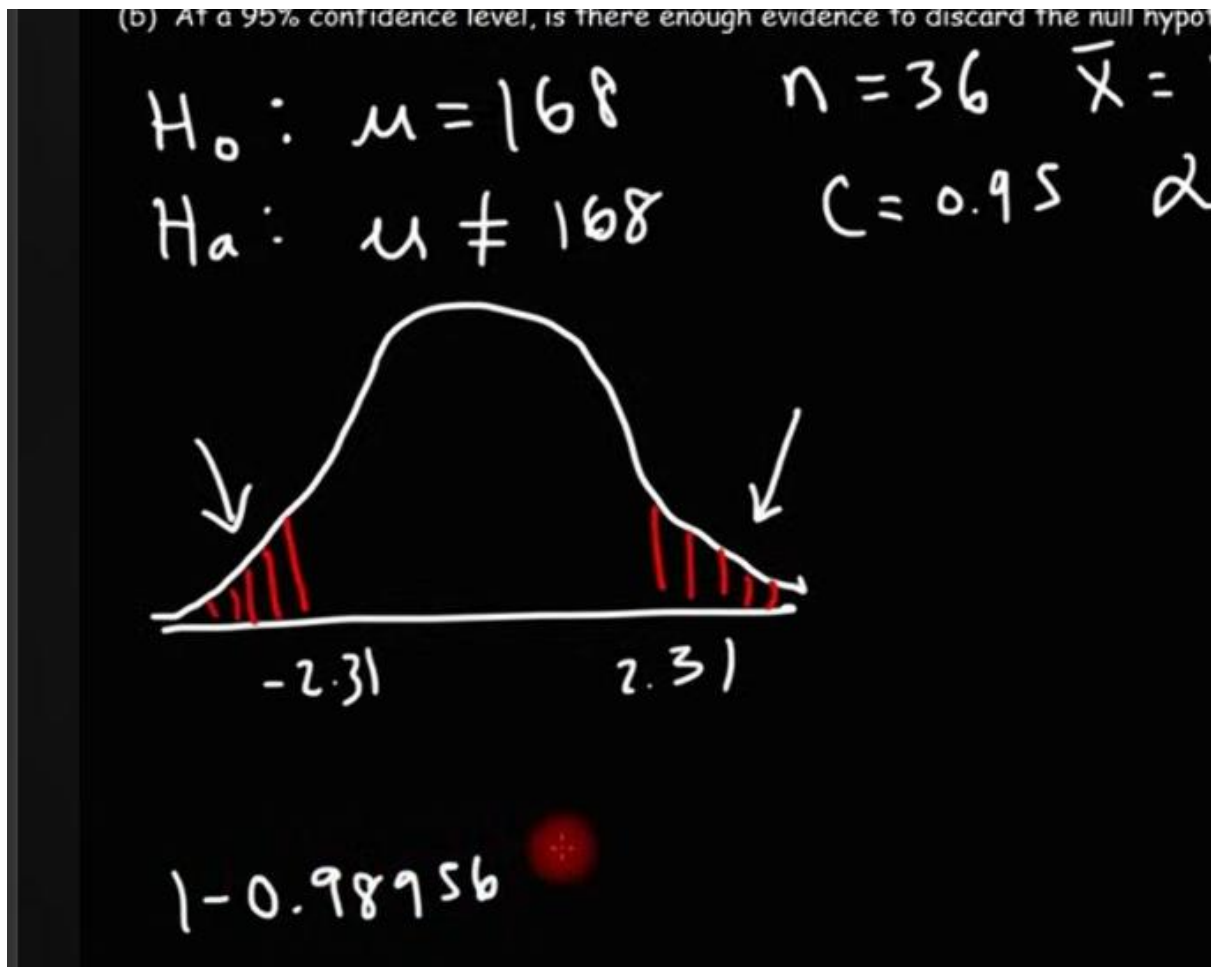
$$z_c = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{169.5 - 168}{3.9/\sqrt{36}}$$

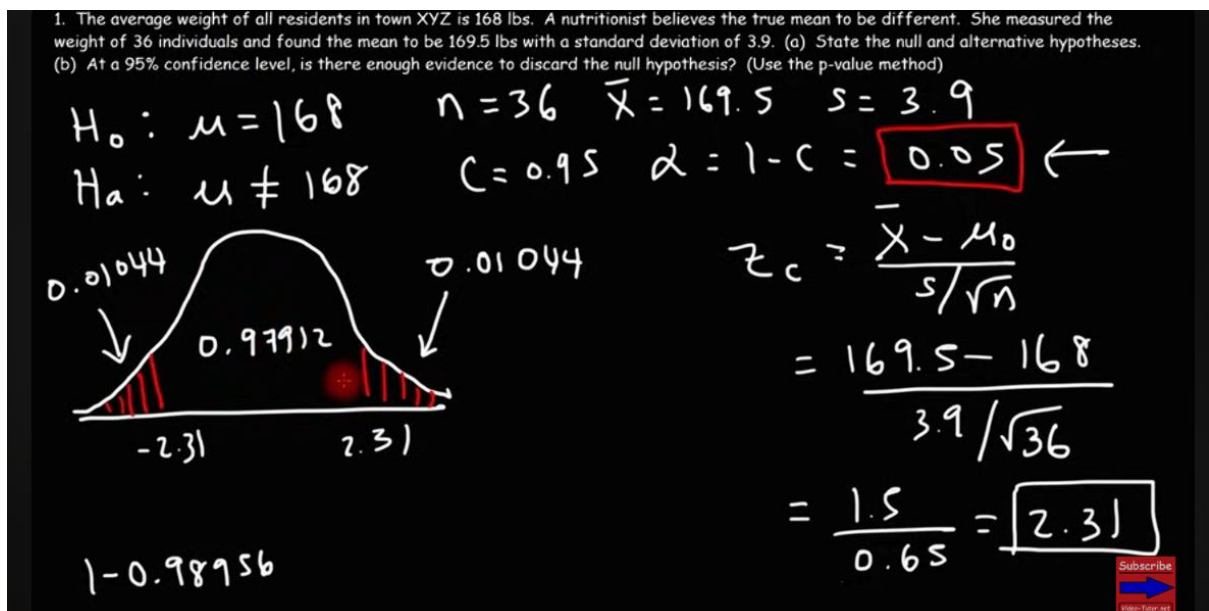
$$= \frac{1.5}{0.65} = 2.31$$



That area value is sum of these two spaces



The area on right will be



Now the p value is sum of shaded region left and right

1. The average weight of all residents in town XYZ is 168 lbs. A nutritionist believes the true mean to be different. She measured the weight of 36 individuals and found the mean to be 169.5 lbs with a standard deviation of 3.9. (a) State the null and alternative hypotheses. (b) At a 95% confidence level, is there enough evidence to discard the null hypothesis? (Use the p-value method)

$H_0: \mu = 168$      $n = 36$      $\bar{x} = 169.5$      $s = 3.9$   
 $H_a: \mu \neq 168$      $C = 0.95$      $\alpha = 1 - C = 0.05$  ←



$$z_c = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{169.5 - 168}{3.9/\sqrt{36}}$$

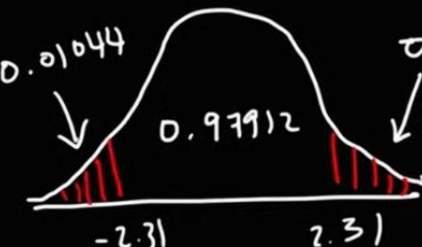
$$= \frac{1.5}{0.65} = 2.31$$

$p\text{-value} = 0.02082$      $\alpha = 0.05$

Now we have p value , compare it to alpha which is 0.05

1. The average weight of all residents in town XYZ is 168 lbs. A nutritionist believes the true mean to be different. She measured the weight of 36 individuals and found the mean to be 169.5 lbs with a standard deviation of 3.9. (a) State the null and alternative hypotheses. (b) At a 95% confidence level, is there enough evidence to discard the null hypothesis? (Use the p-value method)

$H_0: \mu = 168$      $n = 36$      $\bar{x} = 169.5$      $s = 3.9$   
 $H_a: \mu \neq 168$      $C = 0.95$      $\alpha = 1 - C = 0.05$  ←



$$z_c = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{169.5 - 168}{3.9/\sqrt{36}}$$

$$= \frac{1.5}{0.65} = 2.31$$

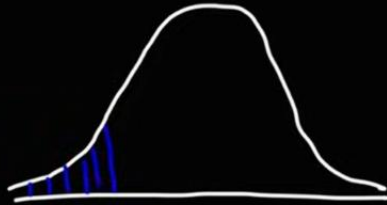
$p\text{-value} = 0.02082$      $\alpha = 0.05$

$p\text{-value} < \alpha \rightarrow \text{reject } H_0$

2. A factory manufactures cars with a warranty of 5 years on the engine and transmission. An engineer believes that the engine or transmission will malfunction in less than 5 years. He tests a sample of 40 cars and find the average time to be 4.8 years with a standard deviation of 0.50. (a) State the null and alternative hypotheses. (b) At a 2% significance level, is there enough evidence to support the idea that the warranty should be revised?

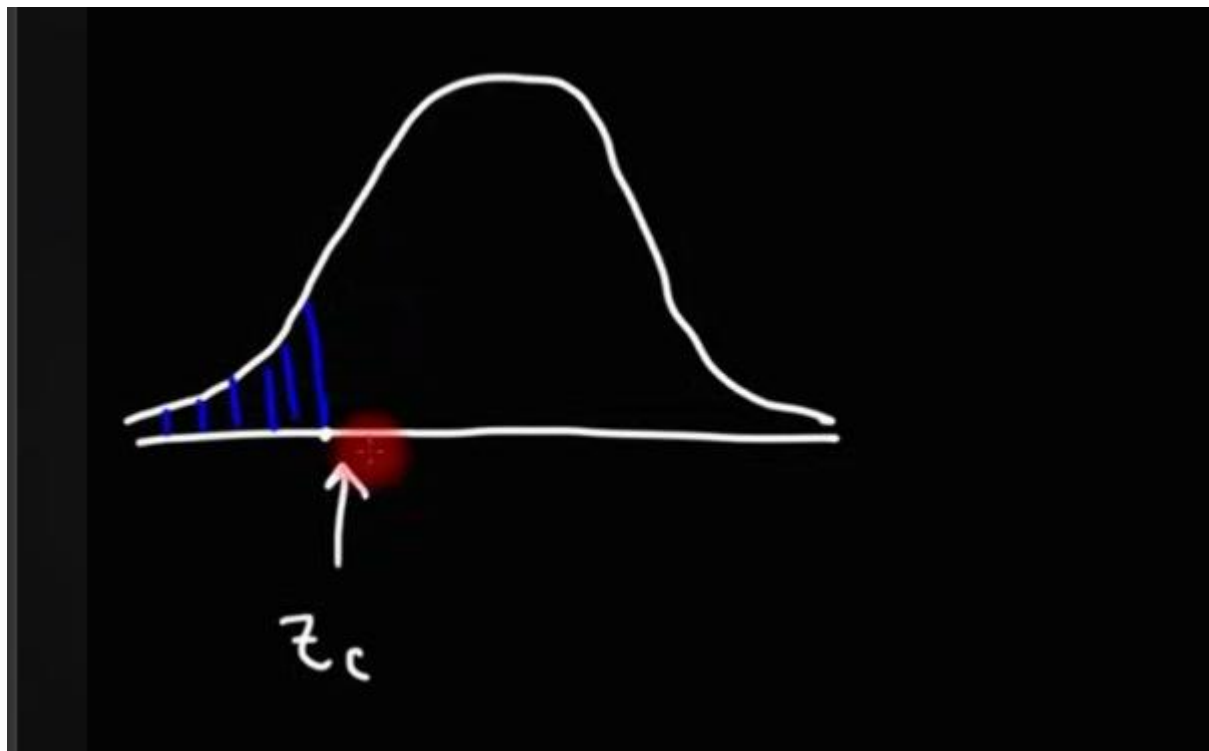
$$H_0: \mu \geq 5 \quad n = 40 \quad \bar{x} = 4.8 \quad s = 0.50 \quad \mu_0 = 5$$

$$H_a: \mu < 5 \quad \alpha = 0.02 \quad C = 0.98$$

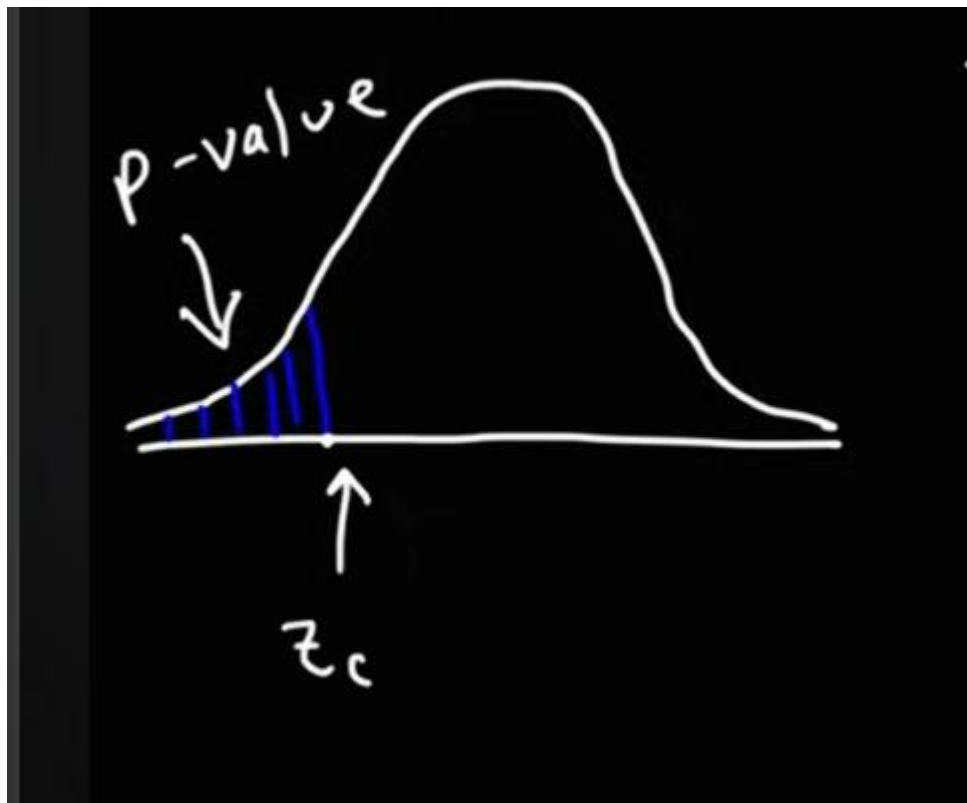


Now we need to calculate p value and compare it to alpha

For that , we must calculate z value , that calculated z value will correspond to



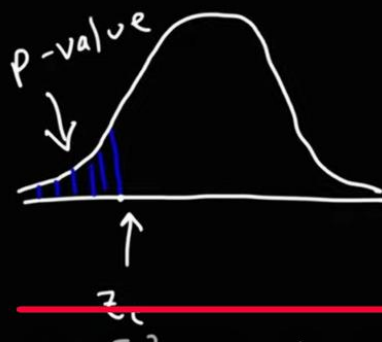
And then using that z value , we can get the area , which will represent the p value



2. A factory manufactures cars with a warranty of 5 years on the engine and transmission. An engineer believes that the engine or transmission will malfunction in less than 5 years. He tests a deviation of 0.50. (a) State the null and alternative hypotheses. (b) find the average time to be 4.8 years with a standard deviation of 0.50. (c) find the average time to be 4.8 years with a standard deviation of 0.50. (d) find the average time to be 4.8 years with a standard deviation of 0.50. (e) find the average time to be 4.8 years with a standard deviation of 0.50. (f) find the average time to be 4.8 years with a standard deviation of 0.50. (g) find the average time to be 4.8 years with a standard deviation of 0.50. (h) find the average time to be 4.8 years with a standard deviation of 0.50. (i) find the average time to be 4.8 years with a standard deviation of 0.50. (j) find the average time to be 4.8 years with a standard deviation of 0.50. (k) find the average time to be 4.8 years with a standard deviation of 0.50. (l) find the average time to be 4.8 years with a standard deviation of 0.50. (m) find the average time to be 4.8 years with a standard deviation of 0.50. (n) find the average time to be 4.8 years with a standard deviation of 0.50. (o) find the average time to be 4.8 years with a standard deviation of 0.50. (p) find the average time to be 4.8 years with a standard deviation of 0.50. (q) find the average time to be 4.8 years with a standard deviation of 0.50. (r) find the average time to be 4.8 years with a standard deviation of 0.50. (s) find the average time to be 4.8 years with a standard deviation of 0.50. (t) find the average time to be 4.8 years with a standard deviation of 0.50. (u) find the average time to be 4.8 years with a standard deviation of 0.50. (v) find the average time to be 4.8 years with a standard deviation of 0.50. (w) find the average time to be 4.8 years with a standard deviation of 0.50. (x) find the average time to be 4.8 years with a standard deviation of 0.50. (y) find the average time to be 4.8 years with a standard deviation of 0.50. (z) find the average time to be 4.8 years with a standard deviation of 0.50.

$$H_0: \mu \geq 5 \quad n = 40 \quad \bar{x} = 4.8 \quad s = 0.50 \quad \mu_0 = 5$$

$$H_a: \mu < 5 \quad \alpha = 0.02 \quad C = 0.98$$

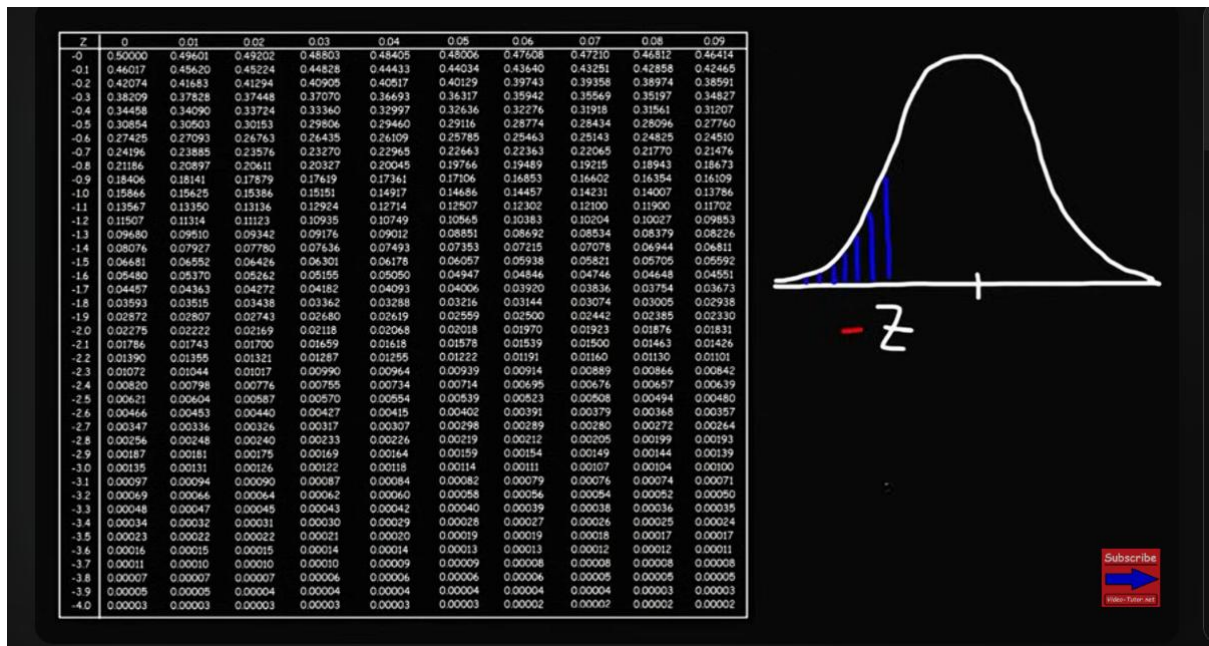


$$z_c = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{4.8 - 5}{0.50 / \sqrt{40}}$$

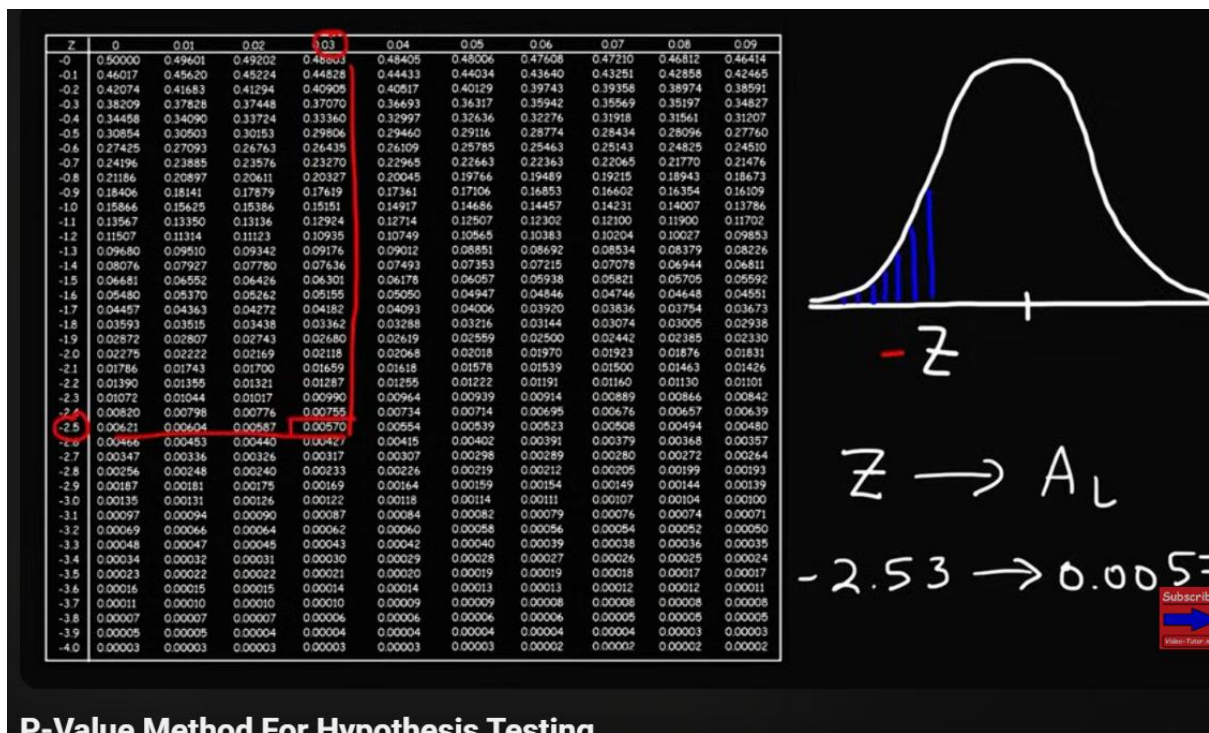
$$= \frac{-0.2}{0.07906} \approx -2.53$$







Negative z score table

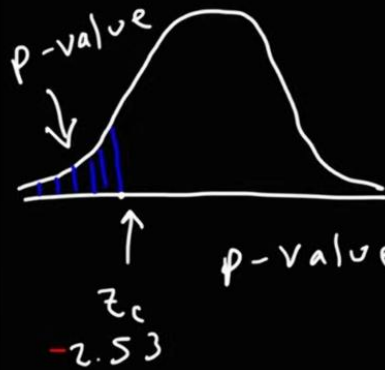


P-Value Method For Hypothesis Testing

2. A factory manufactures cars with a warranty of 5 years on the engine and transmission. An engineer believes that the engine or transmission will malfunction in less than 5 years. He tests a sample of 40 cars and find the average time to be 4.8 years with a standard deviation of 0.50. (a) State the null and alternative hypotheses. (b) At a 2% significance level, is there enough evidence to support the idea that the warranty should be revised?

$$H_0: \mu \geq 5 \quad n = 40 \quad \bar{x} = 4.8 \quad s = 0.50 \quad \mu_0 = 5$$

$$H_a: \mu < 5 \quad \alpha = 0.02 \quad C = 0.98$$



$$z_c = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{4.8 - 5}{0.50 / \sqrt{40}}$$

$$= \frac{-0.2}{0.07906} \approx -2.53$$

$$p\text{-value} = 0.0057 < 0.02$$

$$p\text{-value} < \alpha \rightarrow \text{reject } H_0$$

1. A tech company believes that the percentage of residents in town XYZ that own a cell phone is 70%. A marketing manager believes this value to be different. He conducts a survey of 200 individuals and found that 130 responded yes to owning a cell phone. (a) State the null and alternative hypotheses. (b) At a 95% confidence level, is there enough evidence to reject the null hypothesis?

$$H_0: p = 0.70 \quad n = 200 \quad x = 130$$

$$H_a: p \neq 0.70$$

With this we can calculate sample proportion which is x over n

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$$H_0: p = 0.70 \quad n = 200 \quad x = 130 \quad \hat{p} = 0.65$$

$$H_a: p \neq 0.70$$

$$\hat{p} = \frac{x}{n} = \frac{130}{200} =$$

$$< -5$$



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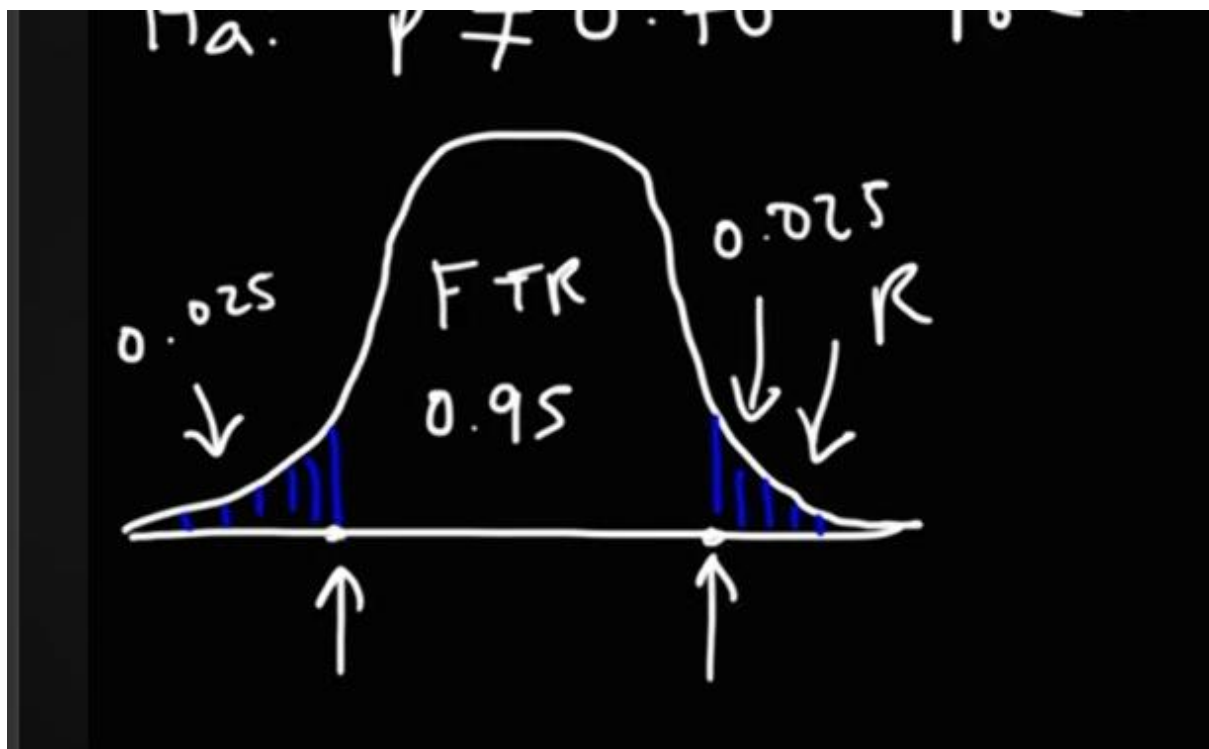
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We have to calculate our z values

We need calculated z value and critical z value and compare



We find this z value which is critical z value using the table

$A_c = 0.975 \rightarrow$

Hypothesis testing diagram showing a normal distribution curve with a shaded area under the curve labeled  $p\text{-value} < \alpha$ .

Hypothesis testing diagram showing a normal distribution curve with a shaded area under the curve labeled  $p\text{-value} < \alpha$ .

| z    | 0       | 0.01    | 0.02    | 0.03    | 0.04    | 0.05    | 0.06    | 0.07    | 0.08    | 0.09    |
|------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| -0.1 | 0.50000 | 0.50399 | 0.50798 | 0.51197 | 0.51596 | 0.51994 | 0.52391 | 0.52790 | 0.53188 | 0.53586 |
| -0.2 | 0.53983 | 0.54380 | 0.54776 | 0.55172 | 0.55567 | 0.55962 | 0.56356 | 0.56749 | 0.57142 | 0.57535 |
| -0.3 | 0.57926 | 0.58317 | 0.58706 | 0.59095 | 0.59483 | 0.59871 | 0.60257 | 0.60642 | 0.61026 | 0.61409 |
| -0.4 | 0.61791 | 0.62172 | 0.62552 | 0.62930 | 0.63307 | 0.63683 | 0.64058 | 0.64431 | 0.64803 | 0.65173 |
| -0.5 | 0.65542 | 0.65910 | 0.66276 | 0.66640 | 0.67003 | 0.67364 | 0.67724 | 0.68082 | 0.68439 | 0.68793 |
| -0.6 | 0.69146 | 0.69497 | 0.69847 | 0.70194 | 0.70540 | 0.70884 | 0.71226 | 0.71566 | 0.71904 | 0.72240 |
| -0.7 | 0.72575 | 0.72907 | 0.73237 | 0.73565 | 0.73891 | 0.74215 | 0.74537 | 0.74857 | 0.75175 | 0.75490 |
| -0.8 | 0.75804 | 0.76115 | 0.76424 | 0.76730 | 0.77035 | 0.77337 | 0.77637 | 0.77935 | 0.78230 | 0.78524 |
| -0.9 | 0.78814 | 0.79103 | 0.79389 | 0.79673 | 0.79955 | 0.80234 | 0.80511 | 0.80785 | 0.81057 | 0.81327 |
| -1.0 | 0.81594 | 0.81859 | 0.82121 | 0.82381 | 0.82639 | 0.82894 | 0.83147 | 0.83398 | 0.83646 | 0.83891 |
| -1.1 | 0.84134 | 0.84375 | 0.84614 | 0.84849 | 0.85083 | 0.85314 | 0.85543 | 0.85769 | 0.85993 | 0.86214 |
| -1.2 | 0.86433 | 0.86650 | 0.86864 | 0.87076 | 0.87286 | 0.87493 | 0.87698 | 0.87900 | 0.88100 | 0.88298 |
| -1.3 | 0.88493 | 0.88686 | 0.88877 | 0.89065 | 0.89251 | 0.89435 | 0.89617 | 0.89796 | 0.89973 | 0.90147 |
| -1.4 | 0.90320 | 0.90490 | 0.90658 | 0.90824 | 0.90988 | 0.91149 | 0.91308 | 0.91466 | 0.91621 | 0.91774 |
| -1.5 | 0.91924 | 0.92073 | 0.92220 | 0.92364 | 0.92507 | 0.92647 | 0.92785 | 0.92922 | 0.93056 | 0.93189 |
| -1.6 | 0.93319 | 0.93448 | 0.93574 | 0.93699 | 0.93822 | 0.93943 | 0.94062 | 0.94179 | 0.94295 | 0.94408 |
| -1.7 | 0.94520 | 0.94638 | 0.94753 | 0.94865 | 0.94975 | 0.95083 | 0.95189 | 0.95294 | 0.95398 | 0.95499 |
| -1.8 | 0.95603 | 0.95705 | 0.95805 | 0.95903 | 0.95999 | 0.96094 | 0.96187 | 0.96279 | 0.96370 | 0.96459 |
| -1.9 | 0.96547 | 0.96633 | 0.96718 | 0.96802 | 0.96885 | 0.96966 | 0.97046 | 0.97125 | 0.97202 | 0.97278 |
| -2.0 | 0.97353 | 0.97428 | 0.97501 | 0.97573 | 0.97644 | 0.97714 | 0.97783 | 0.97850 | 0.97916 | 0.97981 |
| -2.1 | 0.98045 | 0.98109 | 0.98171 | 0.98232 | 0.98292 | 0.98351 | 0.98409 | 0.98466 | 0.98522 | 0.98577 |
| -2.2 | 0.98631 | 0.98688 | 0.98743 | 0.98797 | 0.98850 | 0.98902 | 0.98954 | 0.99005 | 0.99055 | 0.99104 |
| -2.3 | 0.99152 | 0.99200 | 0.99246 | 0.99291 | 0.99335 | 0.99378 | 0.99420 | 0.99461 | 0.99501 | 0.99540 |
| -2.4 | 0.99578 | 0.99615 | 0.99651 | 0.99686 | 0.99720 | 0.99753 | 0.99785 | 0.99816 | 0.99846 | 0.99875 |
| -2.5 | 0.99904 | 0.99933 | 0.99961 | 0.99988 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| -2.6 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| -2.7 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| -2.8 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| -2.9 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| -3.0 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| -3.1 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
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| -3.3 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| -3.4 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| -3.5 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
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| -3.7 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| -3.8 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| -3.9 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| -4.0 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |

$A_c \rightarrow z$   
 $0.975 \rightarrow 1.96$

Normal distribution curve diagram showing the area under the curve between  $-1.96$  and  $1.96$  is  $0.95$ , and the area in the tails is  $0.025$  each.

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$H_0: p = 0.70$   $n = 200$   $X = 130$   $\hat{p} = 0.65$

$H_a: p \neq 0.70$   $p_0 = 0.70$   $q_0 = 0.30$

$A_c = 0.975 \rightarrow$

Normal distribution curve diagram showing the area under the curve between  $-1.96$  and  $1.96$  is  $0.95$ , and the area in the tails is  $0.025$  each.

$Z = \frac{\text{sample proportion} - \text{proportion associated with null hypothesis}}{\sqrt{\frac{p_0 q_0}{n}}}$

$$n = 200 \quad X = 130 \quad \hat{p} = 0.65$$

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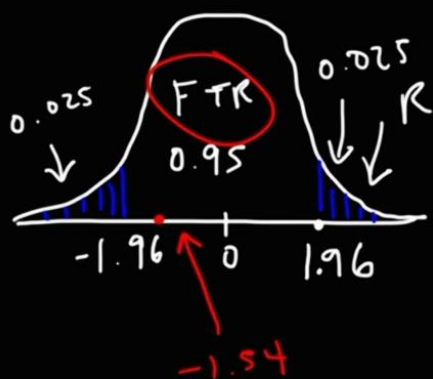
$$A_c = 0.975 \rightarrow$$

$$Z_c = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = 0.$$

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$$A_c = 0.975 \rightarrow$$

$$Z_c = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{0.65 - 0.70}{\sqrt{\frac{0.7(0.3)}{200}}}$$

$$Z_c = \frac{-0.05}{0.0324} \approx -1.54$$



At 95 percent confidence interval there is not enough evidence to reject the null hypothesis

2. A car company believes that the percentage of residents in city ABC that own a vehicle is 60% or less. A sales manager disagrees with this. He conducts a hypothesis test surveying 250 residents and found that 170 responded yes to owning a vehicle. (a) State the null and alternative hypotheses. (b) At a 10% significance level, is there enough evidence to support the idea that the vehicle ownership in city ABC is 60% or less?

$$H_0: p \leq 0.60 \quad n = 250 \quad X = 170 \quad \hat{p} = 0.68$$

$$H_a: p > 0.60$$

$$\hat{p} = \frac{170}{250}$$

2. A car company believes that the percentage of residents in city ABC that own a vehicle is 60% or less. A sales manager disagrees with this. He conducts a hypothesis test surveying 250 residents and found that 170 responded yes to owning a vehicle. (a) State the null and alternative hypotheses. (b) At a 10% significance level, is there enough evidence to support the idea that the vehicle ownership in city ABC is 60% or less?

$$H_0: p \leq 0.60$$

$$H_a: p > 0.60$$

$$n = 250$$

$$X = 170$$

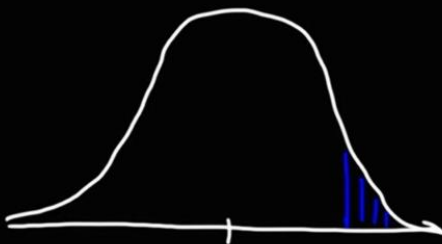
$$\hat{p} = 0.68$$

$$p_0 = 0.60$$

$$q_0 = 0.40$$

$$\alpha = 0.10$$

$$C = 0.90$$



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$$H_0: p \leq 0.60$$

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$$n = 250$$

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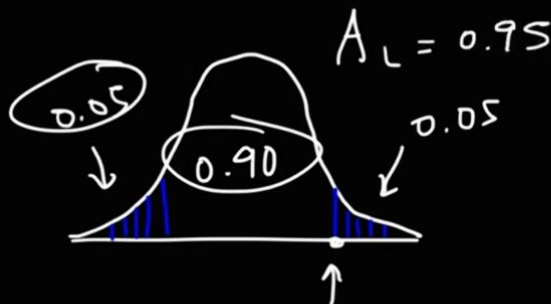
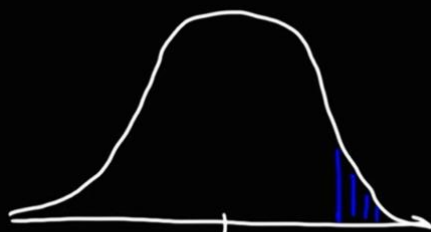
$$\hat{p} = 0.68$$

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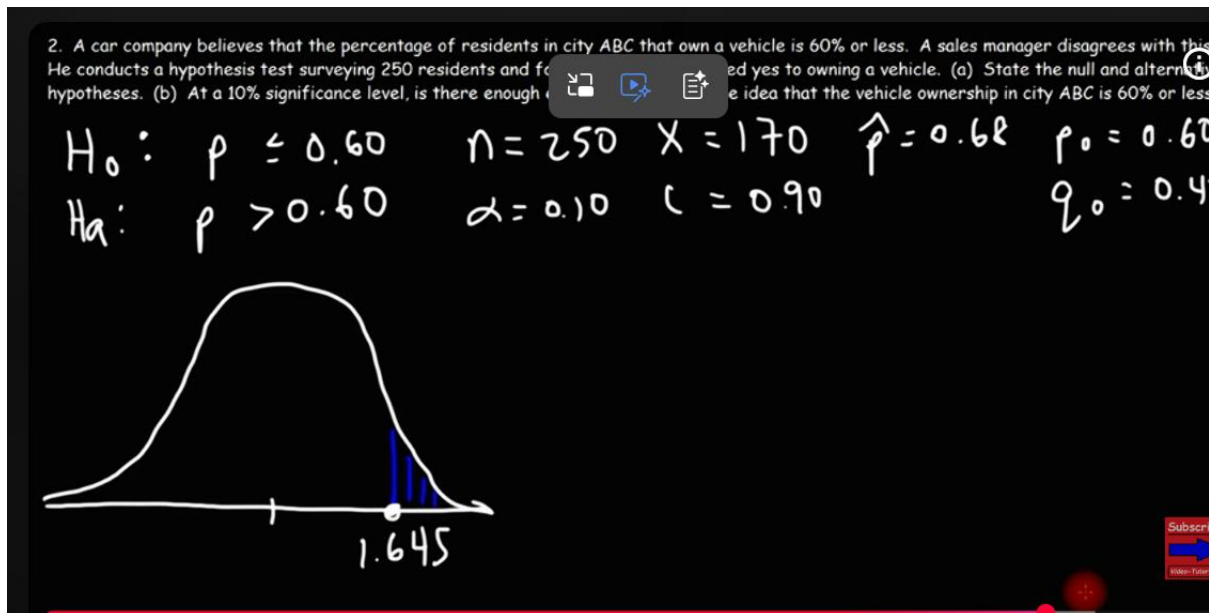
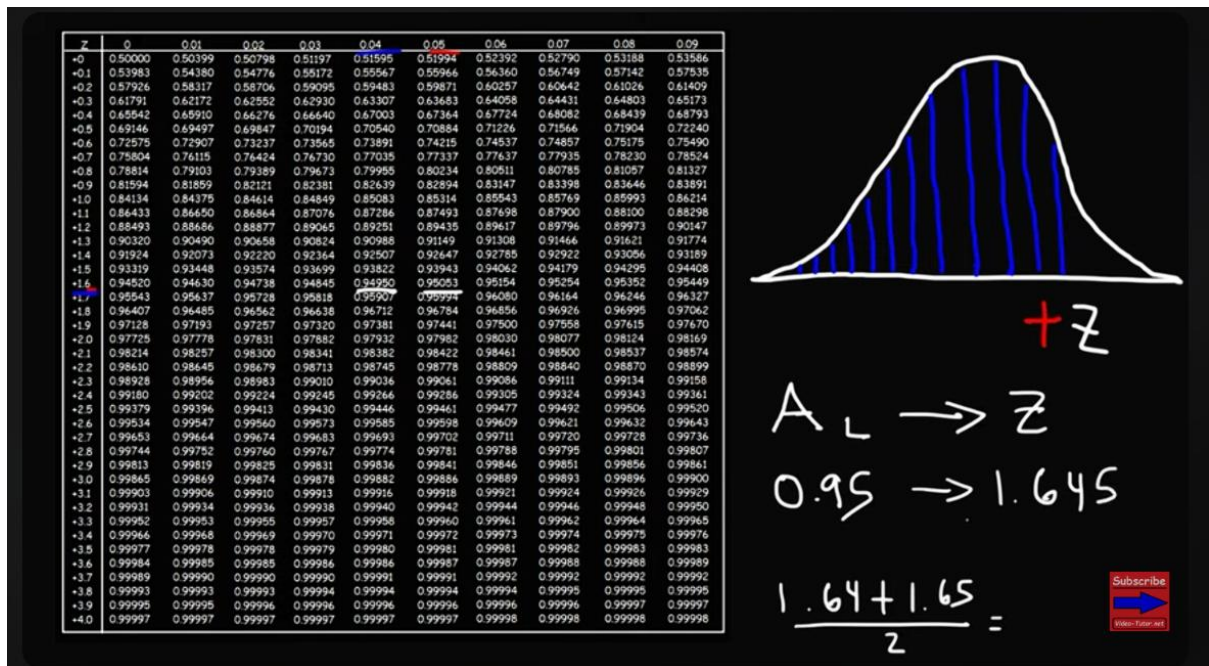
$$q_0 = 0.40$$

$$\alpha = 0.10$$

$$C = 0.90$$







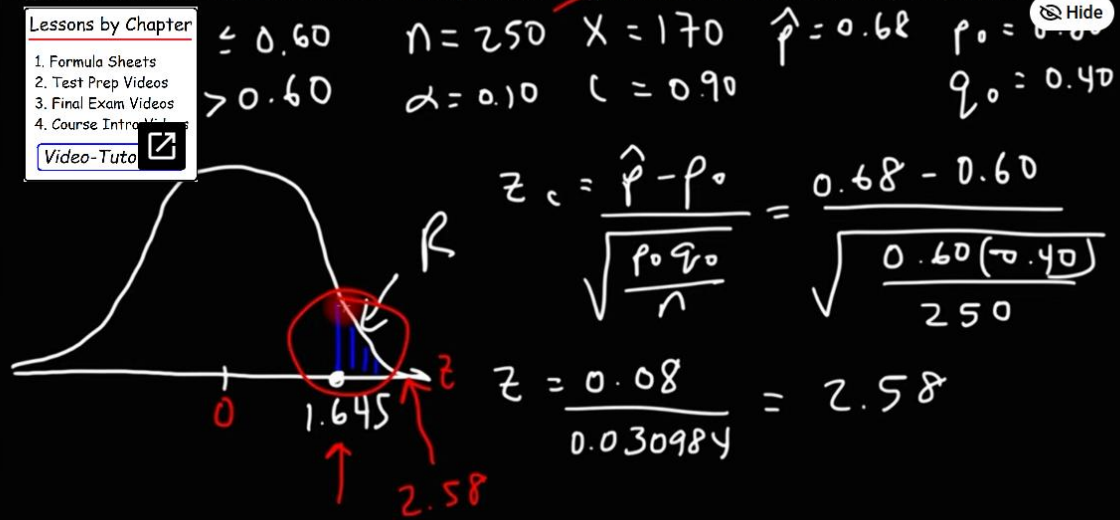
We used that second image to calculate z critical

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Lessons by Chapter

1. Formula Sheets
2. Test Prep Videos
3. Final Exam Videos
4. Course Introduction

Video-Tutorial



## DIFFERENCE OF TWO MEANS

1. A test was conducted on two different classes to see if there was any significant difference between the performance of two teachers. The final exam scores of 15 students were sampled in the 1st class yielding a mean score of 82 with a standard deviation of 2.4. The mean final exam score of the 2nd class was 84 with a standard deviation of 1.7 using a sample of 12 students. Determine if there is any major difference at a 5% significance level.

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$1^{st}: n_1 = 15 \quad \bar{X}_1 = 82 \quad s_1 = 2.4 \quad \alpha = 0.05$   
 $2^{nd}: n_2 = 12 \quad \bar{X}_2 = 84 \quad s_2 = 1.7$

$H_0:$

$H_a: \mu_1 \neq \mu_2$



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$$H_0: \mu_1 = \mu_2$$

$$H_a: \mu_1 \neq \mu_2$$

final exam scores of 15 students were sampled in the 1st class yielding a mean score of 82 with a standard deviation of 2.4. The mean final exam score of the 2nd class was 84 with a standard deviation of 1.7 using a sample of 12 students. Determine if there is any major difference at a 5% significance level.

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$$H_0: \mu_1 = \mu_2 \quad \mu_1 - \mu_2 = 0$$

$$H_a: \mu_1 \neq \mu_2 \quad \mu_1 - \mu_2 \neq 0$$

Now we decide , whether we use t distribution or normal distribution

Sample size is less than 30 so we use t distribution



1. A test was conducted on two different classes to see if there was any significant difference between the performance of two teachers. The final exam scores of 15 students were sampled in the 1st class yielding a mean score of 82 with a standard deviation of 2.4. The mean final exam score of the 2nd class was 84 with a standard deviation of 1.7 using a sample of 12 students. Determine if there is any major difference at a 5% significance level.

$$\begin{aligned} 1^{st}: n_1 &= 15 & \bar{X}_1 &= 82 & S_1 &= 2.4 & \alpha &= 0.05 & \alpha/2 &= 0.025 \\ 2^{nd}: n_2 &= 12 & \bar{X}_2 &= 84 & S_2 &= 1.7 & \mu_1 - \mu_2 &= 0 \leftarrow H_0 \end{aligned}$$

$$df = \frac{\left[ \frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} \right]^2}{\frac{(S_1^2/n_1)^2}{n_1-1} + \frac{(S_2^2/n_2)^2}{n_2-1}} = \frac{\left[ \frac{2.4^2}{15} + \frac{1.7^2}{12} \right]^2}{\frac{(2.4^2/15)^2}{14} + \frac{(1.7^2/12)^2}{11}} = \frac{0.390417}{0.0105326 + 0.00577779}$$

$$df = \frac{0.390417}{0.0158054} = 24.7 \approx 24$$

Now determine t critical value so that we can compare it with calculated t value

T is found from table using df and alpha over 2 value . we already divided alpha to two

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$$t_{25, 0.025}$$

7:57 / 18:35 Solution >

| df | 0.10   | 0.05   | 0.025   | 0.01    | 0.005   | 0.0005   |
|----|--------|--------|---------|---------|---------|----------|
| 1  | 3.0777 | 6.3138 | 12.7062 | 31.8208 | 63.6577 | 636.6192 |
| 2  | 1.8856 | 2.9200 | 4.3027  | 6.9646  | 9.9248  | 31.5991  |
| 3  | 1.6377 | 2.3534 | 3.1824  | 4.5407  | 5.8409  | 12.9240  |
| 4  | 1.5332 | 2.1318 | 2.7764  | 3.7469  | 4.6001  | 8.6103   |
| 5  | 1.4759 | 2.0150 | 2.5706  | 3.3649  | 4.0321  | 6.8688   |
| 6  | 1.4398 | 1.9432 | 2.4469  | 3.1427  | 3.7074  | 5.9588   |
| 7  | 1.4149 | 1.8946 | 2.3646  | 2.9980  | 3.4995  | 5.4079   |
| 8  | 1.3968 | 1.8595 | 2.3060  | 2.8965  | 3.3554  | 5.0413   |
| 9  | 1.3830 | 1.8331 | 2.2622  | 2.8214  | 3.2498  | 4.7809   |
| 10 | 1.3722 | 1.8125 | 2.2281  | 2.7638  | 3.1693  | 4.5869   |
| 11 | 1.3634 | 1.7959 | 2.2010  | 2.7181  | 3.1058  | 4.4370   |
| 12 | 1.3562 | 1.7823 | 2.1788  | 2.6810  | 3.0545  | 4.3178   |
| 13 | 1.3502 | 1.7709 | 2.1604  | 2.6503  | 3.0123  | 4.2208   |
| 14 | 1.3450 | 1.7613 | 2.1448  | 2.6245  | 2.9768  | 4.1405   |
| 15 | 1.3406 | 1.7531 | 2.1314  | 2.6025  | 2.9467  | 4.0728   |
| 16 | 1.3368 | 1.7459 | 2.1199  | 2.5835  | 2.9208  | 4.0150   |
| 17 | 1.3334 | 1.7396 | 2.1098  | 2.5669  | 2.8982  | 3.9651   |
| 18 | 1.3304 | 1.7341 | 2.1009  | 2.5524  | 2.8784  | 3.9216   |
| 19 | 1.3277 | 1.7291 | 2.0930  | 2.5395  | 2.8609  | 3.8834   |
| 20 | 1.3253 | 1.7247 | 2.0860  | 2.5280  | 2.8453  | 3.8495   |
| 21 | 1.3232 | 1.7207 | 2.0796  | 2.5176  | 2.8314  | 3.8193   |
| 22 | 1.3212 | 1.7171 | 2.0739  | 2.5083  | 2.8188  | 3.7921   |
| 23 | 1.3195 | 1.7139 | 2.0687  | 2.4999  | 2.8073  | 3.7676   |
| 24 | 1.3178 | 1.7109 | 2.0639  | 2.4922  | 2.7969  | 3.7454   |
| 25 | 1.3163 | 1.7081 | 2.0595  | 2.4851  | 2.7874  | 3.7251   |
| 26 | 1.3150 | 1.7056 | 2.0555  | 2.4786  | 2.7787  | 3.7066   |
| 27 | 1.3137 | 1.7033 | 2.0518  | 2.4727  | 2.7707  | 3.6896   |
| 28 | 1.3125 | 1.7011 | 2.0484  | 2.4671  | 2.7633  | 3.6739   |
| 29 | 1.3114 | 1.6991 | 2.0452  | 2.4620  | 2.7564  | 3.6594   |
| 30 | 1.3104 | 1.6973 | 2.0423  | 2.4573  | 2.7500  | 3.6460   |

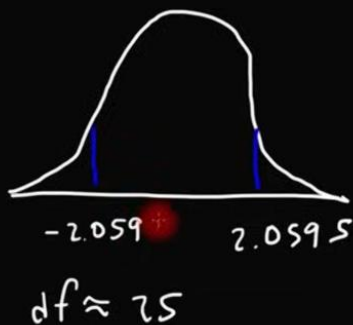


## Hypothesis Testing – Difference of Two Means – Student's Distribution & Normal

1. A test was conducted on two different classes to see if there was any significant difference between the performance of two teachers. The final exam scores of 15 students were sampled in the 1st class yielding a mean score of 82 with a standard deviation of 2.4. The mean final exam score of the 2nd class was 84 with a standard deviation of 1.7 using a sample of 12 students. Determine if there is any major difference at a 5% significance level.

$$\begin{aligned}
 1^{st}: n_1 &= 15 & \bar{X}_1 &= 82 & S_1 &= 2.4 & \alpha &= 0.05 & \alpha/2 &= 0.025 \\
 2^{nd}: n_2 &= 12 & \bar{X}_2 &= 84 & S_2 &= 1.7 & \mu_1 - \mu_2 &= 0 \leftarrow H_0
 \end{aligned}$$

$$t_{25, 0.025} = 2.0595$$



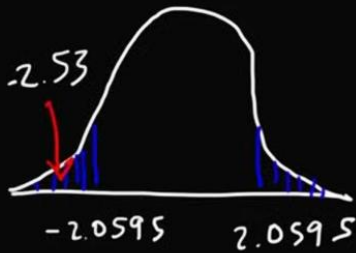
## Hypothesis Testing – Difference of Two Means – Student's Distribution & Normal

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$$1^{st}: n_1 = 15 \quad \bar{X}_1 = 82 \quad S_1 = 2.4 \quad \alpha = 0.05 \quad \alpha/2 = 0.025$$

$$2^{nd}: n_2 = 12 \quad \bar{X}_2 = 84 \quad S_2 = 1.7 \quad \mu_1 - \mu_2 = 0 \leftarrow H_0$$

$$t_{25, 0.025} = 2.0595 \quad t_c = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$



$$df \approx 25$$

$$t_c = \frac{(82 - 84) - (0)}{\sqrt{\frac{2.4^2}{15} + \frac{1.7^2}{12}}} = \frac{-2}{0.79046} = -2.53$$

There is major significance difference at 5 % significance level

2. A business owner is in the process of deciding whether or not to invest in a new factory that refines oil in order to meet the high demand for that commodity. A test showed that the old factory refines oil at a mean rate of 3.1L per second at a standard deviation of 1.0 using a sample size of 40. The new factory was measured to refine oil at a mean rate of 3.8L per second at a standard deviation of 1.5 using a sample size of 36. Determine if there is any major difference at a 10% significance level.

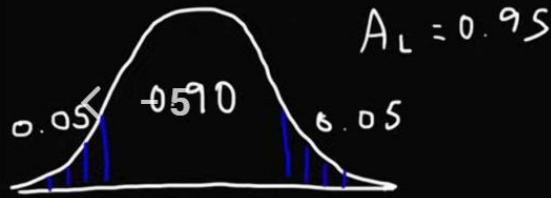
$$\text{old: } n_1 = 40 \quad \bar{X}_1 = 3.1 \quad S_1 = 1$$

$$\text{new: } n_2 = 36 \quad \bar{X}_2 = 3.8 \quad S_2 = 1.5 \quad H_a: \mu_1 \neq \mu_2$$

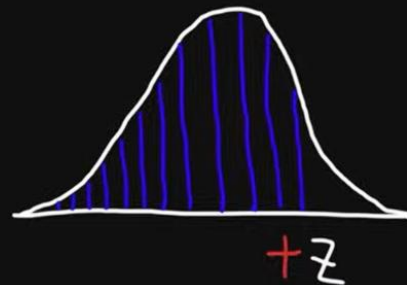
If those two were not different you won't be doing the test, that's why its alternative  $H_a$

2. A business owner is in the process of deciding whether or not to invest in a new factory that refines oil in order to meet the high demand for that commodity. A test showed that the old factory refines oil at a mean rate of 3.1L per second at a standard deviation of 1.0 using a sample size of 40. The new factory was measured to refine oil at a mean rate of 3.8L per second at a standard deviation of 1.5 using a sample size of 36. Determine if there is any major difference at a 10% significance level.

old:  $n_1 = 40$   $\bar{X}_1 = 3.1$   $S_1 = 1$   $H_0: \mu_1 = \mu_2$   $\mu_1 - \mu_2 = 0$   
 new:  $n_2 = 36$   $\bar{X}_2 = 3.8$   $S_2 = 1.5$   $H_a: \mu_1 \neq \mu_2$   $\mu_1 - \mu_2 \neq 0$

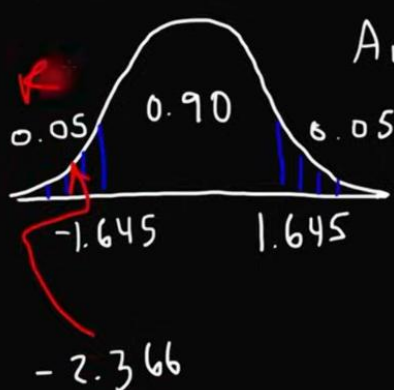


| z    | 0       | 0.01    | 0.02    | 0.03    | 0.04    | 0.05    | 0.06    | 0.07    | 0.08    | 0.09    |
|------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| +0   | 0.50000 | 0.50399 | 0.50798 | 0.51197 | 0.51596 | 0.51994 | 0.52392 | 0.52790 | 0.53188 | 0.53586 |
| +0.1 | 0.53983 | 0.54380 | 0.54776 | 0.55172 | 0.55567 | 0.55966 | 0.56360 | 0.56749 | 0.57142 | 0.57535 |
| +0.2 | 0.57926 | 0.58317 | 0.58706 | 0.59095 | 0.59483 | 0.59871 | 0.60257 | 0.60642 | 0.61026 | 0.61409 |
| +0.3 | 0.61791 | 0.62172 | 0.62552 | 0.62930 | 0.63307 | 0.63683 | 0.64058 | 0.64431 | 0.64803 | 0.65173 |
| +0.4 | 0.65542 | 0.65910 | 0.66276 | 0.66640 | 0.67003 | 0.67364 | 0.67724 | 0.68082 | 0.68439 | 0.68793 |
| +0.5 | 0.69146 | 0.69497 | 0.69847 | 0.70194 | 0.70540 | 0.70884 | 0.71226 | 0.71566 | 0.71904 | 0.72240 |
| +0.6 | 0.72575 | 0.72907 | 0.73237 | 0.73565 | 0.73891 | 0.74215 | 0.74537 | 0.74857 | 0.75175 | 0.75490 |
| +0.7 | 0.75804 | 0.76115 | 0.76424 | 0.76730 | 0.77035 | 0.77337 | 0.77637 | 0.77935 | 0.78230 | 0.78524 |
| +0.8 | 0.78814 | 0.79103 | 0.79389 | 0.79673 | 0.79955 | 0.80234 | 0.80511 | 0.80785 | 0.81057 | 0.81327 |
| +0.9 | 0.81594 | 0.81859 | 0.82121 | 0.82381 | 0.82639 | 0.82894 | 0.83147 | 0.83398 | 0.83646 | 0.83891 |
| +1.0 | 0.84134 | 0.84375 | 0.84614 | 0.84849 | 0.85083 | 0.85314 | 0.85543 | 0.85769 | 0.85993 | 0.86214 |
| +1.1 | 0.86433 | 0.86650 | 0.86864 | 0.87076 | 0.87286 | 0.87493 | 0.87698 | 0.87900 | 0.88100 | 0.88298 |
| +1.2 | 0.88493 | 0.88686 | 0.88877 | 0.89065 | 0.89251 | 0.89435 | 0.89617 | 0.89796 | 0.89973 | 0.90147 |
| +1.3 | 0.90320 | 0.90490 | 0.90658 | 0.90824 | 0.90988 | 0.91149 | 0.91308 | 0.91466 | 0.91621 | 0.91774 |
| +1.4 | 0.91924 | 0.92073 | 0.92220 | 0.92364 | 0.92507 | 0.92647 | 0.92785 | 0.92922 | 0.93056 | 0.93189 |
| +1.5 | 0.93319 | 0.93448 | 0.93574 | 0.93699 | 0.93822 | 0.93943 | 0.94062 | 0.94179 | 0.94295 | 0.94408 |
| +1.6 | 0.94520 | 0.94630 | 0.94738 | 0.94845 | 0.94950 | 0.95053 | 0.95154 | 0.95254 | 0.95352 | 0.95449 |
| +1.7 | 0.95543 | 0.95637 | 0.95728 | 0.95818 | 0.95906 | 0.95994 | 0.96080 | 0.96164 | 0.96246 | 0.96327 |
| +1.8 | 0.96407 | 0.96485 | 0.96562 | 0.96638 | 0.96712 | 0.96784 | 0.96854 | 0.96923 | 0.96990 | 0.97056 |
| +1.9 | 0.97125 | 0.97191 | 0.97257 | 0.97320 | 0.97381 | 0.97441 | 0.97500 | 0.97558 | 0.97615 | 0.97670 |
| +2.0 | 0.97725 | 0.97778 | 0.97831 | 0.97882 | 0.97932 | 0.97982 | 0.98030 | 0.98077 | 0.98124 | 0.98169 |
| +2.1 | 0.98214 | 0.98257 | 0.98300 | 0.98341 | 0.98382 | 0.98422 | 0.98461 | 0.98500 | 0.98537 | 0.98574 |
| +2.2 | 0.98610 | 0.98645 | 0.98679 | 0.98713 | 0.98745 | 0.98778 | 0.98809 | 0.98840 | 0.98870 | 0.98899 |
| +2.3 | 0.98928 | 0.98956 | 0.98983 | 0.99009 | 0.99036 | 0.99061 | 0.99086 | 0.99111 | 0.99134 | 0.99158 |
| +2.4 | 0.99180 | 0.99202 | 0.99224 | 0.99245 | 0.99266 | 0.99286 | 0.99305 | 0.99324 | 0.99343 | 0.99361 |
| +2.5 | 0.99379 | 0.99396 | 0.99413 | 0.99430 | 0.99446 | 0.99461 | 0.99477 | 0.99492 | 0.99506 | 0.99520 |
| +2.6 | 0.99534 | 0.99547 | 0.99560 | 0.99573 | 0.99585 | 0.99598 | 0.99609 | 0.99621 | 0.99632 | 0.99643 |
| +2.7 | 0.99653 | 0.99664 | 0.99674 | 0.99683 | 0.99693 | 0.99702 | 0.99711 | 0.99720 | 0.99728 | 0.99736 |
| +2.8 | 0.99744 | 0.99752 | 0.99760 | 0.99767 | 0.99774 | 0.99781 | 0.99788 | 0.99795 | 0.99801 | 0.99807 |
| +2.9 | 0.99813 | 0.99819 | 0.99825 | 0.99831 | 0.99836 | 0.99841 | 0.99846 | 0.99851 | 0.99856 | 0.99861 |
| +3.0 | 0.99865 | 0.99869 | 0.99874 | 0.99878 | 0.99882 | 0.99886 | 0.99889 | 0.99893 | 0.99896 | 0.99899 |
| +3.1 | 0.99903 | 0.99906 | 0.99909 | 0.99913 | 0.99916 | 0.99918 | 0.99921 | 0.99924 | 0.99926 | 0.99929 |
| +3.2 | 0.99931 | 0.99934 | 0.99936 | 0.99938 | 0.99940 | 0.99942 | 0.99944 | 0.99946 | 0.99948 | 0.99950 |
| +3.3 | 0.99952 | 0.99953 | 0.99955 | 0.99957 | 0.99958 | 0.99960 | 0.99961 | 0.99962 | 0.99964 | 0.99965 |
| +3.4 | 0.99966 | 0.99968 | 0.99969 | 0.99970 | 0.99971 | 0.99972 | 0.99973 | 0.99974 | 0.99975 | 0.99976 |
| +3.5 | 0.99977 | 0.99978 | 0.99979 | 0.99980 | 0.99981 | 0.99982 | 0.99983 | 0.99984 | 0.99985 | 0.99986 |
| +3.6 | 0.99987 | 0.99988 | 0.99989 | 0.99990 | 0.99991 | 0.99992 | 0.99993 | 0.99994 | 0.99995 | 0.99996 |
| +3.7 | 0.99997 | 0.99998 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| +3.8 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| +3.9 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |
| +4.0 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 | 0.99999 |



2. A business owner is in the process of deciding whether or not to invest in a new factory that refines oil in order to meet the high demand for that commodity. A test showed that the old factory refines oil at a mean rate of 3.1L per second at a standard deviation of 1.0 using a sample size of 40. The new factory was measured to refine oil at a mean rate of 3.8L per second at a standard deviation of 1.5 using a sample size of 36. Determine if there is any major difference at a 10% significance level.

old:  $n_1 = 40$   $\bar{X}_1 = 3.1$   $S_1 = 1$   $H_0: \mu_1 = \mu_2$   $\mu_1 - \mu_2 = 0$   
 new:  $n_2 = 36$   $\bar{X}_2 = 3.8$   $S_2 = 1.5$   $H_a: \mu_1 \neq \mu_2$   $\mu_1 - \mu_2 \neq 0$



$$A_L = 0.95 \quad z_c = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$z_c = \frac{3.1 - 3.8}{\sqrt{\frac{1^2}{40} + \frac{1.5^2}{36}}} = \frac{-0.7}{0.2958} = -2.366$$



## Hypothesis Testing With Two Proportions

1. Company XYZ manufactures laptops. For quality control, two sets of laptops were tested. In the first group, 32 out of 800 were found to contain some sort of defect. In the second group, 30 out of 500 were found to have a defect. Is the difference between the two groups significant? (use a significance level of 0.05)

$$n_1 = 800 \quad \hat{p}_1 = 0.04$$
$$n_2 = 500 \quad \hat{p}_2 = 0.06$$

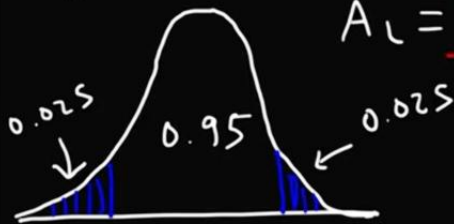
$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{32}{800} = 0.04$$

$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{30}{500} = 0.06$$

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$$n_1 = 800 \quad \hat{p}_1 = 0.04 \quad H_0: p_1 = p_2$$
$$n_2 = 500 \quad \hat{p}_2 = 0.06 \quad H_a: p_1 \neq p_2$$

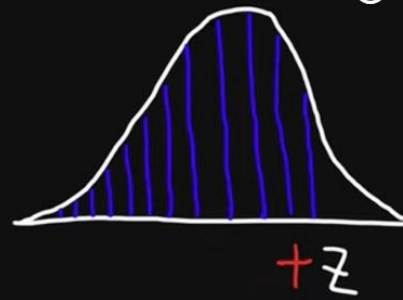
$$A_L = 0.975$$



Find critical z value from table



| Z    | 0       | 0.01    | 0.02    | 0.03    | 0.04    | 0.05    | 0.06    | 0.07    | 0.08    | 0.09    |
|------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| -0   | 0.50000 | 0.50399 | 0.50798 | 0.51197 | 0.51595 | 0.51993 | 0.52391 | 0.52789 | 0.53186 | 0.53586 |
| -0.1 | 0.53983 | 0.54380 | 0.54776 | 0.55172 | 0.55567 | 0.55962 | 0.56357 | 0.56752 | 0.57146 | 0.57540 |
| -0.2 | 0.57934 | 0.58329 | 0.58724 | 0.59119 | 0.59514 | 0.59909 | 0.60304 | 0.60699 | 0.61094 | 0.61489 |
| -0.3 | 0.61791 | 0.62172 | 0.62552 | 0.62930 | 0.63307 | 0.63683 | 0.64058 | 0.64431 | 0.64803 | 0.65173 |
| -0.4 | 0.65542 | 0.65910 | 0.66276 | 0.66640 | 0.67003 | 0.67364 | 0.67724 | 0.68082 | 0.68439 | 0.68793 |
| -0.5 | 0.69146 | 0.69497 | 0.69847 | 0.70194 | 0.70540 | 0.70884 | 0.71226 | 0.71566 | 0.71904 | 0.72240 |
| -0.6 | 0.72575 | 0.72907 | 0.73237 | 0.73565 | 0.73891 | 0.74215 | 0.74537 | 0.74857 | 0.75175 | 0.75490 |
| -0.7 | 0.75804 | 0.76115 | 0.76424 | 0.76730 | 0.77035 | 0.77337 | 0.77637 | 0.77935 | 0.78230 | 0.78524 |
| -0.8 | 0.78814 | 0.79103 | 0.79389 | 0.79673 | 0.79955 | 0.80234 | 0.80511 | 0.80788 | 0.81067 | 0.81342 |
| -0.9 | 0.81594 | 0.81859 | 0.82121 | 0.82381 | 0.82639 | 0.82894 | 0.83147 | 0.83398 | 0.83646 | 0.83891 |
| -1.0 | 0.84134 | 0.84375 | 0.84614 | 0.84849 | 0.85083 | 0.85314 | 0.85543 | 0.85769 | 0.85993 | 0.86214 |
| -1.1 | 0.86433 | 0.86650 | 0.86864 | 0.87076 | 0.87286 | 0.87493 | 0.87698 | 0.87900 | 0.88100 | 0.88298 |
| -1.2 | 0.88493 | 0.88686 | 0.88877 | 0.89065 | 0.89251 | 0.89435 | 0.89617 | 0.89796 | 0.89973 | 0.90147 |
| -1.3 | 0.90320 | 0.90490 | 0.90658 | 0.90824 | 0.90988 | 0.91149 | 0.91308 | 0.91466 | 0.91621 | 0.91774 |
| -1.4 | 0.91924 | 0.92073 | 0.92220 | 0.92364 | 0.92507 | 0.92647 | 0.92786 | 0.92922 | 0.93056 | 0.93189 |
| -1.5 | 0.93319 | 0.93448 | 0.93574 | 0.93699 | 0.93822 | 0.93943 | 0.94062 | 0.94179 | 0.94295 | 0.94408 |
| -1.6 | 0.94520 | 0.94630 | 0.94738 | 0.94845 | 0.94950 | 0.95053 | 0.95154 | 0.95254 | 0.95352 | 0.95449 |
| -1.7 | 0.95543 | 0.95637 | 0.95728 | 0.95815 | 0.95900 | 0.95984 | 0.96068 | 0.96149 | 0.96226 | 0.96302 |
| -1.8 | 0.96377 | 0.96450 | 0.96520 | 0.96588 | 0.96653 | 0.96717 | 0.96780 | 0.96841 | 0.96900 | 0.96958 |
| -1.9 | 0.97015 | 0.97073 | 0.97129 | 0.97183 | 0.97235 | 0.97286 | 0.97335 | 0.97382 | 0.97428 | 0.97473 |
| -2.0 | 0.97517 | 0.97560 | 0.97601 | 0.97641 | 0.97680 | 0.97717 | 0.97753 | 0.97788 | 0.97822 | 0.97855 |
| -2.1 | 0.97887 | 0.97919 | 0.97949 | 0.97978 | 0.98006 | 0.98033 | 0.98059 | 0.98084 | 0.98108 | 0.98131 |
| -2.2 | 0.98154 | 0.98176 | 0.98197 | 0.98217 | 0.98236 | 0.98254 | 0.98271 | 0.98287 | 0.98303 | 0.98318 |
| -2.3 | 0.98332 | 0.98347 | 0.98361 | 0.98375 | 0.98389 | 0.98402 | 0.98415 | 0.98428 | 0.98440 | 0.98452 |
| -2.4 | 0.98464 | 0.98476 | 0.98488 | 0.98499 | 0.98510 | 0.98521 | 0.98532 | 0.98543 | 0.98554 | 0.98564 |
| -2.5 | 0.98574 | 0.98584 | 0.98594 | 0.98604 | 0.98613 | 0.98623 | 0.98632 | 0.98641 | 0.98650 | 0.98658 |
| -2.6 | 0.98667 | 0.98675 | 0.98683 | 0.98691 | 0.98699 | 0.98707 | 0.98715 | 0.98723 | 0.98730 | 0.98737 |
| -2.7 | 0.98744 | 0.98751 | 0.98758 | 0.98765 | 0.98772 | 0.98779 | 0.98785 | 0.98792 | 0.98798 | 0.98804 |
| -2.8 | 0.98810 | 0.98816 | 0.98821 | 0.98827 | 0.98832 | 0.98837 | 0.98842 | 0.98847 | 0.98852 | 0.98856 |
| -2.9 | 0.98861 | 0.98865 | 0.98869 | 0.98873 | 0.98877 | 0.98881 | 0.98885 | 0.98889 | 0.98893 | 0.98896 |
| -3.0 | 0.98900 | 0.98903 | 0.98906 | 0.98909 | 0.98912 | 0.98915 | 0.98918 | 0.98921 | 0.98924 | 0.98927 |
| -3.1 | 0.98930 | 0.98933 | 0.98936 | 0.98938 | 0.98940 | 0.98942 | 0.98944 | 0.98946 | 0.98948 | 0.98950 |
| -3.2 | 0.98952 | 0.98953 | 0.98955 | 0.98957 | 0.98958 | 0.98960 | 0.98961 | 0.98962 | 0.98964 | 0.98965 |
| -3.3 | 0.98966 | 0.98968 | 0.98969 | 0.98970 | 0.98971 | 0.98972 | 0.98973 | 0.98974 | 0.98975 | 0.98976 |
| -3.4 | 0.98977 | 0.98978 | 0.98979 | 0.98980 | 0.98981 | 0.98982 | 0.98983 | 0.98984 | 0.98985 | 0.98986 |
| -3.5 | 0.98987 | 0.98988 | 0.98989 | 0.98990 | 0.98991 | 0.98992 | 0.98993 | 0.98994 | 0.98995 | 0.98996 |
| -3.6 | 0.98997 | 0.98998 | 0.98999 | 0.99000 | 0.99001 | 0.99002 | 0.99003 | 0.99004 | 0.99005 | 0.99006 |
| -3.7 | 0.99007 | 0.99008 | 0.99009 | 0.99010 | 0.99011 | 0.99012 | 0.99013 | 0.99014 | 0.99015 | 0.99016 |
| -3.8 | 0.99017 | 0.99018 | 0.99019 | 0.99020 | 0.99021 | 0.99022 | 0.99023 | 0.99024 | 0.99025 | 0.99026 |
| -3.9 | 0.99027 | 0.99028 | 0.99029 | 0.99030 | 0.99031 | 0.99032 | 0.99033 | 0.99034 | 0.99035 | 0.99036 |
| -4.0 | 0.99037 | 0.99038 | 0.99039 | 0.99040 | 0.99041 | 0.99042 | 0.99043 | 0.99044 | 0.99045 | 0.99046 |



1. Company XYZ manufactures laptops. For quality control, two sets of laptops were tested. In the first group, 32 out of 800 were found to contain some sort of defect. In the second group, 30 out of 500 were found to have a defect. Is the difference between the two groups significant? (use a significance level of 0.05)

$$n_1 = 800 \quad \hat{p}_1 = 0.04$$

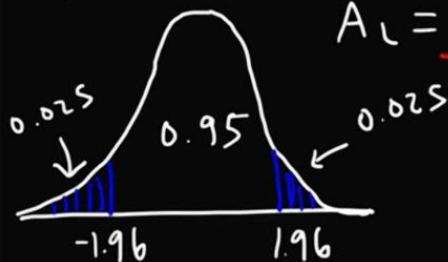
$$n_2 = 500 \quad \hat{p}_2 = 0.06$$

$$H_0: p_1 = p_2 \quad p_1 - p_2 = 0$$

$$H_a: p_1 \neq p_2$$

$$A_c = 0.975$$

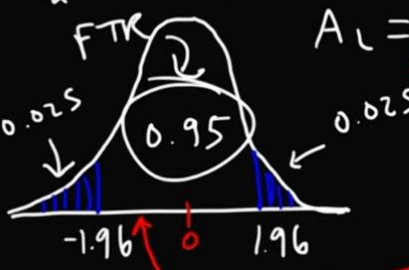
$$Z_c = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$



$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{32 + 30}{800 + 500} = \frac{62}{1300} = 0.04769$$

We calculate generic sample proportion also

1. Company XYZ manufactures laptops. For quality control, two sets of laptops were tested. In the first group, 32 out of 800 were found to contain some sort of defect. In the second group, 30 out of 500 were found to have a defect. Is the difference between the two groups significant? (use a significance level of 0.05)

$n_1 = 800$   $\hat{p}_1 = 0.04 \rightarrow H_0: p_1 = p_2$   $p_1 - p_2 = 0$  Show  
 $n_2 = 500$   $\hat{p}_2 = 0.06$   ~~$H_a: p_1 \neq p_2$~~   
 $\alpha = 0.05$   $A_c = 0.975$   $z_c = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$   
  
 $z_c = \frac{(0.04 - 0.06) - (0)}{\sqrt{0.04769(1-0.04769)(\frac{1}{800} + \frac{1}{500})}} = \frac{-0.02}{0.012149} = -1.646$

There is not a significant difference between two proportion

## ANOVA

### 1. What is ANOVA?

ANOVA = Analysis of Variance

It is used to compare more than 2 population means.

Instead of doing many t-tests (which increases error), ANOVA tests everything in one go.

### 2. Basic Idea

ANOVA checks:

Are the differences between group means real or just due to random variation?

It does this by comparing:

- Variation between groups
- Variation within groups



### ◆ 3. Hypotheses (VERY IMPORTANT — always asked)

From your slide:

#### Null Hypothesis ( $H_0$ )

All population means are equal

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

#### Alternative Hypothesis ( $H_1$ )

At least one mean is different

$H_1$  : At least two means are not equal

### ◆ 4. Key Terms (from your slide)

- $n$  = total number of observations
- $n_j$  = size of  $j$ -th sample
- $T_j$  = sum of  $j$ -th sample

### ◆ 5. Types of Variation

ANOVA splits total variation into:

#### 1. Total Sum of Squares (SST)

Overall variation

#### 2. Sum of Squares Between Groups (SSA)

Variation due to **differences between means**

#### 3. Sum of Squares Within Groups (SSE)

Variation due to **random error**

## ◆ 5. Types of Variation

ANOVA splits total variation into:

### 1. Total Sum of Squares (SST)

Overall variation

### 2. Sum of Squares Between Groups (SSA)

Variation due to **differences between means**

### 3. Sum of Squares Within Groups (SSE)

Variation due to **random error**

## ◆ 6. Core Relationship

$$SST = SSA + SSE$$

## ◆ 7. Your Example (from slides)

You are given 4 samples:

| Sample 1 | Sample 2 | Sample 3 | Sample 4 |
|----------|----------|----------|----------|
| 11       | 9        | 16       | 5        |
| 6        | 2        | 10       | 1        |
| 7        | 4        | 10       | 3        |

Totals:

- $T_1 = 24$
- $T_2 = 15$
- $T_3 = 36$
- $T_4 = 9$
- Grand total = 84

Means:

- 8, 5, 12, 3

## ◆ 8. Step-by-Step Calculations

### ◆ Step 1: Calculate SST

From slide:

$$SST = \sum y_{ij}^2 - \frac{T^2}{n}$$
$$SST = 798 - \frac{84^2}{12} = 210$$



## ◆ 1. What is $y_{ij}$ ?

👉  $y_{ij}$  means:

the i-th observation in the j-th group (sample)

### ◆ Break it down:

- j = group number (sample 1, sample 2, etc.)
- i = observation inside that group

### ◆ From your data:

| Sample (j)     | Values (i) |
|----------------|------------|
| Sample 1 (j=1) | 11, 6, 7   |
| Sample 2 (j=2) | 9, 2, 4    |
| Sample 3 (j=3) | 16, 10, 10 |
| Sample 4 (j=4) | 5, 1, 3    |

## ◆ 2. What is $\sum y_{ij}^2$ ?

◆ Get Plus ×

This means:

Take every observation, square it, then add them all

### ◆ From your slide:

We compute:

$$\sum y_{ij}^2$$

Step-by-step:

Sample 1:

$$11^2 + 6^2 + 7^2 = 121 + 36 + 49 = 206$$

Sample 2:

$$9^2 + 2^2 + 4^2 = 81 + 4 + 16 = 101$$

Sample 3:

$$16^2 + 10^2 + 10^2 = 256 + 100 + 100 = 456$$

Sample 4:

$$5^2 + 1^2 + 3^2 = 25 + 1 + 9 = 35$$

### ◆ Total:

$$\sum y_{ij}^2 = 206 + 101 + 456 + 35 = 798$$

✓ This is exactly the **798** in your slide

## ◆ 3. Full SST Formula (Complete Understanding)

Now you can understand the full formula:

$$SST = \sum y_{ij}^2 - \frac{T^2}{n}$$

◆ Each term meaning:

✓  $\sum y_{ij}^2$

Sum of squared observations (we just calculated = 798)

---

✓  $T$

Grand total of all observations

$$T = 84$$

---

✓  $n$

Total number of observations

$$n = 12$$

---

◆ Substitute values:

$$SST = 798 - \frac{84^2}{12}$$

$$SST = 798 - 588 = 210$$

## ◆ 1. Your Statement

✦ Get Plus ✕

You said:

"Isn't  $SST = SSB + SSW$ ?"

✓ Yes — absolutely correct.

Just different naming:

- $SSB$  (Sum of Squares Between) =  $SSA$
- $SSW$  (Sum of Squares Within) =  $SSE$

So:

$$SST = SSA + SSE$$

✓ Same formula, just different notation.

## ◆ 2. Then Why Did I Calculate SST Directly?

Because there are **two valid ways** to compute SST:

✓ **Method 1 (Direct Formula — what your slide used)**

$$SST = \sum y_{ij}^2 - \frac{T^2}{n}$$

From your slide:

- $\sum y_{ij}^2 = 798$
- $T = 84, n = 12$



So:

## ◆ 8. Step-by-Step Calculations

### ◆ Step 1: Calculate SST

From slide:

$$SST = \sum y_{ij}^2 - \frac{T^2}{n}$$
$$SST = 798 - \frac{84^2}{12} = 210$$

### ◆ Step 2: Calculate SSA

$$SSA = \sum \frac{T_j^2}{n_j} - \frac{T^2}{n}$$
$$SSA = \frac{2178}{3} - \frac{84^2}{12} = 138$$

### ◆ Step 3: Calculate SSE

$$SSE = SST - SSA = 210 - 138 = 72$$

K = no of groups

## ◆ 9. Degrees of Freedom (VERY IMPORTANT)

| Source        | Formula              | Copy |
|---------------|----------------------|------|
| Between (SSA) | $k - 1 = 4 - 1 = 3$  |      |
| Within (SSE)  | $n - k = 12 - 4 = 8$ |      |
| Total         | $n - 1 = 11$         |      |

MSA = MEAN SQUARE BETWEEN GROUPS

MSE = MEAN SQUARE WITHIN GROUPS

## ◆ 10. Mean Squares

$$MSA = \frac{SSA}{df_A} = \frac{138}{3} = 46$$
$$MSE = \frac{SSE}{df_E} = \frac{72}{8} = 9$$



### ◆ 11. F-Test Statistic

$$F = \frac{MSA}{MSE}$$

$$F = \frac{46}{9} = 5.11$$

### ◆ 12. Critical Value

From slide:

$$F_{0.05,3,8} = 4.07$$

Decision rule:

👉 Reject  $H_0$  if

$$F > 4.07$$

### ◆ 13. Final Decision

- Calculated  $F = 5.11$
- Critical  $F = 4.07$

✅ Since:

$$5.11 > 4.07$$

👉 Reject  $H_0$

## ◆ What is this?

You saw:

$$F_{0.05,3,8} = 4.07$$

This is called the **critical value** from the F-distribution.

## ◆ 1. What does each number mean?

✓ 0.05 → Significance level ( $\alpha$ )

- This is the **probability of making a Type I error**
- Usually given in question

👉 Here:

$$\alpha = 0.05 = 5\%$$

Meaning:

You allow a 5% chance of wrongly rejecting  $H_0$

✓ 3 → Numerator degrees of freedom ( $df_1$ )

This comes from **between groups (SSA)**

$$df_1 = k - 1$$

- $k$  = number of groups = 4

$$df_1 = 4 - 1 = 3$$

✓ 8 → Denominator degrees of freedom ( $df_2$ )

This comes from **within groups (SSE)**

$$df_2 = n - k$$

- $n$  = total observations = 12
- $k$  = 4

$$df_2 = 12 - 4 = 8$$

## ◆ 2. So what is $F_{0.05,3,8}$ ?

It means:

The **critical cutoff value** of F when:

- $\alpha = 0.05$
- $df_1 = 3$
- $df_2 = 8$

## ◆ 3. How did we get 4.07?

👉 This value is **NOT** calculated manually

It is taken from:

- F-table (in exams)
- Excel / calculator / software

### ◆ From F-table:

You go like this:

1. Find  $df_1 = 3$  (top row)
2. Find  $df_2 = 8$  (side column)
3. Look under  $\alpha = 0.05$

👉 You get:



$$F = 4.07$$

## ◆ 4. What is this value used for?

It defines the **critical region** (rejection region)

**Decision Rule:**

$$\text{Reject } H_0 \text{ if } F_{\text{calculated}} > F_{\text{critical}}$$

**In your case:**

- $F_{\text{calculated}} = 5.11$
- $F_{\text{critical}} = 4.07$

👉 Since:

$$5.11 > 4.07$$

✅ Reject  $H_0$

## ◆ 5. Intuition (VERY IMPORTANT)

Think of it like this:

- If  $F$  is **small** → groups are similar
- If  $F$  is **large** → groups are very different

👉 4.07 is the **cutoff point**

## ◆ 14. Conclusion (Exam Style)

👉 There is sufficient evidence to conclude that:

At least one population mean is different

## ◆ 15. ANOVA Table (VERY IMPORTANT FOR EXAMS)

| Source        | SS  | df | MS | F    |
|---------------|-----|----|----|------|
| Between (SSA) | 138 | 3  | 46 | 5.11 |
| Within (SSE)  | 72  | 8  | 9  |      |
| Total (SST)   | 210 | 11 |    |      |



## ◆ 16. Full Procedure (How to Answer ANY ANOVA Question)

Memorize this flow:

Step 1: State hypotheses

Step 2: Choose  $\alpha$  (e.g., 0.05)

Step 3: Compute SST, SSA, SSE

Step 4: Find degrees of freedom

Step 5: Compute MSA and MSE

Step 6: Calculate  $F$

Step 7: Find  $F$ -critical

Step 8: Compare

Step 9: Conclusion

## ◆ 17. Key Concepts They May [Get Plus](#) ×

? Why use ANOVA instead of t-test?

Because multiple t-tests increase error rate.

? What does large F mean?

👉 Group means are very different.

? What does small F mean?

👉 Differences are due to randomness.

? When do we reject  $H_0$ ?

👉 When:

$$F_{\text{calculated}} > F_{\text{critical}}$$

? What is SSE?

👉 Random error (within groups)

? What is SSA?

👉 Variation due to treatment/groups



## Example # 1

- A researcher wishes to try three different techniques to lower the blood pressure of individuals diagnosed with high blood pressure. The subjects are randomly assigned to three groups; the first group takes medication, the second group exercises, and the third group follows a special diet. After four weeks, the reduction in each person's blood pressure is recorded. At  $\alpha = 0.05$ , test the claim that there is no difference among the means. The data follow.

| Medication | Exercise | Diet |
|------------|----------|------|
| 10         | 6        | 5    |
| 12         | 8        | 9    |
| 9          | 3        | 12   |
| 15         | 0        | 8    |
| 13         | 2        | 4    |

### ◆ Step 1: Data

Groups:

Medication: 10, 12, 9, 15, 13

Exercise: 6, 8, 3, 0, 2

Diet: 5, 9, 12, 8, 4

### ◆ Step 2: Hypotheses

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

$H_1$  : At least one mean is different

$$\alpha = 0.05$$

### ◆ Step 3: Basic Values

| Group      | Sum ( $T_i$ ) | Mean |
|------------|---------------|------|
| Medication | 59            | 11.8 |
| Exercise   | 19            | 3.8  |
| Diet       | 38            | 7.6  |

- Total  $T = 116$
- Total observations  $n = 15$



#### ◆ Step 4: Compute SST

First calculate:

$$\sum y_{ij}^2$$

- Medication: 719
- Exercise: 113
- Diet: 330

$$\sum y_{ij}^2 = 1162$$

Now:

$$SST = \sum y_{ij}^2 - \frac{T^2}{n}$$

$$SST = 1162 - \frac{116^2}{15}$$

$$SST = 1162 - 897.07 = 264.93$$

#### ◆ Step 5: Compute SSA

$$SSA = \sum \frac{T_j^2}{n_j} - \frac{T^2}{n}$$

$$= \left( \frac{59^2}{5} + \frac{19^2}{5} + \frac{38^2}{5} \right) - \frac{116^2}{15}$$

$$= (696.2 + 72.2 + 288.8) - 897.07$$

$$SSA = 1057.2 - 897.07 = 160.13$$

#### ◆ Step 6: Compute SSE

$$SSE = SST - SSA$$

$$SSE = 264.93 - 160.13 = 104.80$$

#### ◆ Step 7: Degrees of Freedom

- Between:  $k - 1 = 3 - 1 = 2$
- Within:  $n - k = 15 - 3 = 12$
- Total: 14



### ◆ Step 8: Mean Squares

$$MSA = \frac{SSA}{2} = \frac{160.13}{2} = 80.07$$

$$MSE = \frac{SSE}{12} = \frac{104.80}{12} = 8.73$$

### ◆ Step 9: F-Test

$$F = \frac{MSA}{MSE} = \frac{80.07}{8.73} \approx 9.17$$

### ◆ Step 10: Critical Value

$$F_{0.05,2,12} \approx 3.89$$

### ◆ Step 11: Decision

- Calculated F = 9.17
- Critical F = 3.89

$$9.17 > 3.89$$

✓ Reject  $H_0$

### ◆ Step 12: Final Conclusion (EXAM STYLE)

👉 There is sufficient evidence at  $\alpha = 0.05$  to conclude that:

There is a significant difference among the three techniques (medication, exercise, and diet).

### ◆ Step 13: ANOVA Table

| Source  | SS     | df | MS    | F    |
|---------|--------|----|-------|------|
| Between | 160.13 | 2  | 80.07 | 9.17 |
| Within  | 104.80 | 12 | 8.73  |      |
| Total   | 264.93 | 14 |       |      |

## ANOVA (Analysis of Variance)

→ F-Test, F-statistic

→ F-value.

→ compare more than two groups.

$H_0$  :- All samples mean are equal

$H_a$  :- At least one <sup>pair of</sup> sample is significantly different.

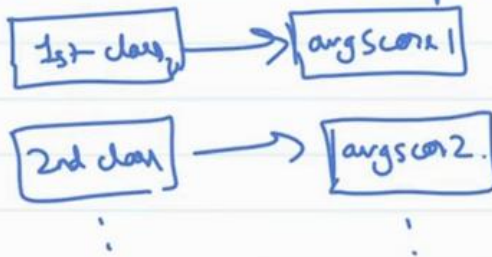
↳ One way ANOVA

↳ Two way ANOVA.

### ↳ One Way ANOVA



3 class. → 3 sample.



In two way anova , two factors are influencing the response variable



$$MSW = \frac{SSW}{df_w}$$

$df_w = N - K$  ,  $N$  = total no. of observations.

$$SSW = \sum (x_{ij} - \bar{x}_i)^2$$

$x_{ij}$  = each individual data point within group

$\bar{x}_i$  = mean of  $i$ th group.